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DEPARTMENT OF MECHANICAL ENGINEERING

ENGINEERING MATERIALS I
(ME 281)

FOR ENGINEERING STUDENTS

(MECHANICAL ENG. LEVEL II & AEROSPACE ENG. LEVEL I)

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OCTOBER, 2020

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CHAPTER 1. INTRODUCTION

This introductory chapter discusses the historical concept of materials and the importance of studying material science and engineering. It also covers the classification of materials.

Learning objectives

After studying this chapter, you should be able to do the following:

- Describe the concept of material science from historical viewpoint.
- State the importance of studying engineering materials.
- Identify the four main components of material science and their interrelation.
- Explain the various material classifications with examples.

1.1 Background and Material Science

1.1.1 Historical Background

Materials are very important to development that the historians have identified early periods of civilization by the name of most significantly used material such as Stone Age, Bronze Age and Iron Age. This is just an observation made to showcase the importance of materials and their impact on human civilization. It is obvious that materials have affected and controlled a broad range of human activities through thousands of decades.

From the historical point of view, it can be said that human civilization started with Stone Age where people used only natural materials, like stone, clay, skin, and wood for the purposes of making weapons, instruments, shelter, etc. Thus the sites of deposits for better quality stones became early colonies of human civilization. However, the increasing need for better quality tools brought forth exploration that led to Bronze Age, followed by Iron Age. When people found copper and how to make it harder by alloying, the Bronze Age started about 3000 BC. The use of iron and steel, a stronger material that gave advantage in wars started at about 1200 BC. Iron was abundant and thus availability was not limited to the privileged. This common material affected every person in many aspects, gaining the name democratic material.

The next big step in human civilization was the discovery of a cheap process to make steel around 1850 AD, which enabled the railroads and the building of the modern infrastructure of the industrial world. One of the most significant features of the democratic material is that the number of users just exploded. Thus there has been a need for human and material resources for centuries, still going strong. It is being said and agreed that we are presently in Space Age marked by many technological advancement towards materials development resulting in stronger and light materials like composites, electronic materials like semiconductors, materials for space voyage like high temperature ceramics, biomaterials, etc.

In summary, foundation of technology rest on materials. The history of human civilization evolved from the Stone Age to the Bronze Age, Iron Age, Steel Age, and to the Space Age (synchronous with the Electronic Age). Each age is marked by the advent of certain materials. The Iron Age brought tools and utensils whereas the Steel Age brought railroads, instruments, and the Industrial Revolution. The Space Age brought about the materials for stronger and light structures (e.g., composite materials) while the Electronic Age brought semiconductors, and thus many varieties of electronic gadgets.

1.1.2 Material Science

As engineering materials constitute foundation of technology, it is not only important but a must to understand how materials behave the way they do and why they differ in properties. This is only possible with the atomistic understanding allowed by quantum mechanics that first explained atoms and then solids starting in the 1930s. The combination of physics, chemistry, and the focus on the relationship between the properties of a material and its microstructure is the domain of Materials Science. The development of this science allowed designing materials and provided a knowledge base for the engineering applications.

Important components of the materials science subject are *structure*, *properties*, *processing*, and *performance*. The structure of a material usually relates to the arrangement of its internal components. On an atomic level, structure encompasses the organization of atoms or molecules relative to one another. The property refers to the material kind and magnitude of response to a specific imposed stimulus. In general, definitions of properties are made independent of the material shape and size. In addition to structure and properties are processing and performance. With regard to the relationships between these four components, the structure of a material will depend on how it is processed. Furthermore, a material's performance will be a function of its properties. Thus, a schematic interrelation between these four components is shown in Figure 1.1.



Figure 1.1: Interrelation between four components of Materials Science.

1.2 Importance and Classification of Materials

1.2.1 Why Study Material Science and Engineering?

All engineers need to know about materials because even software or system engineering depend on the development of new materials, which in turn alter the economics, such as the software-hardware trade-offs. Increasing applications of system engineering are in materials manufacturing (industrial engineering) and complex environmental systems.

Innovation in engineering often means the clever use of a new material for a specific application. For example: plastic containers in place of age-old metallic containers. It is well learnt lesson that engineering disasters are frequently caused by the misuse of materials. So it is vital that the professional engineer should know how to select materials which best fit the demands of the design - economic and aesthetic demands, as well as demands of strength and durability. Beforehand the designer must understand the properties of materials, and their limitations. Thus it is very important that every engineer study and understand the concepts of materials science and engineering. This enables the engineer

- To select a material for a given use based on considerations of cost and performance.
- To understand the limits of materials and the change of their properties with use.
- To be able to create a new material that will have some desirable properties.
- To be able to use the material for different application.

1.2.2 Classification of Materials

Like many other things, materials are classified into groups. The classification is based on many criteria, for example crystal structure (arrangement of atoms and bonds between them), properties, or use. Metals, ceramics, polymers, composites, semiconductors, and biomaterials constitute the main classes of present engineering materials.

1.2.2.1 Metals

These materials are characterized by high thermal and electrical conductivity; strong yet deformable under applied mechanical loads; opaque to light (shiny if polished). These characteristics are due to valence electrons that are detached from atoms, and spread in an electron sea that glues the ions together, i.e. atoms are bound together by metallic bonds and weaker Van der Waals forces. Pure metals are not good enough for many applications, especially structural applications. Thus metals are used in alloy form i.e. a metal mixed with another metal to improve the desired qualities. E.g.: aluminium, steel, brass, gold.

1.2.2.2 Ceramics

These are inorganic compounds, and usually made either of oxides, carbides, nitrides, or silicates of metals. Ceramics are typically partly crystalline and partly amorphous. Atoms (ions often) in ceramic materials behave mostly like either positive or negative ions, and are bound by very strong Coulomb forces between them. These materials are characterized by very high strength under compression, low ductility; usually insulators to heat and electricity. Examples: glass, porcelain, many minerals.

1.2.2.3 Polymers

Polymers in the form of thermo-plastics (nylon, polyethylene, polyvinyl chloride, rubber, etc.) consist of molecules that have covalent bonding within each molecule and van der Waals forces between them. Polymers in the form of thermo-sets (e.g., epoxy, phenolics, etc.) consist of a network of covalent bonds. They are based on H, C and other non-metallic elements. Polymers are amorphous, except for a minority of thermoplastics. Due to the kind of bonding, polymers are typically electrical and thermal insulators. However, conducting polymers can be obtained by doping, and conducting polymer-matrix composites can be obtained by the use of conducting fillers. They decompose at moderate temperatures of about 100 to 400 °C, and are lightweight. Other properties vary greatly.

1.2.2.4 Composite Materials

Composite materials are multiphase materials obtained by artificial combination of different materials to attain properties that the individual components cannot attain. An example is a lightweight brake disc obtained by embedding SiC particles in Al-alloy matrix. Another example is reinforced cement concrete, a structural composite obtained by combining cement (the matrix, i.e., the binder, obtained by a reaction known as hydration, between cement and water), sand (fine aggregate), gravel (coarse aggregate), and, thick steel fibers. However, there are some natural composites available in nature, for example wood. In general, composites are classified according to their matrix materials. The main classes of composites are metal-matrix, polymer-matrix, and ceramic-matrix.

1.2.2.5 Semiconductors

Semiconductors are covalent in nature. Their atomic structure is characterized by the highest occupied energy band (the valence band, where the valence electrons reside energetically) full such that the energy gap between the top of the valence band and the bottom of the empty energy band (the conduction band) is small enough for some fraction of the valence electrons to be excited from the valence band to the conduction band by thermal, optical, or other forms of energy. Their electrical properties depend extremely strongly on minute proportions of contaminants. They are usually doped in order to enhance electrical conductivity. They are used in the form of single crystals without dislocations because the grain boundaries and dislocations would degrade electrical behavior. They are opaque to visible light but transparent to the infrared. Examples: silicon (Si), germanium (Ge), and gallium arsenide (GaAs, which is a compound semiconductor).

1.2.2.6 Biomaterials

These are any type material that can be used for replacement of damaged or diseased human body parts. Primary requirement of these materials is that they must be biocompatible with body tissues, and must not produce toxic substances. Other important material factors are: ability to support forces; low friction, wear, density, cost; reproducibility. Typical applications involve heart valves, hip joints, dental implants, intraocular lenses. Examples: Stainless steel, Co-28Cr-6Mo, Ti-6Al-4V, ultra-high molecular weight poly-ethylene, high purity dense Al-oxide, etc.

1.3 Advanced, Future and Modern Materials

1.3.1 Advanced Materials

These are materials used in High-Tech devices those operate based on relatively intricate and sophisticated principles (e.g. computers, air/space-crafts, electronic gadgets, etc.). These materials are either traditional materials with enhanced properties or newly developed materials with high-performance capabilities. Hence these are relatively expensive. Typical applications: integrated circuits, lasers, fibre optics, thermal protection for space shuttle, etc. Examples: Metallic foams, inter-metallic compounds, multi-component alloys, magnetic alloys, special ceramics and high temperature materials, etc.

1.3.2 Future Materials

Group of new and state-of-the-art materials now being developed, and expected to have significant influence on present-day technologies, especially in the fields of medicine, manufacturing and defence. Smart/Intelligent material system consists some type of sensor (detects an input) and an actuator (performs responsive and adaptive function). Actuators may be called upon to change shape, position, natural frequency, mechanical characteristics in response to changes in temperature, electric/magnetic fields, moisture, pH, etc.

Four types of materials used as actuators: Shape memory alloys, Piezo-electric ceramics, Magneto-strictive materials, Electro-/Magneto-rheological fluids. Materials / Devices used as sensors: Optical fibers, Piezo-electric materials, Micro-electro-mechanical systems (MEMS), etc.

Typical applications: By incorporating sensors, actuators and chip processors into system, researchers are able to stimulate biological human-like behaviour; Fibres for bridges, buildings, and wood utility poles; They also help in fast moving and accurate robot parts, high speed helicopter rotor blades; Actuators that control chatter in precision machine tools; Small microelectronic circuits in machines ranging from computers to photolithography prints; Health monitoring detecting the success or failure of a product.

1.3.3 Modern Materials

Though there has been tremendous progress over the decades in the field of materials science and engineering, innovation of new technologies, and need for better performances of existing technologies demands much more from the materials field. Moreover it is evident that new materials/technologies are needed to be environmental friendly. Some typical needs, thus, of modern materials needs are listed in the following:

- Engine efficiency increases at high temperatures: requires high temperature structural materials
- Use of nuclear energy requires solving problem with residues, or advances in nuclear waste processing.
- Hypersonic flight requires materials that are light, strong and resist high temperatures.
- Optical communications require optical fibers that absorb light negligibly.
- Civil construction – materials for unbreakable windows.
- Structures: materials that are strong like metals and resist corrosion like plastics.

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Chapter 2. ATOMIC STRUCTURE AND INTERATOMIC BONDING

This chapter considers several fundamental and important concepts namely atomic structure, electron configurations in atoms and the various types of primary and secondary interatomic bonds that hold together the atoms that compose a solid.

Learning objectives

After studying this chapter, you should be able to do the following:

- Explain the concept of atomic structure.
- Describe Bohr atomic model in relation to quantum mechanics.
- Identify the principal quantum numbers and their energy levels.
- State the Pauli Exclusion Principle and list the electron configuration of elements.
- Explain the types of primary bonding in solids
- Define van der Waals or secondary bonding

2.1 Atomic Structure

2.1.1 Fundamental Concepts

Atoms are composed of electrons, protons, and neutrons. Electrons and protons are negative and positive charged particles respectively. The magnitude of each charged particle in an atom is 1.6×10^{-19} Coulombs. The mass of the electron is negligible compared to those of the proton and the neutron, which form the nucleus of the atom. The unit of mass is an atomic mass unit (*amu*) which is equal to 1.66×10^{-27} kg, and equals one-twelfth the atomic mass (A) of the most common isotope of carbon, carbon 12 (^{12}C) (A = 12.00000). The nucleus of carbon 12 has the number of protons (Z) equal to 6, and the number of neutrons (N) equal to 6. The neutrons and protons have very similar masses, roughly equal to 1 *amu* each. A neutral atom has the same number of electrons and protons.

A mole (*mol*) is the amount of matter that has a mass in grams equal to the atomic mass in *amu* of the atoms. Thus, a mole of carbon has a mass of 12 grams. The number of atoms in one mole is called the Avogadro number, $N_{av} = 6.023 \times 10^{23}$. Note that $N_{av} = 1$ gram/1 amu.
Calculating n, the number of atoms per cm^3 of a material of density δ (g/cm^3):

$$n = N_{av}\frac{\delta}{M} \qquad (2.1)$$

Where *M* is the atomic mass in *amu* (*grams* per *mol*). Thus, for graphite (carbon) with a density $\delta = 1.8$ g/cm^3, *M* =12, we get 6×10^{23} atoms/mol × 1.8 g/cm^3 / 12 g/mol) = 9×10^{22} C atoms/cm^3.

For a molecular solid like ice, one uses the molecular mass, $M_{(H2O)} = 18$. With a density of 1 g/cm^3, one obtains n = 3.3×10^{22} H_2O molecules/cm^3. Note that since the water molecule contains 3 atoms, this is equivalent to 9.9×10^{22} atoms/cm^3.

Most solids have atomic densities around 6×10^{22} atoms/cm^3. The cube root of that number gives the number of atoms per centimeter, about 39 million. The mean distance between atoms is the inverse of that, *or* 0.25 nm. This is an important number that gives the scale of atomic structures in solids.

2.1.2 Electrons in Atoms

2.1.2.1 Bohr atomic model

An understanding of the behavior of electrons in atoms necessarily involves the discussion of quantum-mechanical concepts. One projection of quantum mechanics was the simplified Bohr atomic model, in which electrons are assumed to revolve around the atomic nucleus in discrete orbitals, and the position of any particular electron is more or less well defined in terms of its orbital. This model of the atom is represented in Figure 2.1.

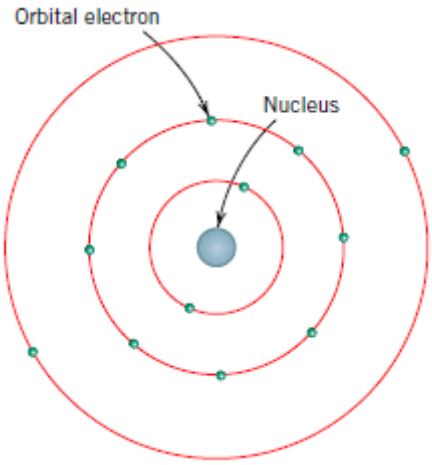


Figure 2.1 Schematic representation of Bohr atom

Another important quantum-mechanical principle stipulates that the energies of electrons are quantized; thus, electrons are permitted to have only specific values of energy. An electron may change energy, but in so doing, it must make a quantum jump either to an allowed higher energy (by absorbing energy) or to a lower energy (with emission of energy). Usually, it is convenient to think of these allowed electron energies as being associated with energy levels or states. These states do not vary continuously with energy; thus, adjacent states are separated by finite energies.

The Bohr model represents an early attempt to describe electrons in atoms, in terms of both electron orbitals and quantized energy levels. This Bohr model was eventually found to have some significant limitations because of its inability to explain several phenomena involving electrons. To resolve this, the wave-mechanical model was developed, which considered an electron to exhibit both wavelike and particle-like characteristics. With this model, an electron is no longer treated as a particle moving in a discrete orbital; rather, position is considered to be the probability of an electron's being at various locations around the nucleus.

2.1.2.2 Quantum numbers

By applying wave mechanics, every electron in an atom is characterized by four parameters called quantum numbers. The size, shape, and spatial orientation of an electron's probability density are specified by three of these quantum numbers. Furthermore, Bohr energy levels separate into electron subshells, and quantum numbers dictate the number of states within each subshell. Shells are specified by a principal quantum number *n*, which may take on integral values beginning with unity; sometimes these shells are designated by the letters K, L, M, N, O, etc., which correspond, respectively, to $n = 1, 2, 3, 4, 5, \ldots$, as indicated in Table 2.1. Note also that this quantum number, is also associated with the Bohr model. This quantum number is related to the distance of an electron from the nucleus, or its position.

The second quantum number, *l*, signifies the subshell, which is denoted by a lowercase letter, *s*, *p*, *d*, or *f* and it is related to the shape of the electron subshell. Additionally, the number of these subshells is restricted by the magnitude of *n*. Allowable subshells for the several *n* values are also presented in Table 2.1.The number of energy states for each subshell is determined by the third quantum number, *ml*. For an *s* subshell, there is a single energy state, whereas for *p*, *d*, and *f* subshells, three, five, and seven states exist, respectively. In the absence of an external magnetic field, the states within each subshell are identical. However, when a magnetic field is applied, these subshell states split, with each state assuming a slightly different energy.

Table 2. 1 The Number of Available Electron States in some of the Electron Shells and Subshells

Principal Quantum Number n	*Shell Designation*	*Subshells*	*Number of States*	*Number of Electrons*	
				Per Subshell	*Per Shell*
1	*K*	*s*	1	2	2
2	*L*	*s*	1	2	8
		p	3	6	
3	*M*	*s*	1	2	18
		p	3	6	
		d	5	10	
4	*N*	*s*	1	2	32
		p	3	6	
		d	5	10	
		f	7	14	

A complete energy level diagram for the various shells and subshells using the wave-mechanical model is shown in Figure 2.2. Several features of the diagram are worth noting. First, the smaller the principal quantum number, the lower the energy level; for example, the energy of a 1*s* state is less than that of a 2*s* state, which in turn is lower than the 3*s*. Secondly, within each shell, the energy of a subshell level increases with the value of the *l* quantum number. For example, the energy of a 3*d* state is greater than a 3*p*, which is larger than 3*s*. Finally, there may be overlap in energy of a state in one shell with states in an adjacent shell, which is especially true of *d* and *f* states; for example, the energy of a 3*d* state is generally greater than that for a 4*s*.

2.1.2.3 Electron configuration

The previous section has dealt primarily with *electron states*; values of energy that are permitted for electrons. To determine the manner in which these states are filled with electrons, the *Pauli exclusion principle* is used**,** another quantum-mechanical concept. This principle stipulates that each electron state can hold at most two electrons, which must have opposite spins. Thus, *s*, *p*, *d*, and *f* subshells may each accommodate, respectively, a total of 2, 6, 10, and 14 electrons. Table 2.1 summarizes the maximum number of electrons that may occupy each of the first four shells. Of course, not all possible states in an atom are filled with electrons. For most atoms, the electrons fill up the lowest possible energy states in the electron shells and subshells, two electrons (having opposite spins) per state. When all the electrons occupy the lowest possible energies in accordance with the foregoing restrictions, an atom is said to be in its *ground state***.**

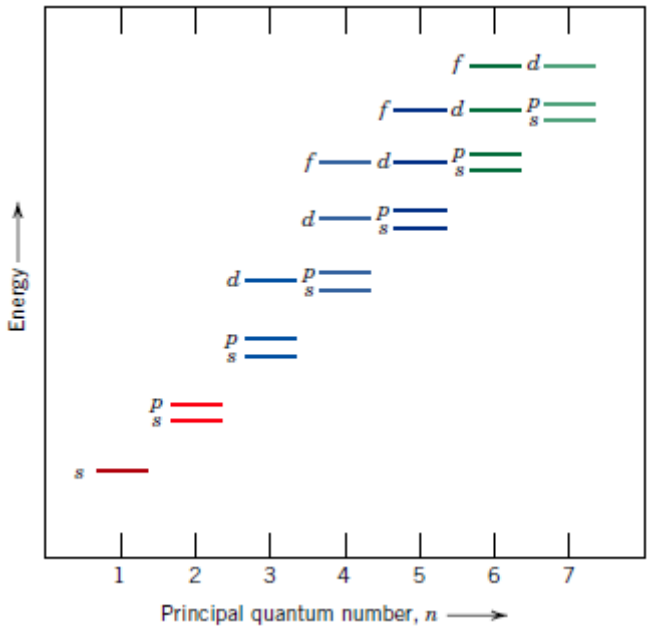


Figure 2.2 Schematic representation of the relative energies of the electrons for the various shells and subshells.

The electron configuration or structure of an atom represents the manner in which these states are occupied. In the conventional notation, the number of electrons in each subshell is indicated by a superscript after the shell. For example, the electron configurations for hydrogen, helium, and sodium are $1s^1$, $1s^2$, and $1s^2 2s^2 2p^6 3s^1$, respectively. Electron configurations for some of the more common elements are listed in Table 2.2. At this point, comments regarding these electron configurations are necessary.

First, the *valence electrons* are those that occupy the outermost shell. These electrons are extremely important; as will be seen, they participate in the bonding between atoms to form atomic and molecular aggregates. Furthermore, many of the physical and chemical properties of solids are based on these valence electrons. In addition, some atoms have what are termed *stable electron configurations;* that is, the states within the outermost or valence electron shell are completely filled. Normally this corresponds to the occupation of just the *s* and *p* states for the outermost shell by a total of eight electrons, as in neon, argon, and krypton; one exception is helium, which contains only two 1*s* electrons. These elements (Ne, Ar, Kr, and He) are the

inert, or noble, gases, which are virtually unreactive chemically. Some atoms of the elements that have unfilled valence shells assume stable electron configurations by gaining or losing electrons to form charged ions, or by sharing electrons with other atoms. This is the basis for some chemical reactions, and also for atomic bonding in solids, as discussed in section 2.2.

Table 2.2 Electron configuration of some common elements

Element	*Symbol*	*Atomic Number*	*Electron Configuration*
Hydrogen	H	1	$1s^1$
Helium	He	2	$1s^2$
Lithium	Li	3	$1s^2 2s^1$
Beryllium	Be	4	$1s^2 2s^2$
Boron	B	5	$1s^2 2s^2 2p^1$
Carbon	C	6	$1s^2 2s^2 2p^2$
Nitrogen	N	7	$1s^2 2s^2 2p^3$
Oxygen	O	8	$1s^2 2s^2 2p^4$
Fluorine	F	9	$1s^2 2s^2 2p^5$
Neon	Ne	10	$1s^2 2s^2 2p^6$
Sodium	Na	11	$1s^2 2s^2 2p^6 3s^1$
Magnesium	Mg	12	$1s^2 2s^2 2p^6 3s^2$
Aluminum	Al	13	$1s^2 2s^2 2p^6 3s^2 3p^1$
Silicon	Si	14	$1s^2 2s^2 2p^6 3s^2 3p^2$
Phosphorus	P	15	$1s^2 2s^2 2p^6 3s^2 3p^3$
Sulfur	S	16	$1s^2 2s^2 2p^6 3s^2 3p^4$
Chlorine	Cl	17	$1s^2 2s^2 2p^6 3s^2 3p^5$
Argon	Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Potassium	K	19	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
Calcium	Ca	20	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Scandium	Sc	21	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$
Titanium	Ti	22	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
Vanadium	V	23	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
Chromium	Cr	24	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
Manganese	Mn	25	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
Iron	Fe	26	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
Cobalt	Co	27	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
Nickel	Ni	28	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
Copper	Cu	29	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
Zinc	Zn	30	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$
Gallium	Ga	31	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^1$
Germanium	Ge	32	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$

2.2 Atomic Bonding in Solids

In order to understand the why materials behave the way they do and why they differ in properties, it is necessary to observe atomic level. The study primarily concentrates on two issues: what made the atoms to cluster together, and how atoms are arranged. As mentioned in earlier chapter, atoms are bound to each other by number of bonds. These interatomic bonds are primarily of two kinds: primary bonds and secondary bonds. Ionic, covalent and metallic bonds are relatively very strong, and grouped as primary bonding, whereas van der Waals and hydrogen bonds are relatively weak, and termed as secondary bonding. Metals and ceramics are entirely held together by primary bonds; the ionic and covalent bonds in ceramics, and the metallic and covalent bonds in metals. Although much weaker than primary bonds, secondary bonds are very important. They provide the links between polymer molecules in polyethylene (and other polymers) which make them solids. Without them, water would boil at -80°C, and life as we know it on earth would not exist.

2.2.1 Primary Interatomic Bonding

2.2.1.1 Ionic bonding

In ionic bonding, there exist transfer of electrons between two atoms. It is always found in compounds that are composed of both metallic and nonmetallic elements. The atoms of the metallic element easily give up their valence electrons to the nonmetallic atoms. In the process, all the atoms acquire stable configurations and, become electrically charged ions. An example of ionic bond is in sodium chloride (NaCl). In the compound, the sodium atoms donate their valence electrons to become positively charged ions (Na^+) whereas the chlorine atoms accept the electrons from sodium to become negatively charged ions (Cl^-). This is illustrated in Figure 2.3. The attractive bonding forces are *coulombic***;** the positive and negative ions by virtue of their net electrical charge, attract one another.

Basically, ionic bonds are non-directional in nature*;* that is, the magnitude of the bond is equal in all directions around an ion. Hence, for ionic materials to be stable, all positive ions must have as nearest neighbors negatively charged ions in a three-dimensional scheme, and vice versa. The predominant bonding in ceramic materials is ionic. Ionic bonds are the strongest bonds. In real solids, ionic bonding usually exists along with covalent bonding.

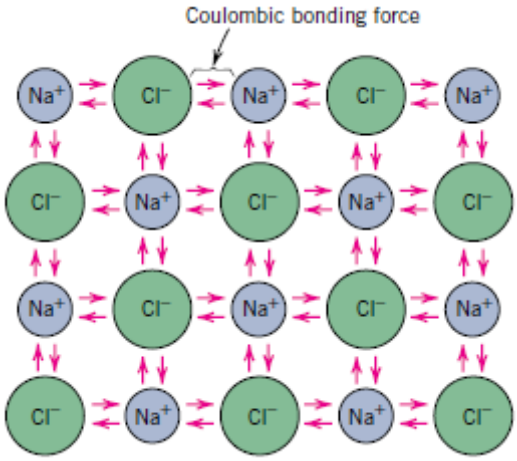


Figure 2.3 Schematic representation of ionic bonding in sodium chloride.

2.2.1.2 Covalent bonding

A covalent bond, also known as molecular bond is a chemical bond that involves the sharing of electron pairs between atoms. The stable balance of attractive and repulsive forces between the atoms, when they share electrons, is known as covalent bonding. Two covalently bonded atoms will each contribute at least one electron to the bond, and the shared electrons may be considered to belong to both atoms. An example of this bond is seen in methane (CH_4) where the carbon atom has four valence electrons and each hydrogen atom has one valence electron making a total of four valence electrons as shown in Figure 2.4. By sharing of valence electrons between the carbon and hydrogen atoms, stable electron configuration is attained.

Usually, covalent bonds are very strong and directional in nature; thus, it is between specific atoms and may exist only in the direction between one atom and another that participates in the electron sharing.

Many nonmetallic elemental molecules (H_2, Cl_2, F_2, etc.) as well as molecules containing dissimilar atoms, such as CH_4, H_2O, HNO_3, and HF, are covalently bonded. Furthermore, this type of bonding is found in elemental solids such as diamond (carbon), silicon, and germanium. The hardness of diamond is a result of the fact that each carbon atom is covalently bonded with four neighboring atoms, and each neighbor is bonded with an equal number of atoms to form a rigid three-dimensional structure.

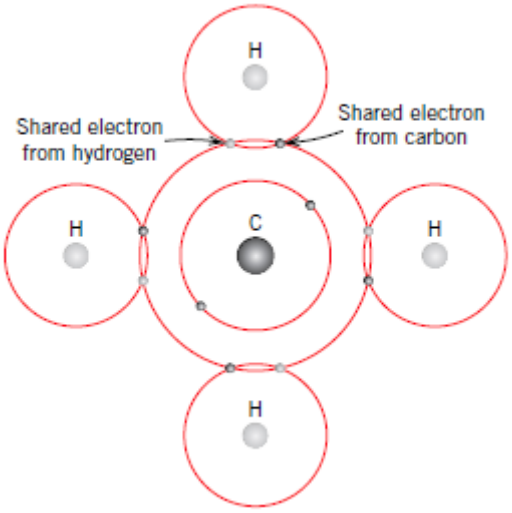


Figure 2.4 Schematic representation of covalent bonding in a molecule of methane.

2.2.1.3 Metallic bonding

This is a type of chemical bonding that rises from the electrostatic attractive force between conduction electrons (in the form of an electron cloud of delocalized electrons) and positively charged metal ions (cations). It can also be seen as sharing of free electrons among a structure of positively charged ions. Metallic bonding accounts for many physical properties of metals, such as strength, ductility, conductivity, thermal and electrical resistivity, etc. Figure 2.5 is a schematic illustration of metallic bonding. The free electrons shield the positively charged ion cores from mutually repulsive electrostatic forces, which they would otherwise exert upon one another. Consequently, the metallic bond is non-directional in character.

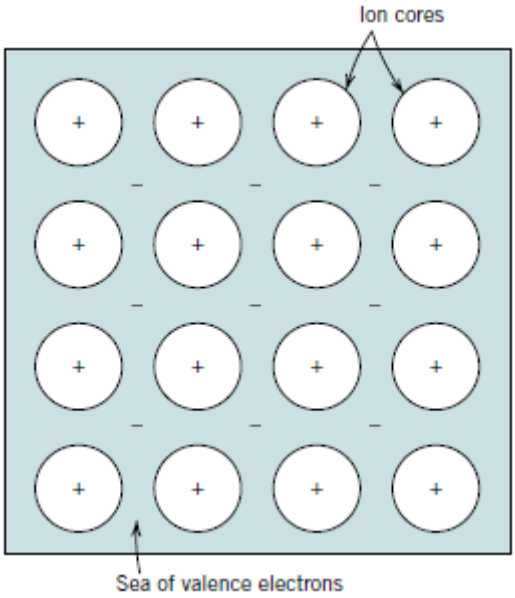


Figure 2.5 Schematic illustration of metallic bonding.

2.2.2 Secondary Bonding

Secondary or van der Waals bonding exists between virtually all atoms or molecules. This is present in inert gases, which have stable electron structures, and, also between molecules in molecular structures that are covalently bonded.

Secondary bonding forces arise from atomic or molecular *dipoles***.** In essence, an electric dipole exists whenever there is some separation of positive and negative portions of an atom or molecule. The bonding results from the coulombic attraction between the positive end of one dipole and the negative region of an adjacent one, as in Figure 2.6. Dipole interactions occur between induced dipoles, between induced dipoles and polar molecules (which have permanent dipoles), and between polar molecules. Hydrogen bonding**,** a special type of secondary bonding, is found to exist between some molecules that have hydrogen as one of the constituents. These bonding mechanisms are now discussed briefly.

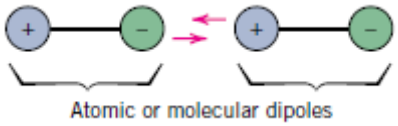


Figure 2.6 Schematic representation of van der Waals bonding between two dipoles.

2.2.2.1 Fluctuating induced dipole bonds

A dipole may be created or induced in an atom or molecule that is normally electrically symmetric; that is, the overall spatial distribution of the electrons is symmetric with respect to the positively charged (+) nucleus, as shown in Figure 2.7*a*. The (+) nucleus at the center and outside electron cloud form a dipole as a result of constant vibrational motion of the atoms as represented in Figure 2.7*b*. As the electron moves, the dipole fluctuates. This fluctuation in one atom, specified as atom X, produces a fluctuating electric field that is felt by the electrons of an adjacent atom, as Y. The atom Y then polarizes so that its outer electrons are on the side of the atom closest to the + side (or opposite to the – side) of the dipole in X.

This is one type of van der Waals bonding. These attractive forces may exist between large numbers of atoms or molecules, which forces are temporary and fluctuate with time.

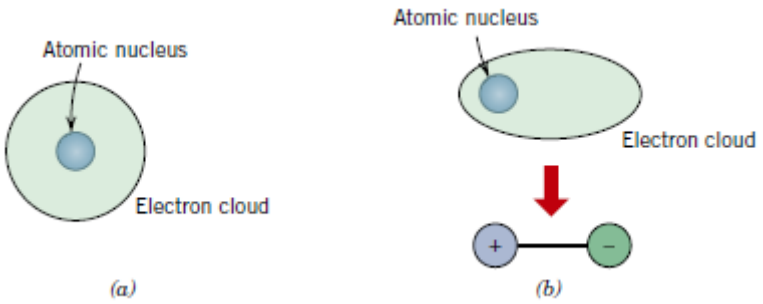


Figure 2.7 Schematic representations of (a) an electrically symmetric atom and (b) an induced atomic dipole.

The liquefaction and, in some cases, the solidification of the inert gases and other electrically neutral and symmetric molecules such as H_2 and Cl_2 are realized because of this type of bonding. Melting and boiling temperatures are extremely low in materials for which induced dipole bonding predominates; of all possible intermolecular bonds, these are the weakest.

2.2.2.2 Polar molecule-induced dipole bonds

Permanent dipole moments exist in some molecules by virtue of asymmetrical arrangement of positively and negatively charged regions. Such molecules are also termed polar molecules. Figure 2.8 is a schematic representation of a hydrogen chloride molecule; a permanent dipole moment arises from net positive and negative charges that are respectively associated with the hydrogen and chlorine ends of the HCl molecule.

Polar molecules can also induce dipoles in adjacent nonpolar molecules, and a bond will form as a result of attractive forces between the two molecules. Furthermore, the magnitude of this bond will be greater than for fluctuating induced dipoles.

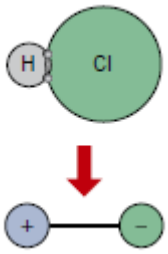


Figure 2.8 Schematic representation of a polar hydrogen chloride (HCl) molecule.

2.2.2.3 Permanent dipole bonds or hydrogen bonding

It occurs between molecules as covalently bonded hydrogen atoms. For example, C-H, O-H, F-H; share single electron with other atom essentially resulting in positively charged proton that is not shielded by any electrons. This highly positively charged end of the molecule is capable of strong attractive force with the negative end of an adjacent molecule, as illustrated in Figure 2.9 for HF.

In essence, this single proton forms a bridge between two negatively charged atoms. The bridges are of sufficient strength, and as a consequence water has the highest melting point of

any molecule of its size. Likewise, its heat of vaporization is very high. Melting and boiling temperatures for hydrogen fluoride (HF) and water (H_2O) are abnormally high in light of their low molecular weights, as a consequence of hydrogen bonding.

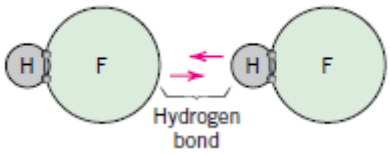


Figure 2.9 Schematic representation of hydrogen bonding in hydrogen fluoride (HF).

References

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2. W. D. Callister, Jr and D. G. Rethwisch, Materials Science and Engineering – An introduction, eighth edition, John Wiley & Sons, Inc. 2010.

CHAPTER 3. THE STRUCTURE OF CRYSTALLINE SOLIDS

The previous chapter discussed the various types of atomic bonding, which are determined by the electron structures of the individual atoms. This chapter is concerned with the structure of solid materials specifically metals. The concept of crystal structure in metals is presented, specified in terms of a unit cell. The most common crystal structures found in metals are then detailed, along with the scheme by which crystallographic points, directions, and planes are expressed. Single crystals, polycrystalline and non-crystalline materials are also considered.

Learning objectives

After studying this chapter, you should be able to do the following:

- Explain the common crystal structures in metals.
- Define the terms coordination number and atomic parking factor
- Identify specific directions and planes in crystals using Miller indices.
- Explain the concept of linear and planar densities
- Explain the terms polymorphism, allotropy, polycrystalline and amorphous solids
- Define Isotropy and Anisotropy

3.1 Crystal Structure

3.1.1 Fundamental Concepts

All metals, a major fraction of ceramics, and certain polymers acquire crystalline structures under normal solidification conditions. Thus, the atoms self-organize to form *crystals* in their solid state. A *crystalline* material possess a long-range order of atomic arrangement through a repeated or periodic array at regular intervals in three dimensions of space, in which each atom is bonded to its nearest neighboring atoms. For those solids that are not crystalline, it is called *non-crystalline or amorphous* material. Examples of crystalline solids are metals, diamond and other precious stones, ice, graphite. Examples of amorphous solids are glass, amorphous carbon (a-C), amorphous Si, most plastics.

Properties of crystalline solids depend on the *crystal structure* of the material, the manner in which atoms, ions, or molecules are spatially arranged. There is an extremely large number of different crystal structures with long-range atomic order. These vary from relatively simple structures for metals to exceedingly complex structures for ceramics and some polymers. This chapter is focused on the common metallic crystal structures.

To discuss crystalline structures, it is useful to consider atoms (or ions) as being hard spheres with well-defined radii. The term *atomic hard-sphere model* is whereby spheres representing nearest-neighbor atoms touch one another. In this scheme, the shortest distance between two like atoms is one diameter. The term *lattice* is used in the context of crystalline structures to represent a three-dimensional periodic array of points coinciding with atom positions.

3.1.2 Unit Cells

The atomic order in crystalline solids indicates that small groups of atoms form a repetitive pattern. The *Unit cell* is smallest repeatable entity that can be used to completely represent a crystal structure. Thus it can be considered that a unit cell is the building block of the crystal structure and defines the crystal structure by virtue of its geometry and the atom positions

within. Unit cells for most crystal structures are parallelepipeds or prisms having three sets of parallel faces; one is drawn within the aggregate of spheres as shown in Figure 3.1 which in this case happens to be a cube. The unit cell is chosen to represent the symmetry of the crystal structure, wherein all the atom positions in the crystal may be generated by translations of the unit cell integral distances along each of its edges. Furthermore, more than a single unit cell may be chosen for a particular crystal structure; however, we generally use the unit cell having the highest level of geometrical symmetry.

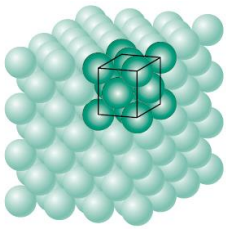


Figure 3.1 Crystal structure with aggregate of spheres

3.1.3 Metallic Crystal Structures

The atomic bonding in this type of materials is metallic and hence non-directional in nature. Consequently, there are minimal restrictions as to the number and position of nearest-neighbor atoms leading to relatively large numbers of nearest neighbors and dense atomic packings for most metallic crystal structures. Also, for metals, each sphere represents an ion core when using the hard-sphere model for the crystal structure. Four relatively simple crystal structures are found for most of the common metals: simple cubic, face-centered cubic, body-centered cubic, and hexagonal close-packed.

Two important characteristics of a crystal structure are its *coordination number* and *atomic packing factor*. For metals, coordination number refers to the number of closest neighboring atoms surrounding a given atom. The atomic parking factor (APF) is the ratio of total atomic volume to the unit cell volume. This is expressed in equation (3.1) as;

$$APF = \frac{Total\ atomic\ volume}{Unit\ cell\ volume} = \frac{N_{atoms\ per\ unit\ cell} \times V_{atom}}{V_{unit\ cell}} = \frac{V_s}{V_c} \qquad (3.1)$$

3.1.3.1 Simple cubic crystal structure

This crystal structure has eight atoms with one at each corner of the cube. The only metal with this structure is Polonium (Po). The reason this crystal structure is so rare is that packing atoms in this way does not lead to a very high packing density. The coordination number of the simple cubic is 6 with atomic parking factor of 0.52.

3.1.3.2 Face-centered cubic crystal structure

The crystal structure of several metals has a unit cell of cubic geometry, with atoms located at each of the corners and the centers of all the cube faces. This is called the *face-centered cubic (FCC)* structure. Some examples of metals with this crystal structure are copper, aluminum, silver, and gold. Figure 3.2*a* shows a hard-sphere model for the FCC unit cell, and in Figure 3.2*b*, the atom centers are represented by small circles to provide a better perspective of atom positions. The aggregate of atoms in Figure 3.2*c* represents a section of crystal consisting of many FCC unit cells. These spheres or ion cores touch one another across a face diagonal; the

cube edge length *a*, and the atomic radius *R*, are related in equation (3.2) obtained through the worked example problem 3.1

$$a = 2R\sqrt{2} \qquad (3.2)$$

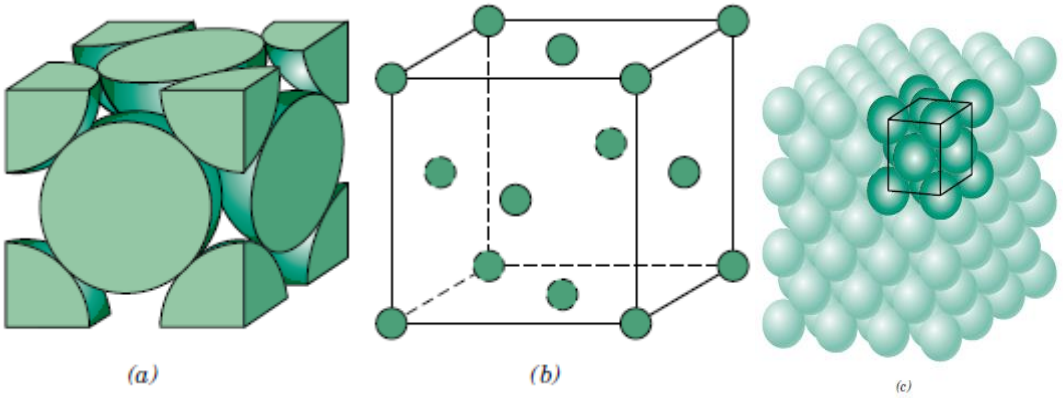


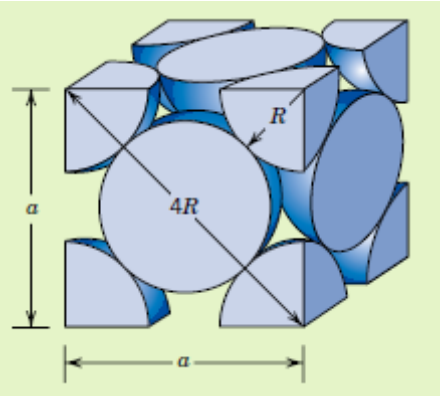
Figure 3.2 For the face-centered cubic crystal structure, (*a*) a hard sphere unit cell representation, (*b*) a reduced-sphere unit cell, and (*c*) an aggregate of many atoms.

Worked example 3.1

Determine the edge length and volume of an FCC unit cell in terms of the atomic radius *R*.

Solution

In the FCC unit cell as illustrated, the atoms touch one another across a face-diagonal the length of which is 4*R*. Because the unit cell is a cube, its volume is a^3, where *a*, is the cell edge length.



From the right angle triangle on the face, the edge length, a, is determined as,

$$a^2 + a^2 = (4R)^2 \Rightarrow a^2 = \frac{1}{2} \times 16\, R^2 = 8R^2 \Rightarrow a = 2R\sqrt{2} \qquad (3.2)$$

For FCC crystal structure, there is an atom on each of the eight corners of the cell and only one-eighth of these atoms can be assigned to a given unit cell. Also, there are six atoms on the faces of the cube and one-half of these atoms form part of the unit cell. Therefore, the FCC crystal structure has a total of four atoms per unit cell. This is depicted in Figure 3.1*a*, where only spherical portions are represented within the confines of the cube. The cell comprises the volume of the cube, which is generated from the centers of the corner atoms as shown in the figure.

The coordination number for the FCC structure is 12. This is examined using Figure 3.2*a*; the front face atom has four corner closest neighbor atoms surrounding it, four face atoms that are in contact from behind, and four other equivalent face atoms residing in the next unit cell to the front, which is not shown. The atomic parking factor (APF) of the FCC structure is 0.74, as shown in example problem 3.2. Metals typically have relatively large APFs to maximize the shielding provided by the free electron cloud.

The mechanical properties of FCC crystal structure are: low young modulus, low yield strength, low hardness, good ductility and high ability for forming.

Worked example 3.2

Show that the atomic packing factor for the FCC crystal structure is 0.74.

Solution

The APF is defined as the fraction of solid sphere volume in a unit cell, or

$$APF = \frac{Total\ atomic\ volume}{Unit\ cell\ volume} = \frac{N_{atoms\ per\ unit\ cell} \times V_{atom}}{V_{unit\ cell}} = \frac{V_S}{V_c}$$

Both the total atom and unit cell volumes may be calculated in terms of the atomic radius *R*. The volume for a sphere is $\frac{4}{3}\pi R^3$, and because there are four atoms per FCC unit cell, the total FCC atom (or sphere) volume is,

$$V_s = (4) \times \frac{4}{3}\pi R^3 = \frac{16}{3}\pi R^3$$

From worked example problem 3.1, the total unit cell volume is

$$V_c = a^3 = \left(2R\sqrt{2}\right)^3 = 16\,R^3\sqrt{2}$$

Therefore, the atomic packing factor is

$$APF = \frac{V_s}{V_c} = \frac{\left(\frac{16}{3}\right)\pi R^3}{16\,R^3\sqrt{2}} = 0.74$$

3.1.3.3 Body-centered cubic crystal structure

Another common metallic crystal structure also has a cubic unit cell with atoms located at all eight corners and a single atom at the cube center. This is called a *body-centered cubic (BCC)* crystal structure as shown in Figure 3.3. The center and corner atoms touch one another along cube diagonals, and unit cell length, *a*, and atomic radius *R*, are related in equation (3.4). Chromium, iron, tungsten, and other metals listed in Table 3.1 exhibit a BCC structure.

$$a = 4R\sqrt{3} \qquad (3.3)$$

Two atoms are associated with each BCC unit cell: the equivalent of one atom from the eight corners, each of which is shared among eight unit cells, and the single center atom, which is wholly contained within its cell. In addition, corner and center atom positions are equivalent. The coordination number for the BCC crystal structure is 8; each body-centered atom has eight closest neighboring corner atoms. Since the coordination number for BCC is less than FCC, the atomic packing factor BCC is also lower, which is 0.68.

Some metals with the BCC structure are chromium, iron and tungsten and their mechanical properties are high yield strength, high young modulus, high hardness, high tensile strength and limited ability to forming.

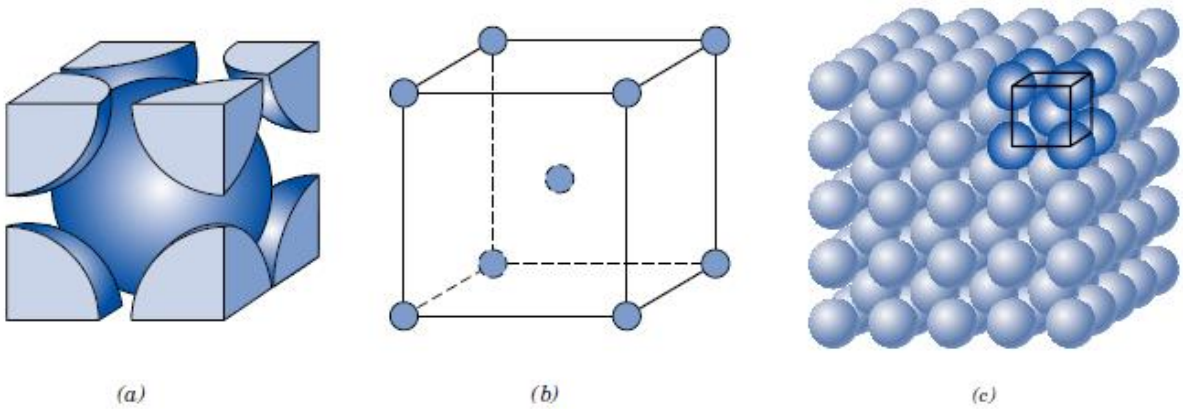


Figure 3.3 For the body-centered cubic crystal structure, (*a*) a hard-sphere unit cell representation, (*b*) a reduced-sphere unit cell, and (*c*) an aggregate of many atoms.

3.1.3.4 Hexagonal closed-packed crystal structure

Some metals have crystal structures with unit cells which are not cubic in symmetry but rather hexagonal. Figure 3.4*a* shows a reduced sphere unit cell for this structure called the *hexagonal close-packed (HCP***)**. An assembly of several HCP unit cells is represented in Figure 3.3*b*. The top and bottom faces of the unit cell consist of six atoms that form regular hexagons and surround a single atom in the center. Another plane that provides three additional atoms to the unit cell is situated between the top and bottom planes. The atoms in this mid plane have nearest neighbor atoms in both of the adjacent two planes. The equivalent of six atoms is contained in each unit cell; one-sixth of each of the 12 top and bottom face corner atoms, one-half of each of the 2 center face atoms, and all 3 mid plane interior atoms. If *a* and *c* represent, respectively, the short and long unit cell dimensions of Figure 3.4*a*, the ratio should be 1.633; however, for some HCP metals this ratio deviates from the ideal value.

The coordination number and atomic packing factor for the HCP crystal structure are the same as for FCC. Metals with HCP structure include cadmium, magnesium, titanium, and zinc. Their mechanical properties are brittle, low yield strength and inability to foaming.

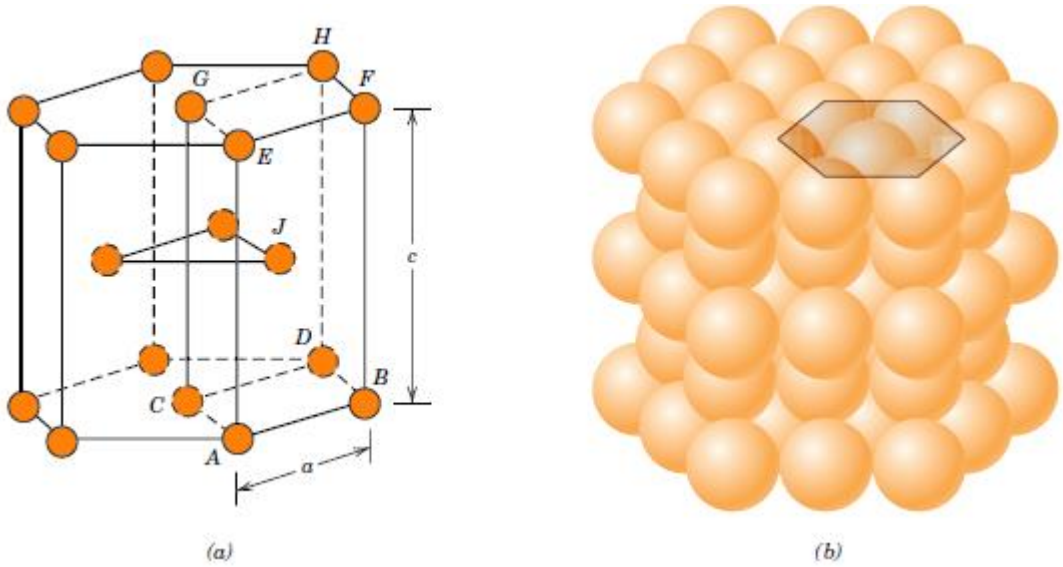


Figure 3.4 For the hexagonal close-packed crystal structure, (*a*) a reduced-sphere unit cell (*a* and *c* represent the short and long edge lengths, respectively), and (*b*) an aggregate of many atoms.

Table 3.1 presents the atomic radius and crystal structure for a number of metals.

Table 3.1 Atomic radii and crystal structures of some metals

Metal	*Crystal Structure*[a]	*Atomic Radius*[b] *(nm)*	*Metal*	*Crystal Structure*	*Atomic Radius (nm)*
Aluminum	FCC	0.1431	Molybdenum	BCC	0.1363
Cadmium	HCP	0.1490	Nickel	FCC	0.1246
Chromium	BCC	0.1249	Platinum	FCC	0.1387
Cobalt	HCP	0.1253	Silver	FCC	0.1445
Copper	FCC	0.1278	Tantalum	BCC	0.1430
Gold	FCC	0.1442	Titanium (α)	HCP	0.1445
Iron (α)	BCC	0.1241	Tungsten	BCC	0.1371
Lead	FCC	0.1750	Zinc	HCP	0.1332

[a]FCC = face-centered cubic; HCP = hexagonal close-packed; BCC = body-centered cubic.

[b]A nanometer (nm) equals 10^{-9} m; to convert from nanometers to angstrom units (Å), multiply the nanometer value by 10.

3.1.4 Calculation of Density for Metals

The crystal structure of a metallic solid permits computation of its theoretical density using the expression in equation (3.5).

$$\rho = \frac{n\,A}{V_c\,N_A} \qquad (3.4)$$

Where n, is the number of atoms associated with each unit cell, A is the atomic weight, V_c is the volume of the unit cell and N_A is the Avogadro's number (16.022 x 10^{23} atoms/mol).

Worked example 3.3

Copper has an atomic radius of 0.128 nm, an FCC crystal structure, and an atomic weight of 63.5 g/mol. Compute its theoretical density and compare answer with its measured density.

Solution

The crystal structure is FCC. Therefore, number of atoms per unit cell, *n*, is 4. Also, atomic weight *A* copper (Cu) is given as 63.5 g/mol. The unit cell volume V_c for FCC was determined in example 3.1 as $16\,R^3\sqrt{2}$ where *R*, the atomic radius, is 0.128 nm.

Substituting the various parameters into equation (3.5) gives

$$\rho = \frac{n\,A}{V_c\,N_A} = \frac{n \times A_{Cu}}{16\,R^3\,\sqrt{2}\,N_A}$$

$$\rho = \frac{\left(4\,\frac{atoms}{unit\ cell}\right) \times \left(63.5\,\frac{g}{mol}\right)}{\left[16\sqrt{2}\,(1.28\ \times\ 10^{-8})\,\frac{cm^3}{unit\ cell}\right] \times \left(6.022\ \times\ 10^{23}\,\frac{atoms}{mol}\right)} = 8.89\ g/cm^3$$

The density of copper in literature is 8.94 g/cm^3, which is in very close to the calculated value obtained.

3.1.5 Polymorphism and Allotropy

Some metals, as well as nonmetals, may have more than one crystal structure, a phenomenon known as *polymorphism*. In elemental solids, this condition is often referred to as *allotropy*.

The crystal structure depends on both the temperature and the external pressure. An example is carbon where graphite becomes a stable polymorph at ambient conditions, whereas diamond formation occur at extremely high pressures. Also, at room temperature, pure iron has a BCC crystal structure which changes to FCC iron at 912 °C (1674 °F). Most often a modification of the density and other physical properties accompanies a polymorphic transformation.

3.1.6 Crystal systems

It is sometimes convenient to divide the different crystal structures into groups according to unit cell configurations and/or atomic arrangements based on the unit cell geometry, that is, the shape of the appropriate unit cell parallelepiped without regard to the atomic positions in the cell. A coordinate system (*xyz*) is established with its origin at one of the unit cell corners; each of the *x*, *y*, and *z* axes coincides with one of the three parallelepiped edges that extend from this corner, as indicated in Figure 3.5. The unit cell geometry is defined completely in terms of six parameters: the three edge lengths *a*, *b*, and *c*, and the three inter-axial angles a, b, and g as indicated in Figure 3.5, and are referred to as the *lattice parameters* of a crystal structure.

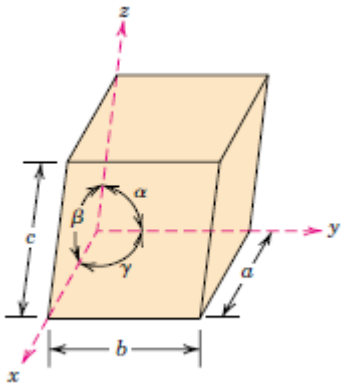


Figure 3.5 A unit cell with *x*, *y*, and *z* coordinate axes, showing axial lengths (*a*, *b*, and *c*) and inter-axial angles (α, β, and γ).

Based on the above explanation, there exist seven different combinations of *a*, *b*, and *c*, and a, b, and g, each representing a distinct *crystal system***.** These seven crystal systems are cubic, tetragonal, hexagonal, orthorhombic, rhombohedral, monoclinic, and triclinic as detailed in the Table 3.2. The cubic system, for which $a = b = c$ and, $\alpha = \beta = \gamma = 90°$, has the greatest degree of symmetry. The triclinic system has the least symmetry because $a \neq b \neq c$ and, $\alpha \neq \beta \neq \gamma$. From the discussion of metallic crystal structures, it should be apparent that both FCC and BCC structures belong to the cubic crystal system, whereas HCP falls within hexagonal.

3.1.7 Crystalline and Non-crystalline materials

3.1.7.1 Single Crystals

Crystals can be *single crystals* where the whole solid is one crystal and therefore has a regular geometric structure with flat faces.

3.1.7.2 Polycrystalline Materials

A solid can be composed of many crystalline grains, not aligned with each other. It is called *polycrystalline*. The grains can be more or less aligned with respect to each other. Where they meet is called a *grain boundary*.

3.1.7.3 Non-Crystalline Solids

In amorphous solids, there is no long-range order. But amorphous does not mean random, since the distance between atoms cannot be smaller than the size of the hard spheres. Also, in many cases there is some form of short-range order. For instance, the tetragonal order of crystalline SiO_2 (quartz) is still apparent in amorphous SiO_2 (silica glass).

3.2 Crystallographic Directions and Planes

It is known that properties of materials depend on their crystal structure, and many of these properties are directional in nature. For example: elastic modulus of BCC iron is greater parallel to the body diagonal than it is to the cube edge. Therefore, for crystalline materials, it becomes necessary to specify a particular point within a unit cell to identify specific direction, or plane of the atoms using *Miller indices* with coordinate axes (*x*, *y*, and *z*) situated at one of the corners and coinciding with the unit cell edges, as shown in Figure 3.5.

To define crystallographic directions in cubic crystal, the steps are as follows;

- A vector of convenient length is placed parallel to the required direction.
- The length of the vector projection on each of the three axes are measured in unit cell dimensions.
- These three numbers are converted to smallest integer values, known as indices, by multiplying or dividing by a common factor.
- The three indices are enclosed in square brackets, *[uvw]*. A family of directions is represented by <*uvw*>.

Some crystallographic directions are indicated in Figure 3.6 as [100], [110], and [111].

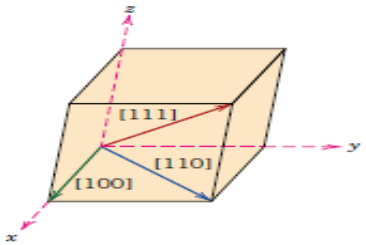


Figure 3.6. Some crystallographic directions within a unit cell

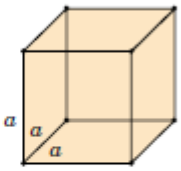
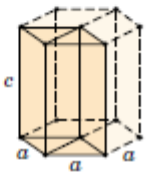
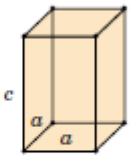
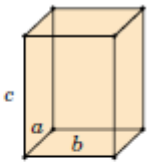
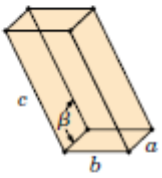
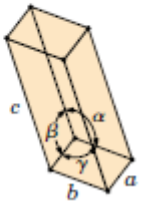
To define crystallographic planes in cubic crystal, the steps are as follows;

- Determine the intercepts of the plane along the crystallographic axes, *in terms of unit cell dimensions*. If plane is passing through origin, there is the need to construct a plane parallel to original plane.
- Take the reciprocals of these intercept numbers.
- Clear fractions.
- Reduce to set of smallest integers.
- The three indices are enclosed in parenthesis, *(hkl)*. A family of planes is represented by *{hkl}*

For example, if the x-, y-, and z- intercepts of a plane are 2, 1, and 3. The Miller indices are calculated as:

- Take reciprocals: 1/2, 1/1, 1/3.
- Clear fractions (multiply by 6): 3, 6, 2.
- Reduce to lowest terms (already there). => Miller indices of the plane are (362).

Table 3.2 Lattice Parameter and Unit Cell Geometries for Crystal Systems

Crystal System	*Axial Relationships*	*Interaxial Angles*	*Unit Cell Geometry*
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
Rhombohedral (Trigonal)	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$	
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	

Some useful conventions of Miller notation:

- If a plane is parallel to an axis, its intercept is at infinity and its Miller index will be zero.
- If a plane has negative intercept, the negative number is denoted by a bar above the number. *Never alter negative numbers.* For example, do not divide -1, -1, -1 by -1 to get 1, 1, 1. This implies symmetry that the crystal may not have!
- The crystal directions of a family are not necessarily parallel to each other. Similarly, not all planes of a family are parallel to each other.
- By changing signs of all indices of a direction, we obtain opposite direction. Similarly, by changing all signs of a plane, a plane at same distance in other side of the origin can be obtained.
- Multiplying or dividing a Miller index by constant has no effect on the orientation of the plane.
- The smaller the Miller index, more nearly parallel the plane to that axis, and vice versa.
- When the integers used in the Miller indices contain more than one digit, the indices must be separated by commas. For example (3,10,13)
- By changing the signs of all the indices of (a) a direction, we obtain opposite direction, and (b) a plane, we obtain a plane located at the same distance on the other side of the origin.

3.2.1 Linear and Planar Densities

3.2.1.1 Linear density

Linear density (LD) is defined as the number of atoms per unit length whose centers lie on the direction vector for a specific crystallographic direction; that is,

$$LD = \frac{\text{number of atoms centered on direction vector}}{\text{length of direction vector}} \qquad (3.5)$$

The units of linear density are reciprocal length (e.g., nm^{-1}, m^{-1}). For example, to determine the linear density of the [110] direction for the FCC crystal structure as shown in Figure 3.6, it is necessary to take into account the sharing of atoms with adjacent unit cells

$$LD_{[110]} = \frac{\frac{1}{2}atom + 1\ atom + \frac{1}{2}atom}{R+R+R+R} = \frac{2\ atoms}{4R} = \frac{1}{2R}$$

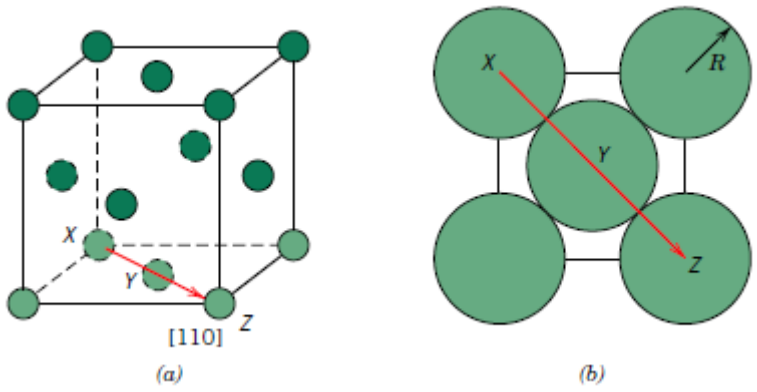


Figure 3. 6 FCC unit cell with atomic spacing in the [110] direction, through atoms labeled *X*, *Y*, and *Z*.

3.2.2.2 Planar density

Planar density (PD) is define as the number of atoms per unit area that are centered on a particular crystallographic plane, or

$$PD = \frac{\text{number of atoms centered on a plane}}{\text{area of plane}} \quad (3.6)$$

The units for planar density are reciprocal area (e.g., nm^{-2}, m^{-2}). For example, consider the section of a (110) plane within an FCC unit cell as represented in Figure 3.7.

$$Area\ of\ plane\ (110) = a \times fd = a \times (a^2 + a^2) = a \times \sqrt{2a} = \sqrt{2}a^2$$

$$Cube\ edge\ length_{FCC}, a = 2R\sqrt{2} \Rightarrow a^2 = 8R^2$$

$$Area\ of\ plane\ (110) = \sqrt{2}a^2 = 8R^2\sqrt{2}$$

$$PD_{[110]} = \frac{\frac{1}{4}+\frac{1}{2}+\frac{1}{4}+\frac{1}{4}+\frac{1}{2}+\frac{1}{4}}{8R^2\sqrt{2}} = \frac{2}{8R^2\sqrt{2}} = \frac{1}{4R^2\sqrt{2}}$$

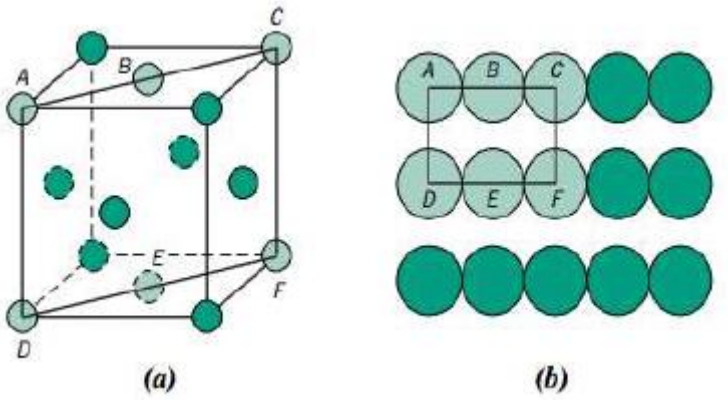


Figure 3. 7 FCC unit cell with atomic spacing in the (110) plane

3.3 Anisotropy

It has been understood that most of the materials properties depend on the crystal structure. However, crystals are not symmetric in all directions, or the crystal planes are not the same with respect to atomic density/packing. The different directions in the crystal have different packing. For instance, atoms along the edge of FCC crystals are more separated than along its face diagonal. This causes properties to be different in different directions. This directionality of properties is termed as *Anisotropy* and it is associated with the variation of atomic or ionic spacing with crystallographic direction. Substances whereby the measured properties are independent of direction of measurement are *isotropic*.

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CHAPTER 4. IMPERFECTIONS AND DIFFUSION IN SOLIDS

The assumption so far has been that there is perfect order throughout crystalline materials on an atomic scale. However, there exist various defects or imperfections. As a matter of fact, many of the properties of materials are profoundly sensitive to deviations from crystalline perfection. Crystalline defect refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter. This chapter discusses the different imperfections in solids including point defects (associated with one or two atomic positions), linear defects (or one-dimensional), and interfacial defects which are two-dimensional. Impurities in solids are also explained, because impurity atoms may exist as point defects. Lastly, diffusion mechanisms in solids are discussed.

Learning objectives

After studying this chapter, you should be able to do the following:

- Explain the various imperfections in solid crystals
- Describe impurities associated with solid atoms
- Determine the concentration of an alloy in terms of its constituent elements.
- Explain the diffusion mechanisms in solids
- State Fick's laws associated with steady and unsteady state diffusion
- Solve problems involving steady and unsteady state diffusion
- Explain the factors that influence diffusion in solids.

4.1 Defects in Solids

4.1.1 Point Defects

Point defects are imperfect point-like regions in the crystal. Typical size of a point defect is about 1-2 atomic diameters. Various point defects occur in solids as shown in Figure 4.1.

A ***vacancy*** is a vacant lattice position from where the atom is missing. It is usually created when the solid is formed by cooling the liquid. There are other ways of making a vacancy, but they also occur naturally as a result of thermal excitation, and these are thermodynamically stable at temperatures greater than zero. At equilibrium, the fraction of lattice sites that are vacant at a given temperature (T) is given by the expression;

$$\frac{n}{N} = e^{\left(-Q/kT\right)} \qquad (4.1)$$

where n is the number of vacant sites in N lattice positions, T is absolute temperature in kelvins, Q is the energy required to move an atom from the interior of a crystal to its surface and k is the gas or Boltzmann's constant. The value of k is 1.38×10^{-23} J/atom K, or 8.62×10^{-5} eV/atom K. It is clear from the equation that there is an exponential increase in number of vacancies with temperature. When the density of vacancies becomes relatively large, there is a possibility for them to cluster together and form voids.

An ***interstitial atom*** is an atom that occupies a place outside the normal lattice position. It may be the same type of atom as the rest surrounding it (self-interstitial) *or* a foreign impurity atom. Interstitial is most probable if the atomic packing factor is low.

Another way by which an impurity atom can be fitted into a crystal lattice is by substitution. A ***substitutional atom*** is a foreign atom occupying original lattice position by displacing the

parent atom. In the case of vacancies and foreign atoms (both interstitial and substitutional), there is a change in the coordination of atoms around the defect. This means the forces are not balanced in the same way as for other atoms in the solid, which results in lattice distortion around the defect.

In ionic crystals, existence of point defects is subjected to the condition of charge neutrality. There are two possibilities for point defects in ionic solids. When an ion is displaced from a regular position to an interstitial position creating a vacancy, the pair of vacancy-interstitial is called ***Frenkel defect***. Cations are usually smaller and thus displaced easily than anions. Closed packed structures have fewer interstitials and displaced ions than vacancies because additional energy is required to force the atoms into the interstitial positions. A pair of one cation and one anion can be missing from an ionic crystal, without violating the condition of charge neutrality when the valency of ions is equal. The pair of vacant sites formed is called ***Schottky defect***. This type of point defect is dominant in alkali halides. These *ion-pair vacancies*, like single vacancies, facilitate atomic diffusion.

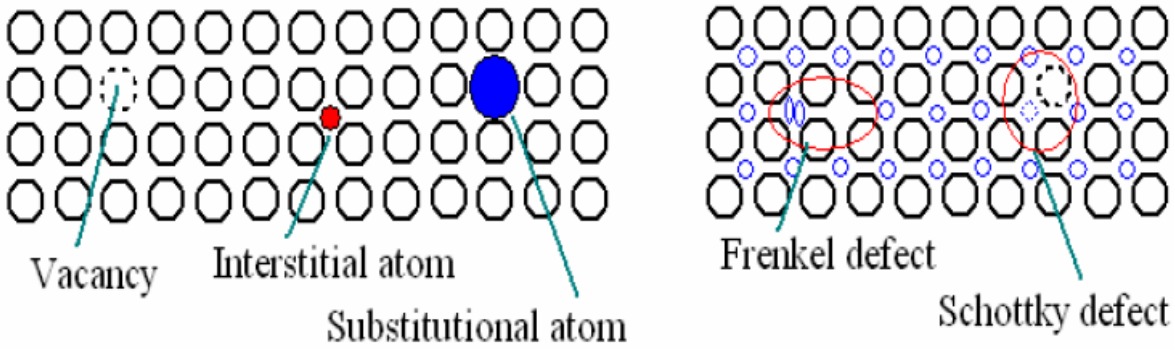


Figure 4.1 Schematic representation of various point defects

Worked example 4.1

Calculate the equilibrium number of vacancies per cubic meter for copper at 1000 °C. The energy for vacancy formation is 0.9 eV/atom; the atomic weight and density (at 1000 °C) for copper are 63.5 g/mol and 8.4 g/cm^3, respectively.

Solution

This problem is solved using Equation 4.1. However, it is necessary to first determine the value of N, the number of atomic sites per cubic meter for copper, from its atomic weight A_{Cu}, its density ρ, and Avogadro's number N_A, according to

$$N = \frac{N_A \cdot \rho}{A_{cu}} = \frac{(6.022 \times 10^{23}\ atoms/mol)(8.4\ g/cm^3)(10^6\ cm^3/m^3)}{63.5\ \ g/mol}$$

$$N = 8.0 \times 10^{28}\ atoms/m^3$$

Thus, the equilibrium number of vacancies at 1000 °C (1273 K) is equal to

$$n = N \cdot exp\left(\frac{-Q}{kT}\right) = 8.0 \times 10^{28}\ atoms/m^3 \cdot exp\left(\frac{-(0.9\ \ eV/atom)}{(8.62 \times 10^{-5}\ eV/atom \cdot K)(1273\ K)}\right)$$

$$n = 2.2 \times 10^{25}\ vacancies/m^3$$

4.1.1.1 Impurities in Solids

Most familiar metals are not highly pure; rather, they are alloys, in which impurity atoms have been added intentionally to impart specific characteristics to the material. The addition of impurity atoms to a metal will result in the formation of a *solid solution* and/or a new second phase, depending on the type of impurity, their concentrations, and the temperature of the alloy.

With alloys, solute and solvent are commonly used terms. *Solvent* and solute represents the element or compound that is present in major and minor concentrations, respectively. On occasion, solvent atoms are also called *host atoms*.

A solid solution is formed as the solute atoms being added to the host material maintains the crystal structure with no new structures. A solid solution is compositionally homogeneous; the impurity atoms are randomly and uniformly dispersed within the solid. The impurity point defects found in solid solutions are of two types namely substitutional and interstitial.

In substitutional solid solution, the solute or impurity atoms replace or substitute for the host atoms. The features of the solute and solvent atoms that determines the degree to which the former dissolves in the latter are as follows:

- *Atomic size factor*: the difference in atomic radii between the two atom types should not be more than 14%.
- *Crystal structure*: the crystal structures for metals of both atom types must be the same.
- *Electronegativity*: The atoms should have similar electronegativity, or compounds will form.
- *Valences*: Other factors being equal, then the metal of lower valence will dissolve more in crystal structure of the higher valence metal than vice versa.

Note: Not all alloys systems that fit these rules will form appreciable solid solutions.

For interstitial solid solutions, impurity atoms fill the voids or interstices among the host atoms. For metallic materials that have relatively high atomic packing factors, these interstitial positions are relatively small. Consequently, the atomic diameter of an interstitial impurity must be substantially smaller than that of the host atoms.

4.1.1.2 Specification of Alloy Composition

Most often, the composition (or *concentration*) of an alloy is expressed in terms of its constituent elements. The two most common ways to specify composition are weight (or mass) percent and atom percent. The weight percent (wt%) is the weight of a particular element relative to the total alloy weight. For an alloy that contains two hypothetical atoms denoted by x and y, the concentration of x in wt%, C_x, is defined as

$$C_x = \frac{m_x}{m_x + m_y} \times 100\% \qquad (4.2)$$

Where m_x and m_y represent the weight or mass of elements x and y, respectively. The concentration of y would be determined in an analogous manner.

The concentration in terms of atom percent (at%) of an element x in an alloy containing elements x and y atoms, is defined by

$$C'_x = \frac{(m_x/A_x)}{[(m_x/A_x)+(m_y/A_y)]} \times 100\% = \frac{n_x}{n_x + n_y} \times 100\% \qquad (4.3)$$

Where m_x and m_y represent the weight or mass, A_x and A_y represent the atomic weight and n_x and n_y represent number of moles of the elements x and y, respectively.

Given an alloy having x and y elements with concentrations in weight percent (wt%), the concentrations in atom percent for x and y elements can be determined using equations (4.4) and (4.5) respectively.

$$C'_x = \frac{C_x A_y}{C_x A_y + C_y A_x} \times 100\% \qquad (4.4)$$

$$C'_y = \frac{C_y A_x}{C_y A_x + C_x A_y} \times 100\% \qquad (4.5)$$

Worked example 4.1

Determine the composition, in atom percent, of an alloy that consists of 97 wt% aluminum and 3 wt% copper. The atomic weight (molar mass) of copper and aluminum are 63.55 g/mol and 26.98 g/mol respectively.

Solution

Given C_{Al} to be 97 wt% and C_{Cu} to be 3 wt%, the compositions of aluminum and copper in terms of at% are determined respectively using equations (4.4) and (4.5).

$$C'_{Al} = \frac{C_{Al} A_{Cu}}{C_{Al} A_{Cu} + C_{Cu} A_{Al}} \times 100\% = \frac{97 \times 63.55\ g/mol}{(97 \times 63.55\ g/mol) + (3 \times 26.98\ g/mol)} \times 100\%$$

$$C'_{Al} = 98.7\ at\%$$

$$C'_{Cu} = \frac{C_{Cu} A_{Al}}{C_{Cu} A_{Al} + C_{Al} A_{Cu}} \times 100\% = \frac{3 \times 26.98\ g/mol}{(3 \times 26.98\ g/mol) + (97 \times 63.55\ g/mol)} \times 100\%$$

$$C'_{Cu} = 1.30\ at\%$$

4.1.2 Line Defects

Line imperfections (one-dimensional defects) are also called Dislocations. They are abrupt changes in the regular ordering of atoms along a line (dislocation line) in the solid. They occur in high densities and strongly influence the mechanical properties of material. They are characterized by the Burgers vector (***b***), whose direction and magnitude can be determined by constructing a loop around the disrupted region and noticing the extra inter-atomic spacing needed to close the loop. The Burgers vector in metals points in a close packed lattice direction. It is unique to a dislocation.

Dislocations occur when an extra incomplete plane is inserted. The dislocation line is at the end of the plane. Dislocations can be best understood by referring to two limiting cases - *Edge dislocation* and *Screw dislocation*.

4.1.2.1 Edge Dislocation

Edge dislocation or *Taylor-Orowan dislocation* is characterized by a Burger's vector that is perpendicular to the dislocation line. It may be described as an edge of an extra plane of atoms within a crystal structure. Thus regions of compression and tension are associated with an edge dislocation. Because of extra incomplete plane of atoms, the atoms above the dislocation line are squeezed together and are in state of compression whereas atoms below are pulled apart

and experience tensile stresses. Edge dislocation is considered *positive* when compressive stresses present above the dislocation line, and is represented by ⊥. If the stress state is opposite i.e. compressive stresses exist below the dislocation line, it is considered as negative edge dislocation, and represented by ┬. A schematic view of edge dislocations are shown in Figure 4.2.

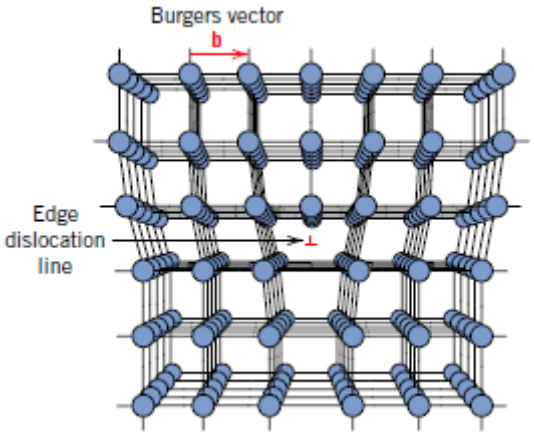


Figure 4.2 Edge dislocation

A pure edge dislocation can glide or slip in a direction perpendicular to its length i.e. along its Burger's vector in the slip plane (made of ***b*** and ***t*** vectors), on which dislocation moves by slip while conserving number of atoms in the incomplete plane. It may move vertically by a process known as climb, if diffusion of atoms or vacancies can take place at appropriate rate.

Atoms are added to incomplete plane for negative climb i.e. the incomplete plane increases in extent downwards, and vice versa. Thus climb motion is considered as non-conservative, the movement by climb is controlled by diffusion process.

4.1.2.2 Screw Dislocation

Screw dislocation or *Burgers dislocation* has its dislocation line parallel to the Burger's vector. A screw dislocation is like a spiral ramp with an imperfection line down its axis. Screw dislocation may be thought of as being formed by a shear stress that is applied to produce the distortion as shown in Figure 4.3. The upper front region of the crystal is shifted one atomic distance to the right relative to the bottom portion. Shear stresses are associated with the atoms adjacent to the screw dislocation; therefore extra energy is involved as it is in the case of edge dislocations. The symbol ↻ is used to designate a screw dislocation. Screw dislocation is considered positive if the unit vector representing the direction of the dislocation line and the Burger's vector are parallel, and vice versa.

Dislocations mostly originate from plastic deformation, solidification, and as a consequence of thermal stresses that result from rapid cooling. Also, unlike point defects, linear defects are not thermodynamically stable. They can be removed by heating to high temperatures where they cancel each other or move out through the crystal to its surface. Virtually all crystalline materials contain some dislocations. Any dislocation in a crystal is a combination of edge and screw types, having varying degrees of edge and screw character.

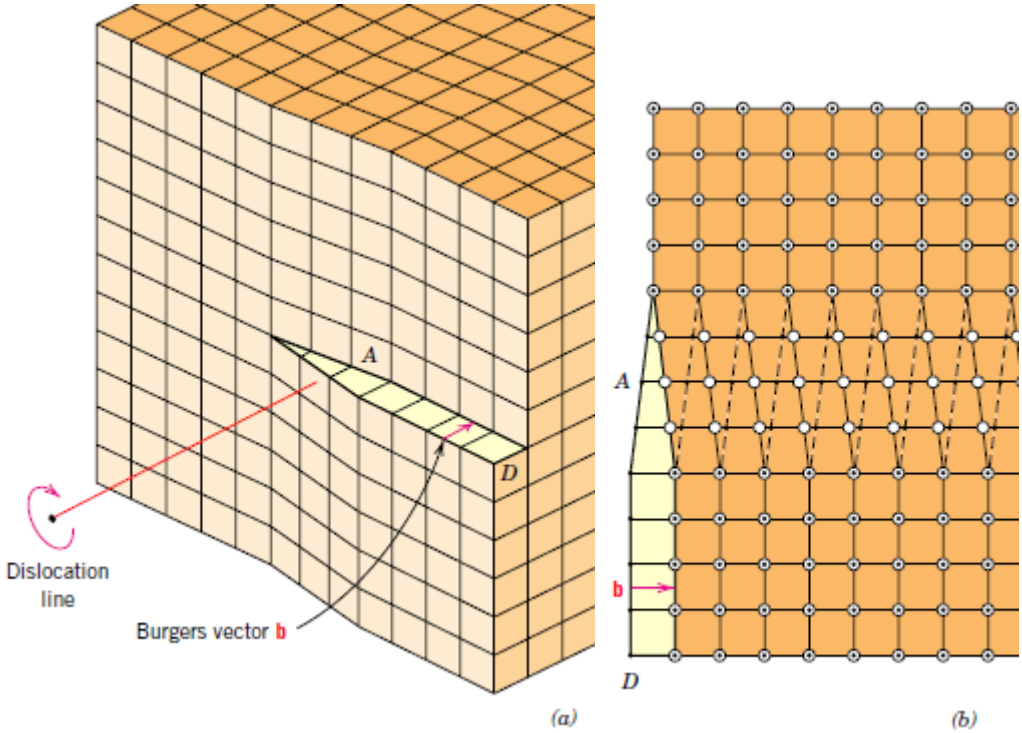


Figure 4.3 (a) Screw dislocation within a crystal. (b) The screw dislocation in (a) as viewed from above.

4.1.3 Interfacial Defects

Interfacial defects are defined as boundaries that have two dimensional imperfections in crystalline solids, and usually separate regions of the materials that have different crystal structures and/or crystallographic orientations. They refer to the regions of distortions that lie about a surface having thickness of a few atomic diameters. These defects include external surfaces and grain boundaries among others. These imperfections are not thermodynamically stable, rather they are metastable imperfections. They arise from the clustering of line defects into a plane.

4.1.3.1 External Surfaces

One of the most obvious boundaries is the external surface, along which the crystal structure terminates. Surface atoms are not bonded to the maximum number of nearest neighbors, and are therefore in a higher energy state than the atoms at interior positions. The bonds of these surface atoms that are not satisfied give rise to a surface energy, expressed in units of energy per unit area (J/m^2 or erg/cm^2). To reduce this energy, materials tend to minimize, if at all possible, the total surface area. For example, liquids assume a shape having a minimum area; the droplets become spherical. This is not possible with solids, which are mechanically rigid.

4.1.3.2 Grain boundaries

Crystalline solids usually consist of number of grains separated by grain boundaries. Grain boundaries are several atoms distances wide, and there is mismatch of orientation of grains on either side of the boundary as shown in Figure 4.4. When this misalignment is slight, on the order of few degrees (< 10°), it is called *low angle grain boundary*. These boundaries can be described in terms of aligned dislocation arrays. If the low grain boundary is formed by edge

dislocations, it is called *tilt boundary*, and *twist boundary* if formed of screw dislocations. Both tilt and twist boundaries are planar surface imperfections in contrast to high angle grain boundaries. For *high angle grain boundaries*, degree of disorientation is of large range (> 15°). Grain boundaries are chemically more reactive because of grain boundary energy. In spite of disordered orientation of atoms at grain boundaries, polycrystalline solids are still very strong as cohesive forces present within and across the boundary.

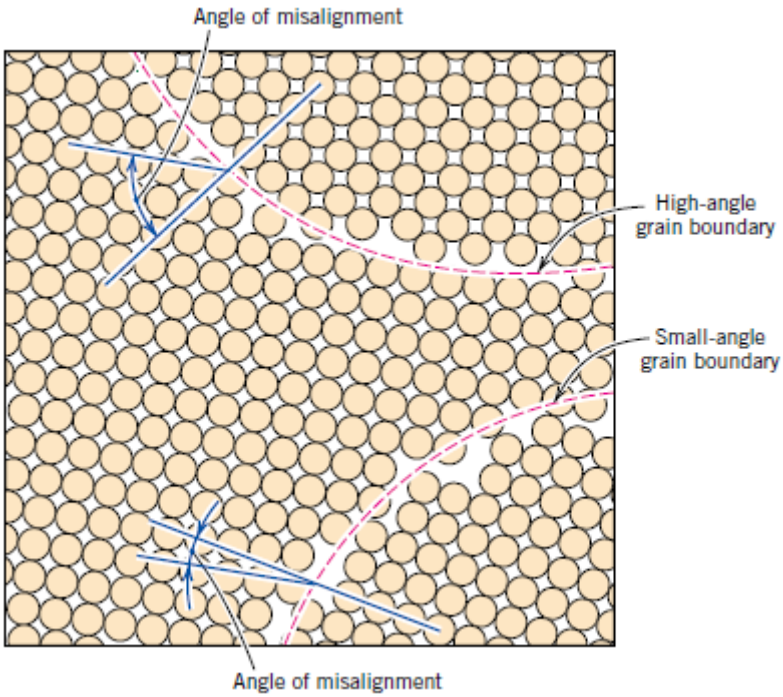


Figure 4.4 Schematic presentation of grain boundaries

4.2 Diffusion in Solids

Diffusion is the process by which atoms move in a material. Many reactions in solids and liquids are diffusion dependent. Structural control in a solid to achieve the optimum properties is also dependent on the rate of diffusion.

Atoms are able to move throughout solids because they are not stationary but execute rapid, small-amplitude vibrations about their equilibrium positions. Such vibrations increase with temperature and at any temperature a very small fraction of atoms has sufficient amplitude to move from one atomic position to an adjacent one. The fraction of atoms possessing this amplitude increases markedly with rising temperature.

In jumping from one equilibrium position to another, an atom passes through a higher energy state since atomic bonds are distorted and broken, and the increase in energy is supplied by thermal vibrations. As might be expected defects, especially vacancies, are quite instrumental in affecting the diffusion process on the type and number of defects that are present, as well as the thermal vibrations of atoms.

Diffusion can be defined as the mass flow process in which atoms change their positions relative to neighbors in a given phase under the influence of temperature and a gradient. The gradient can be a concentration gradient, an electric or magnetic gradient, or stress gradient. In this section we discuss diffusion because of concentration gradient only.

4.2.1 Mechanisms of Diffusion

Self-diffusion occurs in pure metals where there is no net mass transport, but atoms migrate in a random manner throughout the crystal. In alloys, inter-diffusion takes place where the mass transport almost always occurs so as to minimize compositional differences. Various atomic mechanisms for self-diffusion and inter-diffusion have been proposed. Figure 4.5 indicates the schematic view of different atomic diffusion mechanisms.

The most energetic mechanism is ***vacancy diffusion*** which involves the interchange of an atom from a normal lattice position to an adjacent vacant lattice site or vacancy. This process demands not only the motion of vacancies, but also the presence of vacancies. The unit step in vacancy diffusion involves an atom breaking its bonds and jumping into a neighboring vacant site.

In ***interstitial diffusion***, solute atoms which are small enough to occupy interstitial sites diffuse by jumping from one interstitial site to another. The unit step here involves jump of the diffusing atom from one interstitial site to a neighboring site. Hydrogen, Carbon, Nitrogen and Oxygen diffuse interstitially in most metals, and the activation energy for diffusion is only that associated with motion since the number of occupied, adjacent interstitial sites usually is large.

Substitutional diffusion occurs by the movement of atoms from one atomic site to another. It generally proceeds by the vacancy mechanism and that interstitial diffusion is faster than substitutional diffusion. During ***self-diffusion*** or *direct-exchange mechanism*, three or four atoms in the form of a ring move simultaneously round the ring, thereby interchanging their positions. This mechanism is untenable because exceptionally high activation energy would be required. A ***self-interstitial*** is more mobile than a vacancy as only small activation energy is required for self-interstitial atom to move to an equilibrium atomic position and simultaneously displace the neighboring atom into an interstitial site. However, the equilibrium number of self-interstitial atoms present at any temperature is negligible compared to the number of vacancies. This is because the energy to form a self-interstitial is extremely large.

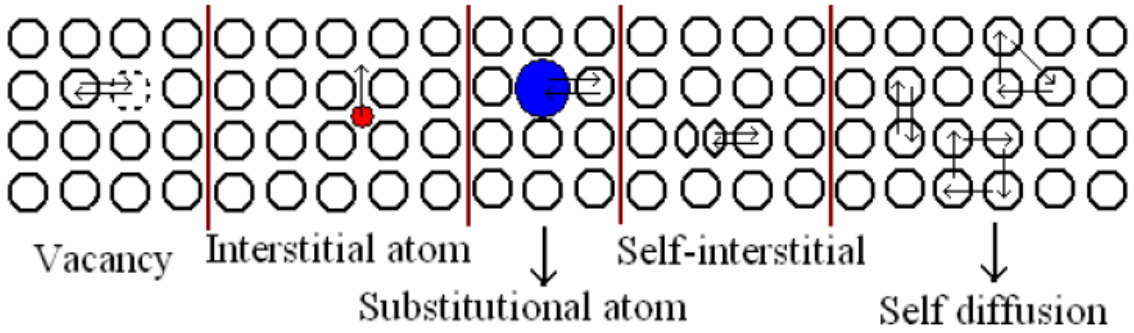


Figure 4.5 Different atomic diffusion mechanisms

Diffusion in most ionic solids occurs by a vacancy mechanism. In ionic crystals, Schottky and Frenkel defects assist the diffusion process. When Frenkel defects (pair of vacancy-interstitial) dominate in an ionic crystal, the cation interstitial of the Frenkel defect carries the diffusion flux. If Schottky defects (pair of vacant sites) dominate, the cation vacancy carries the diffusion flux. In thermal equilibrium, in addition to above defects, ionic crystal may have

defects generated by impurities and by deviation from stoichiometry. Thus imperfections in ionic materials that influence diffusion arise in two ways: (1) intrinsic point defects such as Frenkel and Schottky defects whose number depends on temperature, and (2) extrinsic point defects whose presence is due to impurity ions of different valance than the host ions. The former is responsible for temperature dependence of diffusion similar to that for self-diffusion in metals, while the latter result in a temperature dependence of diffusion which is similar to that for interstitial solute diffusion in metals.

4.2.2 Steady-State Diffusion

Diffusion is a time-dependent process; that is, in a macroscopic sense, the quantity of an element that is transported within another is a function of time. It is often necessary to know how fast diffusion occurs which is characterized by a parameter known as flux. The *diffusion flux (J)*, is defined as the net number of atoms crossing a unit area perpendicular to a given direction per unit time. For steady-state diffusion, the flux is constant with time. One example of steady-state diffusion is the diffusion of atoms of a gas through a plate of metal for which the concentrations (or pressures) of the diffusing species on both surfaces of the plate are held constant as shown in Figure 4.6a. The concentration versus position (or distance) graph within the solid *x* is plotted as shown in Figure 4.6b resulting in the concentration profile curve; the slope at a particular point on this curve is the concentration gradient. In the present treatment, the concentration profile is assumed to be linear.

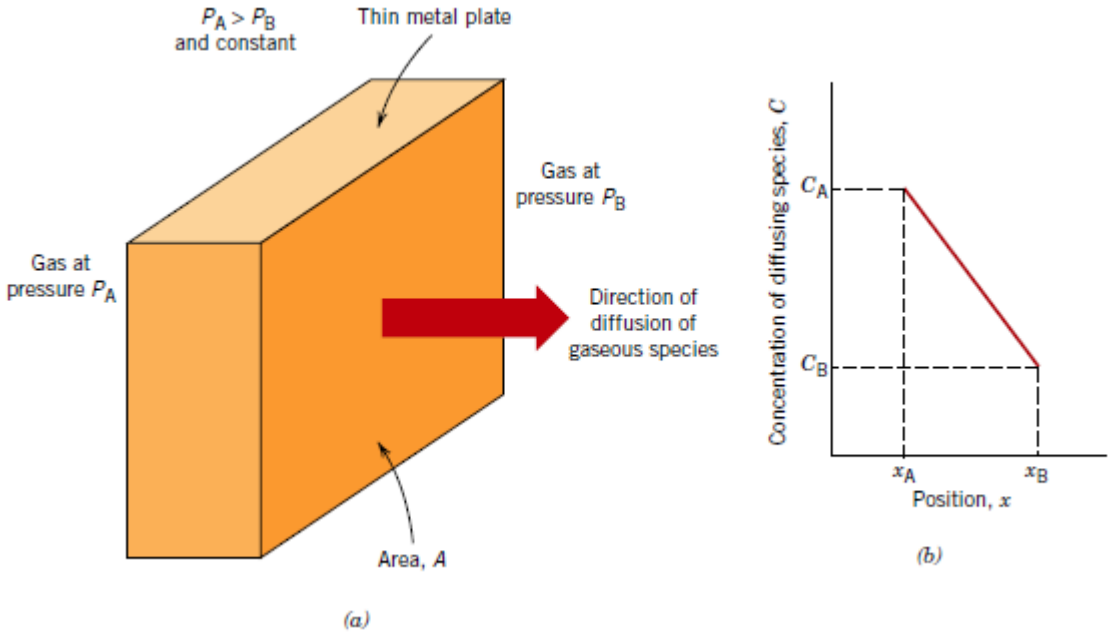


Figure 4.6 (*a*) Steady-state diffusion across a thin plate. (*b*) A linear concentration profile in (*a*).

Steady-state diffusion is described by Fick's first law which states that diffusion flux, *J*, is proportional to the concentration gradient. The constant of proportionality is called diffusion coefficient (diffusivity), *D* (cm^2/sec). Diffusivity is characteristic of the system and depends on the nature of the diffusing species, the matrix in which it is diffusing, and the temperature at which diffusion occurs. Thus under steady-state flow, flux is independent of time and remains the same at any cross-sectional plane along the diffusion direction.

For one-dimensional case, *Fick's first law* is given by

$$J_x = -D\frac{dC}{dx} = \frac{1}{A}\frac{dn}{dt} \qquad (4.6)$$

Where D is the diffusion coefficient, dc/dx is the concentration gradient, dn/dt is the number atoms crossing per unit time a cross-sectional plane of area A. The minus sign in the equation means diffusion occurs down the concentration gradient. Although, the concentration gradient is often called the *driving force* for diffusion, it is more appropriate to consider the reduction in total free energy as the driving force.

An example of steady-state diffusion is provided by the permeation of hydrogen atoms through a sheet of palladium with different imposed hydrogen gas pressures on either side the slab. This process has been used to purify the hydrogen gas as other gases like nitrogen, oxygen and water vapor cannot diffuse through palladium.

Worked example 4.2

A plate of iron is exposed to a carburizing atmosphere on one side and a decarburizing atmosphere on the other side at 700 °C (1300 °F). If a condition of steady state is achieved, calculate the diffusion flux of carbon through the plate if the concentrations of carbon at positions of 5 and 10 mm beneath the carburizing surface are 1.2 and 0.8 kg/m^3, respectively. Assume diffusion coefficient of 3 x 10^{-11} m^2/s at this temperature.

Solution

Fick's first law in equation 4.6 is utilized to determine the diffusion flux. Substitution of the values above into this expression yields

$$J_x = -D\frac{C_A - C_B}{x_A - x_B} = -(3x10^{-11}\,m^2/s)\frac{(1.2-0.8)\,kg/m^3}{(5x10^{-3}-10^{-2})m} = 2.4x10^{-9}\,kg/m^2\cdot s$$

4.2.2 Unsteady-State Diffusion

Most interesting cases of diffusion are non-steady-state processes since the concentration at a given position changes with time, and thus the flux changes with time. This is the case when the diffusion flux depends on time, which means that a type of atoms accumulates in a region or depleted from a region (which may cause them to accumulate in another region). Figure 4.7 shows concentration profiles at three different diffusion times. *Fick's second law* characterizes these processes, which is expressed as:

$$\frac{\partial C}{\partial t} = -\frac{\partial J_x}{\partial x} = \frac{\partial}{x}\left(D\frac{\partial C}{\partial x}\right) \qquad (4.7)$$

Where $\partial C/\partial t$ is the time rate of change of concentration at a particular position, *x*. If *D* is assumed to be a constant, then

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} \qquad (4.8)$$

The solution to the above expression is possible when meaningful boundary conditions are specified. One common set of boundary conditions can be written as

$$For\ t = 0, \qquad C = C_o \ \ at\ \ 0 \le x \le \infty$$
$$For\ t > 0, \qquad C = C_s \ \ at\ \ x = 0$$

$$For\ t > 0, \qquad C = C_o \ \ at\ x = \infty$$

And the solution is

$$\frac{C_x - C_o}{C_s - C_o} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right) \qquad (4.9)$$

Where C_x represents the concentration at depth x after time t. The expression $erf\left(x/2\sqrt{Dt}\right)$ is the Gaussian error function, values of which are given in mathematical tables for various $x/2\sqrt{Dt}$ values; a partial listing is given in Table 4.1.

The above equation demonstrates the relationship between concentration, position, and time. Thus it can be used to explain many practical industrial problems like corrosion resistance of duralumin, carburization and de-carburization of steel, doping of semi-conductors, etc. The concentration parameters that appear in equation 4.9 are noted in Figure 4.7.

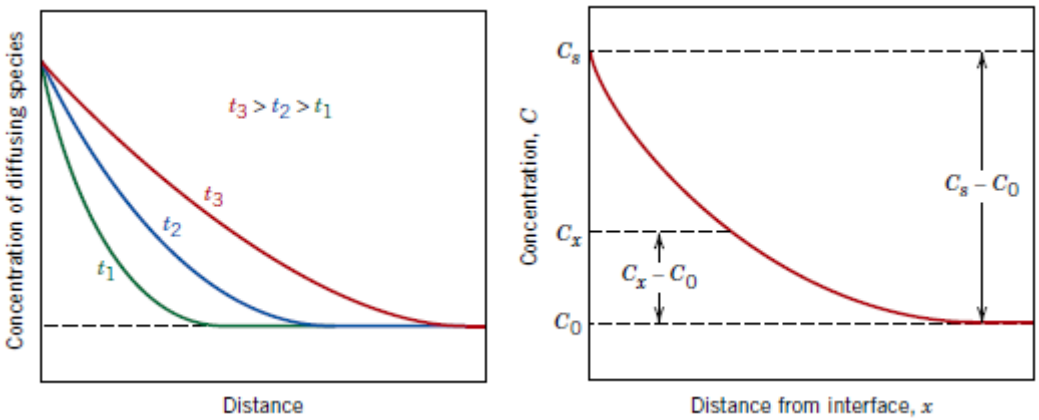


Figure 4.7 Concentration profiles for unsteady-state diffusion

Table 4.1 Tabulation of Error Function Values

z	*erf(z)*	*z*	*erf(z)*	*z*	*erf(z)*
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

Worked example 4.3

Consider one such alloy that initially has a uniform carbon concentration of 0.25 wt% and is to be treated at 950 °C (1750 °F). If the concentration of carbon at the surface is suddenly brought to and maintained at 1.20 wt%, how long will it take to achieve a carbon content of 0.80 wt% at a position 0.5 mm below the surface? The diffusion coefficient for carbon in iron at this temperature is 1.6 x 10^{-11} m^2/s; assume that the steel piece is semi-infinite.

Solution

Because this is an unsteady-state diffusion problem in which the surface composition is held constant, equation 4.9 is used. Values for all the parameters in this expression except time *t* are specified in the problem as follows:

$$C_o = 0.25\ wt\%;\ C_s = 1.2\ wt\%;\ C_x = 0.8\ wt\%; x = 0.5\ mm; D = 1.6 \times 10^{-11}\ m^2/s$$

$$\frac{C_x - C_o}{C_s - C_o} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right) \Rightarrow \frac{0.8-0.25}{1.2-0.25} = 1 - erf\left[\frac{5\times10^{-4}m}{2\sqrt{(1.6\times10^{-11}m^2/s)(t)}}\right]$$

$$0.4210 = erf\left(\frac{62.5\cdot s^{1/2}}{\sqrt{t}}\right)$$

Determine the value of *z* from Table 4.1 which the error function is 0.4210. An interpolation may be necessary as

$$\frac{z - 0.35}{0.40 - 0.35} = \frac{0.4210 - 0.3794}{0.4284 - 0.3794} \Rightarrow z = 0.3920$$

Therefore,

$$\frac{62.5s^{1/2}}{\sqrt{t}} = 0.932 \Rightarrow t = \left(\frac{62.5s^{1/2}}{0.3920}\right)^2 = 25{,}400\ s = 7.1\ \text{h}$$

4.2.3 Factors that Influence Diffusion

Ease of a diffusion process is characterized by the diffusion coefficient or diffusivity, D. The value of diffusivity for a particular system depends on many factors as many mechanisms could be operative.

Diffusing species: If the diffusing species is able to occupy interstitial sites, then it can easily diffuse through the parent matrix. On the other hand, if the size of substitutional species is almost equal to that of parent atomic size, substitutional diffusion would be easier. Thus size of diffusing species will have great influence on diffusivity of the system.

Temperature: Temperature has a most profound influence on the diffusivity and diffusion rates. It is known that there is a barrier to diffusion created by neighboring atoms those need to move to let the diffusing atom pass. Thus, atomic vibrations created by temperature assist diffusion. Empirical analysis of the system resulted in Arrhenius type of relationship between diffusivity and temperature.

$$D = D_o \cdot exp\left(\frac{-Q_d}{RT}\right) \qquad (5.0)$$

Where D_o is a pre-exponential constant, Q_d is the activation energy for diffusion, *R* is gas constant (Boltzmann's constant) and *T* is absolute temperature. From the above equation it can be inferred that large activation energy means relatively small diffusion coefficient. It can also be observed that there exists a linear proportional relation between *(ln D)* and *(1/T)*. Thus by plotting and considering the intercepts, values of Q_d and D_o can be found experimentally. Table 4.2 shows diffusion data of some diffusing species.

Lattice structure: Diffusion is faster in open lattices or in open directions than in closed directions.

Presence of defects: As stated in earlier section, defects like dislocations, grain boundaries act as short-circuit paths for diffusing species, where the activation energy is diffusion is less. Thus the presence of defects enhances the diffusivity of diffusing species.

Table 4.2 Diffusion data

Diffusing Species	*Host Metal*	$D_0(m^2/s)$	*Activation Energy* Q_d		*Calculated Value*	
			kJ/mol	*eV/atom*	$T(°C)$	$D(m^2/s)$
Fe	α-Fe (BCC)	2.8×10^{-4}	251	2.60	500	3.0×10^{-21}
					900	1.8×10^{-15}
Fe	γ-Fe (FCC)	5.0×10^{-5}	284	2.94	900	1.1×10^{-17}
					1100	7.8×10^{-16}
C	α-Fe	6.2×10^{-7}	80	0.83	500	2.4×10^{-12}
					900	1.7×10^{-10}
C	γ-Fe	2.3×10^{-5}	148	1.53	900	5.9×10^{-12}
					1100	5.3×10^{-11}
Cu	Cu	7.8×10^{-5}	211	2.19	500	4.2×10^{-19}
Zn	Cu	2.4×10^{-5}	189	1.96	500	4.0×10^{-18}
Al	Al	2.3×10^{-4}	144	1.49	500	4.2×10^{-14}
Cu	Al	6.5×10^{-5}	136	1.41	500	4.1×10^{-14}
Mg	Al	1.2×10^{-4}	131	1.35	500	1.9×10^{-13}
Cu	Ni	2.7×10^{-5}	256	2.65	500	1.3×10^{-22}

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CHAPTER 5. MECHANICAL PROPERTIES OF METALS

Most of the materials used in engineering are metallic in nature. The prime reason is due to the versatile nature of their properties over a very broad range compared with other kinds of materials. Many engineering materials are subjected to forces or loads when in service. When a force is applied on a solid material, it may result in translation, rotation, or deformation of that material. The present discussion is confined primarily to the mechanical behavior of metals under deformation. This chapter discusses the stress–strain behavior of metals and the related mechanical properties, and also examines other important mechanical characteristics.

Learning objectives

After studying this chapter, you should be able to do the following:

- Explain the concepts of stress and strain.
- Describe elastic deformation using stress-strain diagram.
- State Hooke's law and note the conditions under which it is valid.
- Define Poisson's ratio
- Determine the modulus of elasticity given a stress-strain diagram.
- Explain the tensile properties of materials under plastic deformation
- Compute the working stress for a ductile material.

5.0 Introduction

The mechanical behavior of a material reflects the relationship between its response and deformation to an applied load or force. The key mechanical design properties are stiffness, strength, hardness, ductility, and toughness. Deformation constitutes both change in shape, *distortion*, and change in size/volume, *dilatation*. Solid materials are defined such that change in their volume under applied forces is very small, thus deformation is used as synonymous to distortion. The ability of a material to withstand the applied force without any deformation is expressed in two ways, i.e. strength and hardness. Strength is defined in many ways as per the design requirements, while the hardness may be defined as resistance to indentation or scratch.

Material deformation can be permanent or temporary. Permanent deformation is irreversible i.e. stays even after the applied force is removed, while the temporary deformation disappears after removing the applied force. Thus, deformation is recoverable. Both kinds of deformation can be function of time, or independent of time. Temporary deformation is called *elastic deformation*, while the permanent deformation is called *plastic deformation*. When a material is subjected to applied forces, first the material experiences elastic deformation followed by plastic deformation.

The mechanical properties of materials are ascertained by performing carefully designed laboratory experiments that replicate as nearly as possible the service conditions. The extent of elastic and plastic deformations will primarily depend on the kind of material, rate of load application, ambient temperature, among other factors. As shown in Figure 5.1, it is possible for the load to be tensile, compressive, or shear, and its magnitude may be constant with time, or fluctuate continuously. Application time may be only a fraction of a second, or it may extend over a period of many years. Service temperature may be an important factor.

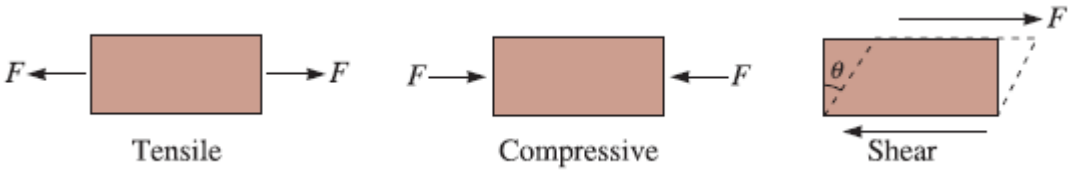


Figure 5.1 Tensile, compressive, and shear stresses. F is the applied load or force

5.1 Concepts of Stress and Strain

Forces applied act on a surface of the material, and thus the force intensity, force per unit area, is used in analysis. Analogous to this, deformation is characterized by percentage change in length per unit length in three distinct directions. Force intensity is also called *engineering stress* (or simply *stress*, σ), and is given by force divided by area on which the force is acting. *Engineering strain* (or simply *strain*, ε) is given by change in length divided by original length. Strain actually indicates an average change in length in a particular direction. According to definitions, *stress* and *strain* are expressed in equations (5.1) and (5.2), respectively.

$$\sigma = \frac{F}{A_o} \qquad (5.1)$$

$$\varepsilon = \frac{\Delta l}{l_o} = \frac{l_i - l_o}{l_o} \qquad (5.2)$$

Where F is the instantaneous load applied perpendicular to the specimen cross section, in units of newton (N) or pounds force (lb_f), A_o is the original cross sectional area before any load is applied (m^2 or $in.^2$), l_o is the original length before any load is applied, l_i is the instantaneous length and Δl and is the deformation elongation or change in length at some instant, relative to the original length.

The SI unit of stress is pascal (Pa) or N/m^2 whereas the customary (U.S) unit is pounds force per square inch (psi). Strain is dimensionless, but meters per meter or inches per inch are often used; the value of strain is obviously independent of the unit system. Sometimes strain is also expressed as a percentage.

When the load is applied continuously, the dimensions of the material change and therefore engineering stress and strain values do not indicate the true deformation characteristics of the material. Thus, the need for measures of stress and strain based on instantaneous dimensions arises. Ludwik first proposed the concept of, and defined the *true strain* or *natural strain* (ε_t) as follows:

$$\varepsilon_t = \int_{l_o}^{l} \frac{dl}{l} = ln\frac{l}{l_o} \qquad (5.3)$$

The material volume is expected to be constant i.e. $A_oL_o = AL$, and therefore,

$$\varepsilon_t = ln\frac{l}{l_o} = ln\frac{A_o}{A} = ln(\varepsilon + 1) \qquad (5.4)$$

There are certain advantages of using true strain over conventional or engineering strain. These include (i) equivalent absolute numerical value for true strains in cases of tensile and compressive for same intuitive deformation and (ii) total true strain is equal to the sum of the incremental strains.

True stress (σ_t) is given as load divided by the cross-sectional area over which it acts at an instant.

$$\sigma_t = \frac{F}{A} = \frac{F}{A_o} \cdot \frac{A_o}{A} = \sigma(\varepsilon + 1) \qquad (5.5)$$

It is to be noted that engineering stress is equal to true stress up to the elastic limit of the material. The same applies to the strains. After the elastic limit i.e. once the material starts deforming plastically, engineering values and true values of stresses and strains differ. The above equation relating engineering and true stress-strains are valid only up to the limit of uniform deformation i.e. up to the onset of necking in tension test. This is because the relations are developed by assuming both constancy of volume and homogeneous distribution of strain along the length of the tension specimen.

5.2 Elastic Deformation

The degree to which a structure deforms or strains depends on the magnitude of an imposed stress. For most metals that are stressed in tension and at relatively low levels, stress and strain are proportional to each other through the relationship

$$\sigma = E\varepsilon \qquad (5.6)$$

This is known as Hooke's law, and the constant of proportionality, E, is the **modulus of elasticity,** or *Young's modulus*. The SI unit for the modulus of elasticity is gigapascal (GPa). For most typical metals the magnitude of this modulus ranges between 45 GPa (6.5 x 10^6 psi), for magnesium, and 407 GPa (59 x 10^6 psi), for tungsten.

Deformation in which stress and strain are proportional is called **elastic deformation;** a plot of stress (ordinate) versus strain (abscissa) results in a linear relationship, as shown in Figure 5.2. The slope of this linear segment corresponds to the modulus of elasticity *E*. This modulus may be thought of as stiffness, or a material's resistance to elastic deformation.

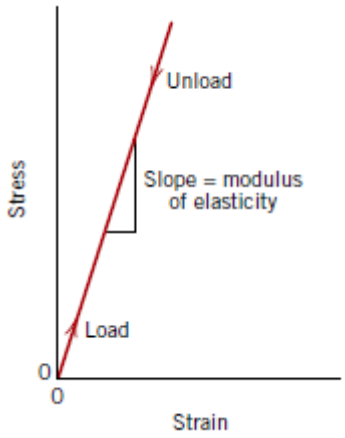


Figure 5.2 Schematic stress–strain diagram showing linear elastic deformation for loading and unloading cycles.

The greater the modulus, the stiffer the material, or the smaller the elastic strain that results from the application of a given stress. The modulus is an important design parameter used for computing elastic deflections.

Elastic deformation is nonpermanent, which means that when the applied load is released, the piece returns to its original shape. As shown in the stress–strain plot, application of the load corresponds to moving from the origin up and along the straight line. When the load is released, the line is traversed in the opposite direction, back to the origin.

There are some materials such as gray cast iron, concrete, etc. for which the elastic portion of the stress–strain curve is not linear. Hence, it is impossible to determine the modulus of elasticity as described previously. For this nonlinear behavior, either *tangent* or *secant modulus* is normally used. Tangent modulus is taken as the slope of the stress–strain curve at some specified level of stress, whereas secant modulus represents the slope of a secant drawn from the origin to some given point of the curve. The determination of these moduli is illustrated in Figure 5.3.

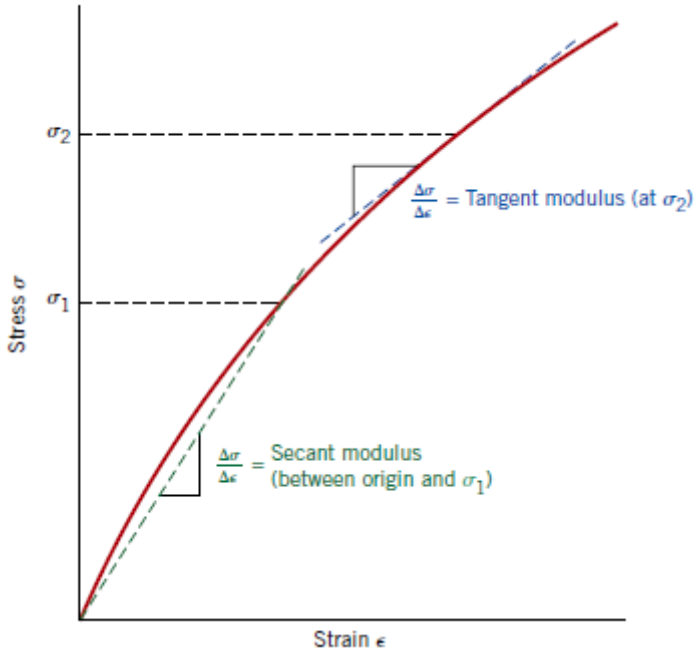


Figure 5.3 Schematic stress–strain diagram showing nonlinear elastic behavior and how secant and tangent moduli are determined

If one dimension of the material changed, other dimensions of the material need to be changed to keep the volume constant. This lateral or transverse strain is related to the applied longitudinal strain by empirical means, and the ratio of transverse strain to longitudinal strain is known as *Poisson's ratio* (v). Transverse strain can be expected to be opposite in nature to longitudinal strain, and both longitudinal and transverse strains are linear strains. For most metals the values of v are close to 0.33, for polymers it is between 0.4 – 0.5, and for ionic solids it is around 0.2.

The stress–strain characteristics at low stress levels are virtually the same for both tensile and compressive situations, to include the magnitude of the modulus of elasticity. Analogous to the relation between normal stress and linear strain defined earlier, shear stress (τ) and shear strain (γ) in elastic range are related through the expression;

$$\tau = G\gamma \qquad (5.7)$$

Where G is known as *Shear modulus* of the material. It is also known as *modulus of elasticity in shear*. It is related with Young's modulus, E, through Poisson's ratio, v, as;

$$G = \frac{E}{2(1+\nu)} \qquad (5.8)$$

Similarly, the *Bulk modulus* or *volumetric modulus of elasticity K*, of a material is defined as the ratio of hydrostatic or *mean stress* (σ_m) to the volumetric strain (Δ). The relation between *E* and *K* is given by

$$K = \frac{\sigma_m}{\Delta} = \frac{E}{3(1-2\nu)} \qquad (5.9)$$

The Modulus of elasticity, shear modulus and Poisson ratio values for some metals at room temperature are presented in Table 5.1.

Table 5.1 Room-Temperature Elastic and Shear Moduli and Poisson's Ratio for Various Metal Alloys

	Modulus of Elasticity		*Shear Modulus*		*Poisson's*
Metal Alloy	***GPa***	***10^6 psi***	***GPa***	***10^6 psi***	***Ratio***
Aluminum	69	10	25	3.6	0.33
Brass	97	14	37	5.4	0.34
Copper	110	16	46	6.7	0.34
Magnesium	45	6.5	17	2.5	0.29
Nickel	207	30	76	11.0	0.31
Steel	207	30	83	12.0	0.30
Titanium	107	15.5	45	6.5	0.34
Tungsten	407	59	160	23.2	0.28

Up to this point, it has been assumed that elastic deformation is time independent. That is, an applied stress produces an instantaneous elastic strain that remains constant over the period of time the stress is maintained. It has also been assumed that upon release of the load the strain is totally recovered; thus, the strain immediately returns to zero. However, in most engineering materials, there exist a time-dependent elastic strain component. That is, elastic deformation will continue after the stress application, and upon load release some finite time is required for complete recovery. This time-dependent elastic behavior is known as *anelasticity***,** and it is due to time-dependent microscopic and atomistic processes that are attendant to the deformation. The anelastic component for metals is normally small and is often neglected.

Worked example 5.1

A piece of copper originally 305 mm (12 in.) long is pulled in tension with a stress of 276 MPa (40,000 psi). If the deformation is entirely elastic, what will be the resultant elongation?

Solution

Since the deformation is elastic, strain is dependent on stress according to equation 5.6. Also, the elongation Δl is related to the original length l_o in Equation 5.2. Combining these two expressions and solving for Δl gives;

$$\sigma = \varepsilon E = \left(\frac{\Delta l}{l_o}\right) E \Rightarrow \Delta l = \frac{\sigma l_o}{E}$$

The values of σ and l_o are given as 276 MPa and 305 mm, respectively, and the magnitude of *E* for copper from Table 5.1 is 110 GPa (16 x 10^6 psi). Elongation is obtained by substitution into the preceding expression as

$$\Delta l = \frac{\sigma l_o}{E} = \frac{(270\ MPa) \cdot (305\ mm)}{(110\ x\ 10^3\ MPa)} = 0.77\ mm\ (0.03\ in.)$$

5.3 Plastic Deformation

For most metallic materials, elastic deformation persists only to strains of about 0.005. As the material is deformed beyond this point, the stress is no longer proportional to strain, thus, Hooke's law in equation 5.6, ceases to be valid, and permanent, non-recoverable, or **plastic deformation** occurs. Figure 5.4*a* shows schematic plot of the tensile stress–strain behavior into the plastic region for a typical metal. The transition from elastic to plastic is a gradual one for most metals; some curvature results at the onset of plastic deformation, which increases more rapidly with rising stress.

From an atomic perspective, plastic deformation corresponds to the breaking of bonds with original atom neighbors and then re-forming bonds with new neighbors as large numbers of atoms or molecules move relative to one another; upon removal of the stress they do not return to their original positions. The mechanism of this deformation is different for crystalline and amorphous materials. For crystalline solids, deformation is accomplished by means of a process called *slip,* which involves the motion of dislocations as discussed in the next chapter.

5.3.1 Tensile properties

Most structures are designed to ensure that only elastic deformation will result when a stress is applied. A structure or component that has plastically deformed, or experienced a permanent change in shape, may not be capable of functioning as intended. It is therefore desirable to know the stress level at which plastic deformation begins, or where the phenomenon of ***yielding*** occurs.

For metals that experience this gradual elastic–plastic transition, the point of yielding may be determined as the initial departure from linearity of the stress–strain curve; this is sometimes called the ***proportional limit,*** as indicated by point *P,* in Figure 5.4*a,* and represents the onset of plastic deformation on a microscopic level. The position of this point *P* is difficult to measure precisely. As a consequence, a convention has been established wherein a straight line is constructed parallel to the elastic portion of the stress–strain curve at some specified strain offset, usually 0.002. The stress corresponding to the intersection of this line and the stress–strain curve as it bends over in the plastic region is defined as the ***yield strength,*** σ_y**.** This is demonstrated in Figure 5.4*a*. The units of yield strength are MPa or psi.

For those materials having a nonlinear elastic region, use of the strain offset method is not possible, and the usual practice is to define the yield strength as the stress required to produce some amount of strain (e.g., ε = 0.005).

Some steels and other materials exhibit the tensile stress–strain behavior shown in Figure 5.4*b*.The elastic–plastic transition is very well defined and occurs abruptly in what is termed a *yield point phenomenon*. At the upper yield point, plastic deformation is initiated with an apparent decrease in engineering stress. Continued deformation fluctuates slightly about some

constant stress value, termed the lower yield point; stress subsequently rises with increasing strain. For metals that display this effect, the yield strength is taken as the average stress that is associated with the lower yield point, because it is well defined and relatively insensitive to the testing procedure. Thus, it is unnecessary to employ the strain offset method for these materials.

The magnitude of the yield strength for a metal is a measure of its resistance to plastic deformation. Yield strengths may range from 35 MPa (5000 psi) for a low strength aluminum to over 1400 MPa (200,000 psi) for high-strength steels.

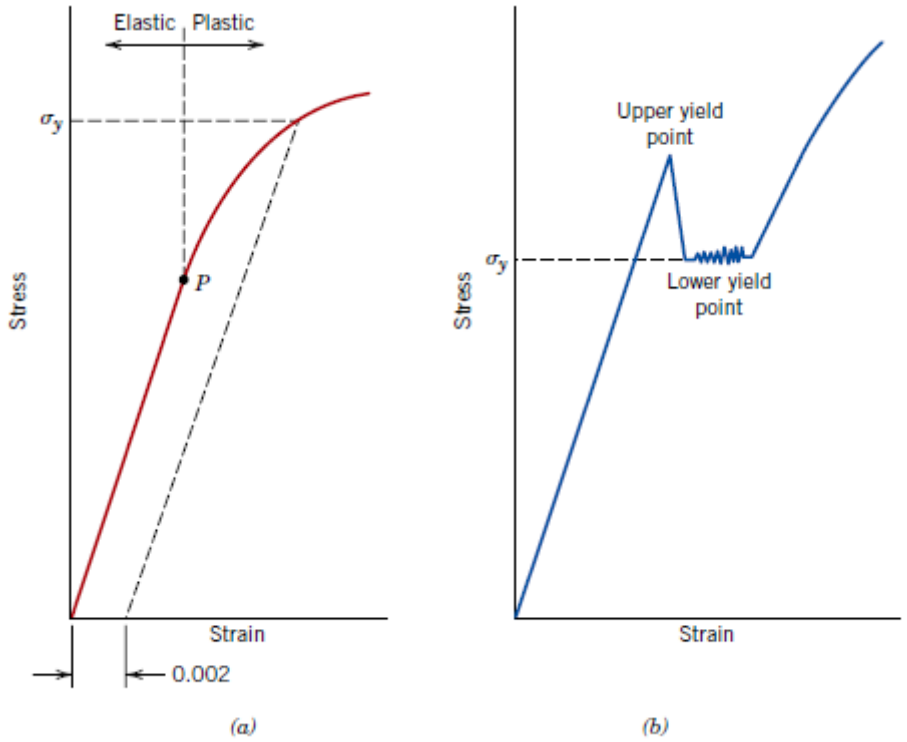


Figure 5.4 (*a*) Typical stress– strain behavior for a metal showing elastic and plastic deformations (*b*) Representative stress–strain behavior found for some steels

5.3.1.1 Tensile strength

After yielding, the stress necessary to continue plastic deformation in metals increases to a maximum, point *M*, as shown in Figure 5.5, and then decreases to the eventual fracture, point *F*. The ***tensile strength,*** *TS* (MPa or psi) is the stress at the maximum on the stress–strain curve. This corresponds to the maximum stress that can be sustained by a structure in tension; if this stress is applied and maintained, fracture will result. All deformation up to this point is uniform throughout the narrow region of the tensile specimen.

However, at this maximum stress, a small constriction or neck begins to form at some point, and all subsequent deformation is confined at this neck, as indicated by the schematic specimen insets in Figure 5.5. This phenomenon is termed *necking*, and fracture ultimately occurs at the neck. The fracture strength corresponds to the stress at fracture.

Tensile strengths may vary anywhere from 50 MPa (7000 psi) for an aluminum to as high as 3000 MPa (450,000 psi) for the high-strength steels. Ordinarily, when the strength of a metal is cited for design purposes, the yield strength is used. This is because by the time a stress corresponding to the tensile strength has been applied, often the structure has experienced so

much plastic deformation that it is useless. Furthermore, fracture strengths are not normally specified for engineering design purposes.

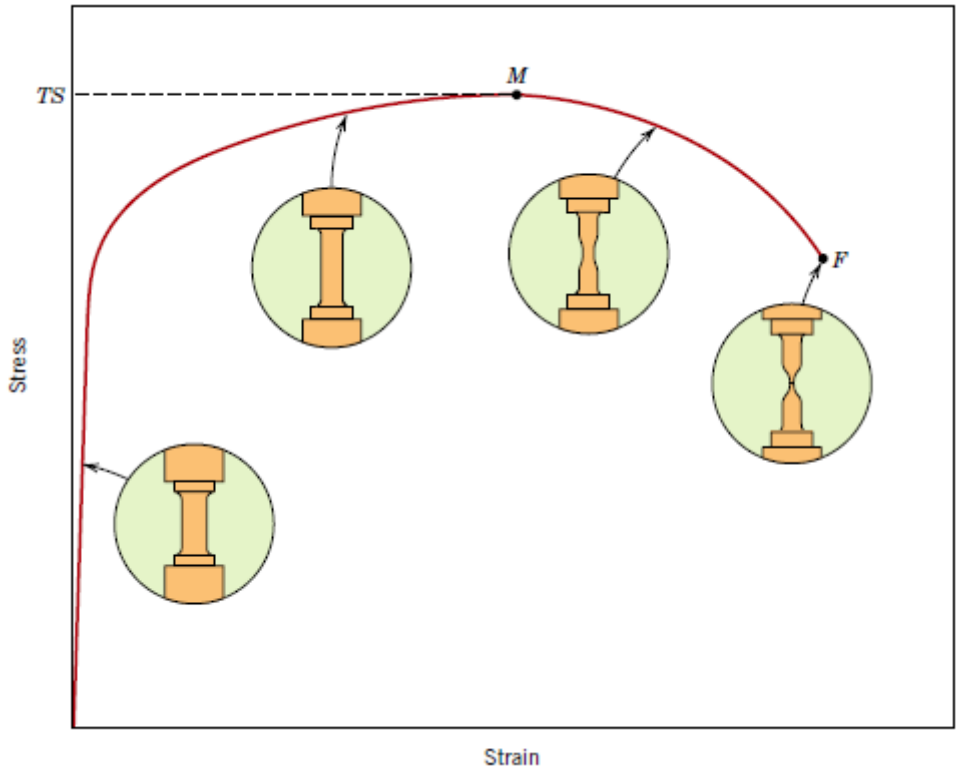


Figure 5.5 Typical stress–strain behavior to fracture, point *F*

Worked example 5.2

Using the tensile stress–strain behavior for the brass specimen in Figure 5.6, determine the following:

(a) The modulus of elasticity

(b) The yield strength at a strain offset of 0.002

(c) The maximum load that can be sustained by a cylindrical specimen having an original diameter of 12.8 mm (0.505 in.)

(d) The change in length of a specimen originally 250 mm (10 in.) long that is subjected to a tensile stress of 345 MPa (50,000 psi)

Solution

(a) The modulus of elasticity is the slope of the elastic or initial linear portion of the stress–strain curve. The slope of this linear region is the rise over the run, or change in stress divided by the corresponding change in strain; in mathematical terms,

$$E = slope = \frac{\Delta\sigma}{\Delta\varepsilon} = \frac{\sigma_2 - \sigma_1}{\varepsilon_2 - \varepsilon_1}$$

It is convenient to take both σ_1 and ε_1 as zero. If σ_2 is arbitrarily taken as 150 MPa, then ε_2 will have a value of 0.0016. Therefore,

$$E = \frac{\sigma_2 - \sigma_1}{\varepsilon_2 - \varepsilon_1} = \frac{150 - 0}{0.0016 - 0} = 93.8\ GPa\ (13.6\ x\ 10^6\ psi)$$

This is very close to the value of 97 GPa (14 x 10^6 psi) given for brass in Table 5.1.

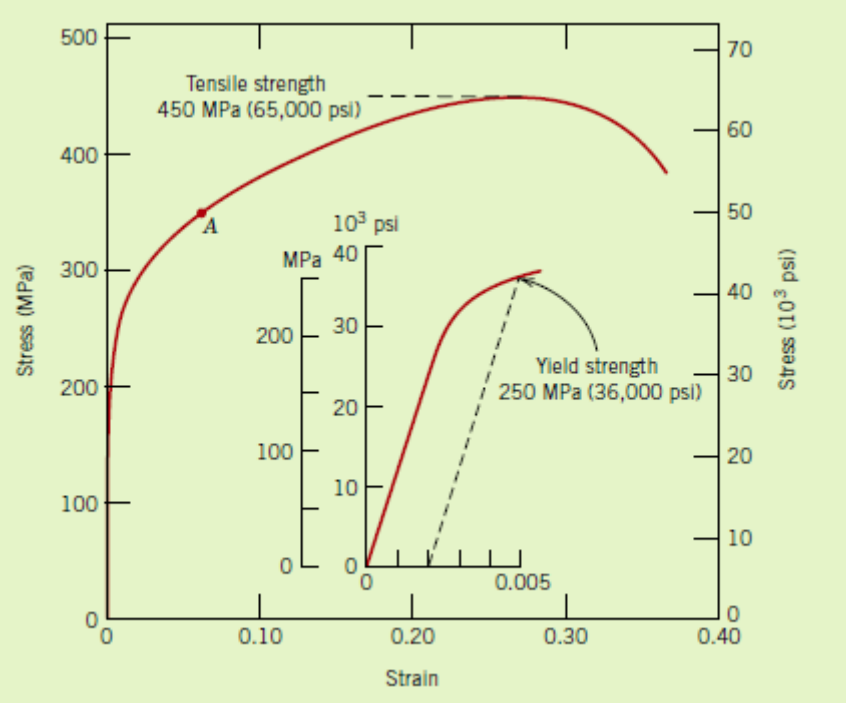


Figure 5.6 The stress–strain behavior for the brass specimen

(b) The 0.002 strain offset line is constructed as shown in the inset; its intersection with the stress–strain curve is at approximately 250 MPa (36,000 psi), which is the yield strength of the brass.

(c) The maximum load that can be sustained by the specimen is calculated by using equation 5.1, in which σ is taken to be the tensile strength, which is 450 MPa (65,000 psi) from Figure 5.6. Solving for *F*, the maximum load, yields

$$F = \sigma \cdot A_o = \sigma \cdot \pi \left(\frac{d_o}{2}\right)^2$$

$$F = (450\ x\ 10^6\ N/m^2) \times \pi \times \left(\frac{0.00128\ m}{2}\right)^2 = 57{,}900\ N$$

(d) To compute the change in length, Δ*l*, in Equation 5.2, it is first necessary to determine the strain that is produced by a stress of 345 MPa. This is accomplished by locating the stress point on the stress–strain curve, point *A*, and reading the corresponding strain from the strain axis, which is approximately 0.06. As l_o is given as 250 mm, the elongation becomes;

$$\Delta l = \varepsilon l_o = 0.06 \times 250\ mm = 15\ mm\ (0.6\ in.)$$

5.3.1.2 Ductility

Ductility is another important mechanical property. It is a measure of the degree of plastic deformation that has been sustained at fracture. A metal that experiences very little or no plastic deformation upon fracture is termed *brittle*. The tensile stress–strain behaviors for both ductile and brittle metals are schematically illustrated in Figure 5.7.

Ductility may be expressed quantitatively as either *percent elongation* or *percent reduction in area*. The percent elongation %EL is the percentage of plastic strain at fracture, or

$$\%\ EL = \left(\frac{l_f - l_o}{l_o}\right) \times 100 \qquad (5.10)$$

Where l_f is the fracture length and l_o is the original gauge length as given earlier. Inasmuch as a significant proportion of the plastic deformation at fracture is confined to the neck region, the magnitude of %EL will depend on specimen gauge length. The shorter l_o, the greater the fraction of total elongation from the neck and, consequently, the higher the value of %EL. Therefore, l_o should be specified when percent elongation values are cited.

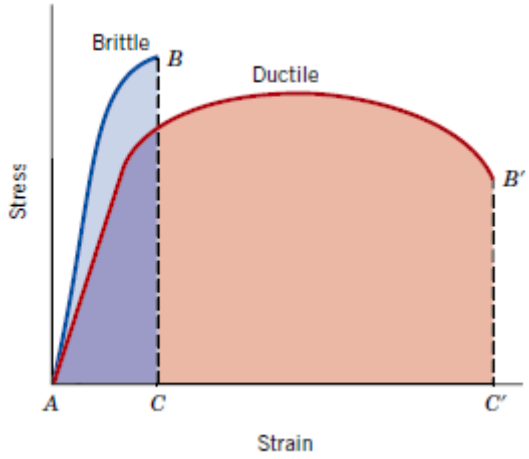


Figure 5.7 Schematic representations of tensile stress–strain behavior for brittle and ductile metals loaded to fracture.

Percent reduction in area %RA is defined as;

$$\%RA = \left(\frac{A_o - A_f}{A_o}\right) \qquad (5.11)$$

Where A_o is the original cross-sectional area and A_f is the cross-sectional area at the point of fracture. Percent reduction in area values are independent of both l_o and A_o. Furthermore, for a given material, the magnitudes of %EL and %RA will, in general, be different. Most metals possess at least a moderate degree of ductility at room temperature; however, some become brittle as the temperature is lowered.

A knowledge of the ductility of materials is important for at least two reasons. First, it indicates to a designer the degree to which a structure will deform plastically before fracture. Second, it specifies the degree of allowable deformation during fabrication operations. Brittle materials are approximately considered to be those having a fracture strain of less than about 5%. Thus, several important mechanical properties of metals may be determined from tensile stress–strain tests. Table 5.2 presents some typical room-temperature values of yield strength, tensile strength, and ductility for some common metals.

These properties are sensitive to any prior deformation, the presence of impurities, and/or any heat treatment to which the metal has been subjected. The modulus of elasticity is one mechanical parameter that is insensitive to these treatments. As with modulus of elasticity, the magnitudes of both yield and tensile strengths decline with increasing temperature whereas for ductility, it usually increases with temperature.

Table 5.2 Typical Mechanical Properties of Some Metals and Alloys in an Annealed State

Metal Alloy	*Yield Strength, MPa (ksi)*	*Tensile Strength, MPa (ksi)*	*Ductility, %EL [in 50 mm (2 in.)]*
Aluminum	35 (5)	90 (13)	40
Copper	69 (10)	200 (29)	45
Brass (70Cu–30Zn)	75 (11)	300 (44)	68
Iron	130 (19)	262 (38)	45
Nickel	138 (20)	480 (70)	40
Steel (1020)	180 (26)	380 (55)	25
Titanium	450 (65)	520 (75)	25
Molybdenum	565 (82)	655 (95)	35

5.3.1.3 Resilience

Resilience is the capacity of a material to absorb energy when it is deformed elastically and then, upon unloading, to have this energy recovered. The associated property is the *modulus of resilience, Ur*, which is the strain energy per unit volume required to stress a material from an unloaded state up to the point of yielding.

The modulus of resilience for a specimen subjected to a uniaxial tension test is just the area under the stress–strain curve taken to yielding as shown in Figure 5.8. It is computed as

$$U_r = \int_0^{\varepsilon_y} \sigma d\varepsilon \qquad (5.12a)$$

Assuming a linear elastic region,

$$U_r = \frac{1}{2}\sigma_y \varepsilon_y \qquad (5.12b)$$

Where ε_y is the strain at yielding.

The SI unit of resilience is joules per cubic meter (J/m^3, equivalent to Pa) which is a unit of energy, and thus area under the stress–strain curve represents energy absorbed per unit volume of material.

Substituting equation 5.6 into equation 5.12b yields

$$U_r = \frac{1}{2}\sigma_y \left(\frac{\sigma_y}{E}\right) = \frac{\sigma_y^2}{2E} \qquad (5.13)$$

Thus, resilient materials are those having high yield strengths and low moduli of elasticity; such alloys would be used in spring applications.

5.3.1.4 Toughness

Toughness or specifically, fracture toughness is a property that is indicative of a material's resistance to fracture when a crack (or other stress-concentrating defect) is present. Since it is nearly impossible to manufacture materials with zero defects, fracture toughness is a major consideration for all structural materials.

Toughness can also be defined as the ability of a material to absorb energy and plastically deform before fracturing. For dynamic (high strain rate) loading conditions and when a point of stress concentration is present, *notch toughness* is assessed by using an impact test.

For static (low strain rate) situation, a measure of toughness in metals (derived from plastic deformation) may be ascertained from the results of a tensile stress–strain test. It is the area

under the stress–strain curve up to the point of fracture. The units are the same as for resilience (i.e., energy per unit volume of material).

For a metal to be tough, it must display both strength and ductility. This is demonstrated in Figure 5.7, in which the stress–strain curves are plotted for both metal types. Hence, even though the brittle metal has higher yield and tensile strengths, it has a lower toughness than the ductile one, as can be seen by comparing the areas *ABC* and *AB'C'* in Figure 5.7.

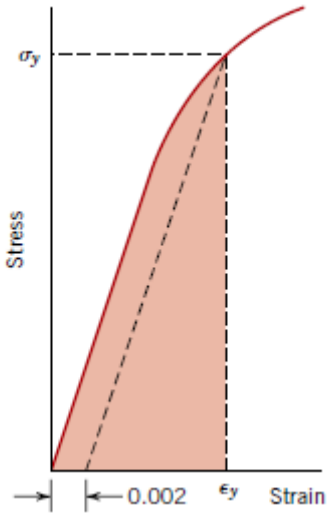


Figure 5.8 Schematic showing determination of modulus of resilience from the tensile stress–strain behavior of a material.

Worked example 5.3

Cylindrical specimen of steel having an original diameter of 12.8 mm (0.505 in.) is tensile tested to fracture and found to have an engineering fracture strength of 460 MPa (67,000 psi). If its cross-sectional diameter at fracture is 10.7 mm (0.422 in.), determine:

(a) The ductility in terms of percent reduction in area

(b) The true stress at fracture

Solution

(a) Ductility is computed using equation 5.11, as

$$\%RA = \left(\frac{A_o - A_f}{A_o}\right) = \frac{\pi\cdot\left(\frac{12.8\ mm}{2}\right)^2 - \pi\cdot\left(\frac{10.7\ mm}{2}\right)^2}{\pi\cdot\left(\frac{12.8\ mm}{2}\right)^2} \times 100 = 30\%$$

(b) True stress is defined by equation 5.5, where in this case the area is taken as the fracture area A_f. However, the load at fracture must first be computed from the fracture strength as

$$F = \sigma_f \cdot A_o = (460\ \times 10^6\ N/m^2)(128.7\ mm^2)\left(\frac{1\ m^2}{10^6\ mm^2}\right) = 59{,}200\ N$$

Thus, the true stress is calculated as

$$\sigma_t = \frac{F}{A_f} = \frac{59{,}200\ N}{(89.9\ mm^2)\left(\frac{1\ m^2}{10^6\ mm^2}\right)} = 6.6\ \times\ 10^8\ N/m^2 = 660\ MPa$$

5.3.2 Hardness

Another mechanical property to consider is hardness**,** which is a measure of a material's resistance to localized plastic deformation (e.g., a small dent or a scratch). Early hardness tests were based on natural minerals with a scale constructed solely on the ability of one material to scratch another that was softer. Quantitative hardness techniques have been developed over the years in which a small indenter is forced into the surface of a material to be tested, under controlled conditions of the load and rate of application. The depth or size of the resulting indentation is measured, which in turn is related to a hardness number; the softer the material, the larger and deeper the indentation, and the lower the hardness index number. Measured hardness are only relative (rather than absolute), and care should be exercised when comparing values determined by different techniques.

Hardness tests are performed more often than any other mechanical test because they are simple and inexpensive. No special specimen need to be prepared, and the testing apparatus is relatively inexpensive. Also, the test is nondestructive and specimen is neither fractured nor excessively deformed; a small indentation is the only deformation. Lastly, other mechanical properties often may be estimated from hardness data, such as tensile strength. Some hardness tests include Rockwell hardness test and Brinell hardness test.

Both tensile strength and hardness indicate a metal's resistance to plastic deformation. Consequently, they are roughly proportional, as shown in Figure 5.9, for tensile strength as a function of the Brinell harness number (HB) for cast iron, steel, and brass. As a rule of thumb for most steels, the HB and the tensile strength are related according to the expression;

$$TS\ (MPa) = 3.45 \times HB \qquad (5.14a)$$

$$TS\ (psi) = 500 \times HB \qquad (5.14b)$$

5.4 Elastic Recovery after Plastic Deformation

Upon load release during a stress–strain test, some fraction of the total deformation is recovered as elastic strain. This behavior is demonstrated in the schematic stress–strain plot in Figure 5.10. During the unloading cycle, the curve traces a near straight-line path from the point of unloading (point *D*), and its slope is virtually identical to the modulus of elasticity, or parallel to the initial elastic portion of the curve. The magnitude of this elastic strain, which is regained during unloading, corresponds to the strain recovery, as shown in Figure 5.10. If the load is reapplied, the curve will traverse essentially the same linear portion in the direction opposite to unloading; yielding will again occur at the unloading stress level where the unloading began. There will also be an elastic strain recovery associated with fracture.

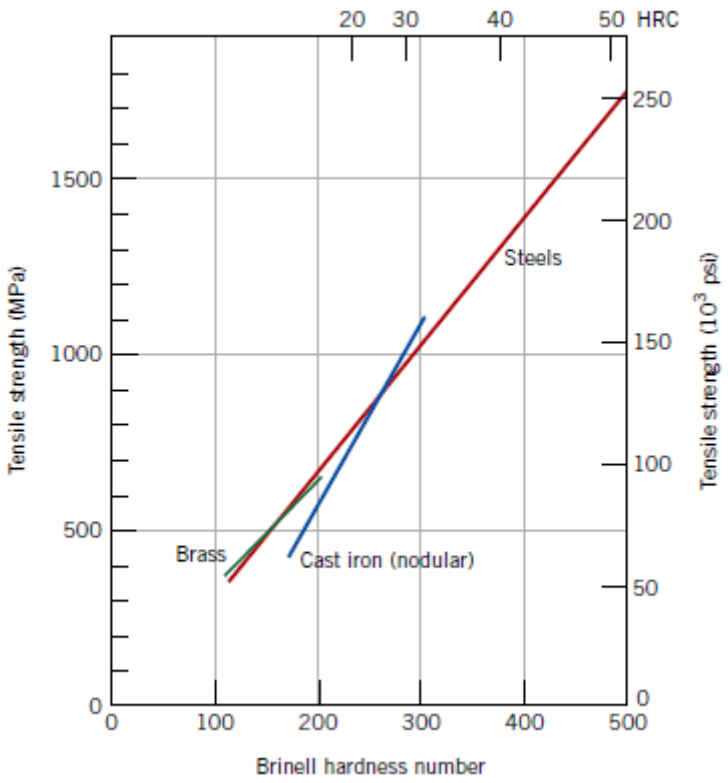


Figure 5.9 Relationships between hardness and tensile strength for steel, brass, and cast iron.

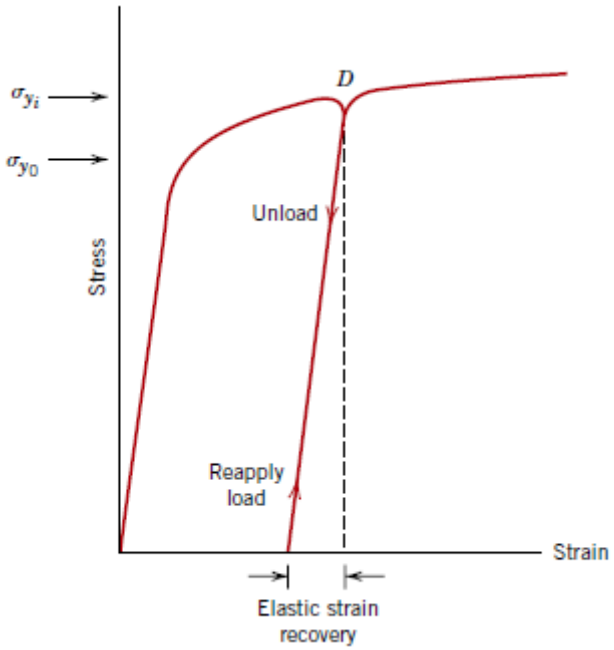


Figure 5.10 Schematic tensile stress–strain diagram showing the phenomena of elastic strain recovery and strain hardening.

5.5 Property Variability and Design Consideration

Scatter in measured properties of engineering materials is inevitable because of number of factors such as test method, specimen fabrication procedure, apparatus calibration, operator

bias, etc. In spite of property variation, some typical value is desirable. Most commonly used typical value is by taking an average of the data. The average (x) of a parameter, x_i, is given as

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad (5.15)$$

Where n is the number of samples, and x_i is the discrete measurement.

In some instances, it is desirable to have an idea about the degree of variability, scatter, of the measured data. The most common degree of variability is the standard deviation, s, which is given by

$$s = \left[\frac{1}{n-1}\cdot\sum_{i=1}^{n}(x_i - \bar{x})^2\right]^{1/2} \qquad (5.16)$$

A large value for s means a high degree of scatter. Scatter is usually represented in graphical form using error bars. If a parameter is averaged to $\bar{x}$, and the corresponding standard deviation is s, the upper error bar limit is given by ($\bar{x}$+s), while the lower error bar is equal to ($\bar{x}$-s).

There are always uncertainties in characterizing the magnitude of applied loads and their associated stress levels for in-service applications. Also, virtually all engineering materials exhibit a variability in their measured mechanical properties, have imperfections and sustained damage during service. Consequently, design approaches must be employed to protect against unanticipated failure. The procedure was to reduce the applied stress by a design safety factor.

For less critical static situations and when tough materials are used, a **design stress,** σ_d, is taken as the calculated stress level σ_c (on the basis of the estimated maximum load) multiplied by a *design factor, N_d*. That is,

$$\sigma_d = N_d \cdot \sigma_c \qquad (5.17)$$

Where N_d is greater than unity. Thus, the material to be used for the particular application is chosen so as to have a yield strength at least as high as this value of σ_d.

Alternatively, a **safe stress** or *working stress, σ_w*, is used instead of design stress. This safe stress is based on the yield strength of the material and is defined as the yield strength divided by a *factor of safety, N_s*, or

$$\sigma_w = \frac{\sigma_y}{N_s} \qquad (5.18)$$

Using the design stress in equation (5.17) is usually preferred because it is based on the anticipated maximum applied stress instead of the yield strength of the material; normally there is a greater uncertainty in estimating this stress level than in the specification of the yield strength. However, in the discussion of this text, we are concerned with factors that influence the yield strengths of metal alloys and not in the determination of applied stresses; therefore, the succeeding discussion will deal with working stresses and factors of safety.

The choice of an appropriate value of N_s is necessary. If N_s is too large, then component will result in overdesign; that is, either too much material or an alloy having a higher-than-necessary strength will be used. Values normally range between 1.2 and 4.0. Selection of N_s depend on a number of factors such as economics, previous experience, accuracy with which mechanical

forces and material properties may be determined, and, most important, the consequences of failure in terms of loss of life and/or property damage. Because large N_s values lead to increased material cost and weight, structural designers are moving toward using tougher materials with redundant designs, where economically feasible.

Worked example 5.4

A tensile-testing apparatus is to be constructed that must withstand a maximum load of 220,000 N. The design calls for two cylindrical support posts, each of which is to support half of the maximum load. Furthermore, plain-carbon (1045) steel ground and polished shafting rounds are to be used; the minimum yield and tensile strengths of this alloy are 310 MPa and 565 MPa, respectively. Specify a suitable diameter for these support posts.

Solution

First, decide on a factor of safety, N_s, which allows determination of a working stress according to equation 5.18. In addition, to ensure that the apparatus will be safe to operate, any elastic deflection of the rods during testing should be minimized; therefore, a relatively conservative factor of safety is to be used, say 5. Thus, the working stress becomes;

$$\sigma_w = \frac{\sigma_y}{N_s} = \frac{310\ MPa}{5} = 62\ MPa$$

From the definition of stress, in equation 5.1,

$$A_o = \left(\frac{d}{2}\right)^2 \cdot \pi = \frac{F}{\sigma_w} \Rightarrow d = 2\sqrt{\frac{F}{\pi\sigma_w}}$$

$$d = 2\sqrt{\frac{110{,}00\ N}{\pi \times 62 \times 10^6\ N/m^2}} = 0.00475\ m = 47.5\ mm$$

Therefore, the diameter of each of the two rods should be 47.5 mm or 1.87 in.

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CHAPTER 6. DISLOCATIONS AND STRENGTHENING MECHANISMS

In chapter 5, it was explained that materials may experience two kinds of deformation: elastic and plastic. Plastic deformation is permanent, and strength and hardness are measures of a material's resistance to this deformation. On a microscopic scale, it corresponds to the net movement of large numbers of atoms in response to an applied stress. In crystal solids, plastic deformation most often involves the motion of dislocations, linear crystalline defects that were introduced in earlier chapter.

This chapter discusses the characteristics of dislocations and their involvement in plastic deformation. Twinning, another process by which some metals plastically deform, is also treated. In addition, techniques for strengthening single-phase metals, the mechanisms of which are described in terms of dislocations are presented. Lastly, recovery and recrystallization processes that occur in plastically deformed metals, normally at elevated temperatures and, in addition, grain growth are discussed.

Learning objectives

After studying this chapter, you should be able to do the following:

- Explain the mechanics of dislocation motion in relation to plastic deformation
- Describe slip and twinning mechanisms of plastic deformation.
- Define slip system and cite examples.
- Explain grain size reduction and solid solution strengthening mechanisms in metals.
- Describe the phenomenon of strain hardening (or cold working) in terms of dislocations and strain field interactions.
- Explain the processes of recovery, and recrystallization grain growth in annealing.

6.1 Dislocations and Plastic Deformation

While some materials are elastic in nature up to fracture point, many engineering materials like metals and thermo-plastic polymers can undergo substantial permanent deformation. This characteristic property of materials makes it feasible to shape them. However, it imposes some limitations on the engineering usefulness of such materials. Permanent deformation is due to process of shear where particles change their neighbors. During this process interatomic or inter-molecular forces and structure plays important roles, although the former are much less significant than they are in elastic behavior. Permanent deformation is broadly two types; plastic deformation and viscous flow. Plastic deformation involves the relative sliding of atomic planes in organized manner in crystalline solids, while the viscous flow involves the switching of neighbors with much more freedom that does not exist in crystalline solids.

It is well known that dislocations can move under applied external stresses. The cumulative movement of dislocations leads to the gross plastic deformation. Dislocation motion, at the microscopic level, involves rupture and reformation of inter-atomic bonds. The necessity of dislocation motion for ease of plastic deformation is well explained by the discrepancy between theoretical strength and real strength of solids. It has been concluded that one-dimensional crystal defects–dislocations, plays an important role in plastic deformation of crystalline solids.

Their importance in plastic deformation is relevant to their characteristic nature of motion in specific directions (slip-directions) on specific planes (slip-planes), where edge dislocation move by slip and climb while screw dislocation can be moved by slip and cross-slip.

The mechanics of dislocation motion are represented in Figure 6.1. When the shear stress is applied as indicated in Figure 6.1*a*, plane *A* is forced to the right; this in turn pushes the top halves of planes *B*, *C*, *D*, and so on, in the same direction. If the applied shear stress is of sufficient magnitude, the interatomic bonds of plane *B* are severed along the shear plane, and the upper half of plane *B* becomes the extra half-plane as plane *A* links up with the bottom half of plane *B* in Figure 6.1*b*. This process is subsequently repeated for the other planes, such that the extra half-plane, by discrete steps, moves from left to right by successive and repeated breaking of bonds and shifting by interatomic distances of upper half-planes.

Before and after the movement of a dislocation through some particular region of the crystal, the atomic arrangement is ordered and perfect; it is only during the passage of the extra half-plane that the lattice structure is disrupted. Ultimately this extra half-plane may emerge from the right surface of the crystal, forming an edge that is one atomic distance wide; this is shown in Figure 6.1*c*. The process by which plastic deformation is produced by dislocation motion is termed **slip;** the crystallographic plane along which the dislocation line traverses is the *slip plane*, as indicated in Figure 6.1.

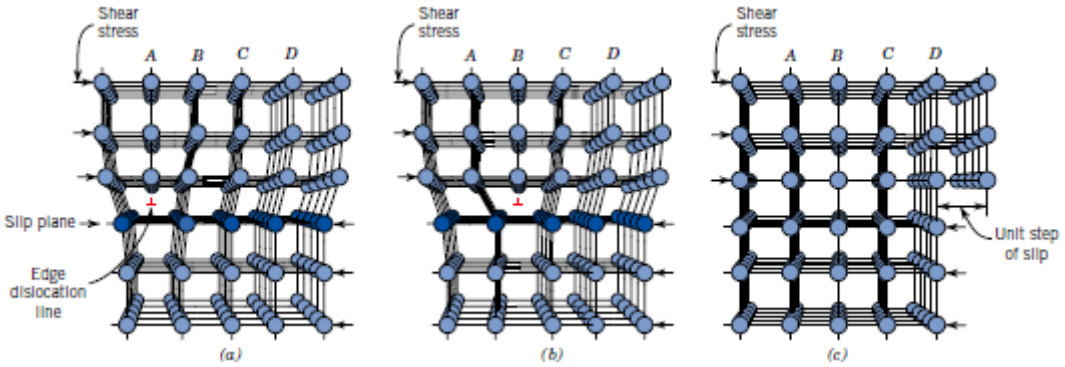


Figure 6.1 Mechanics of dislocation motion

The onset of plastic deformation involves initial motion of existing dislocations in real crystal. In perfect crystal, it can be attributed to the generation of dislocations and motion. During motion, dislocations will tend to interact among themselves. Dislocation interaction is very complex due to a number of dislocations moving on a number of slip planes in various directions. When they are in the same plane, they repel each other if they have the same sign, and annihilate if they have opposite signs (leaving behind a perfect crystal). In general, when dislocations are close and their strain fields add to a larger value, they repel, because being close increases the potential energy (it takes energy to strain a region of the material). When unlike dislocations are on closely spaced neighboring slip planes, complete annihilation cannot occur. In this situation, they combine to form a row of vacancies *or* an interstitial atom.

An important consequence interaction of dislocations that are not on parallel planes is that they intersect each other or inhibit each other's motion. Intersection of two dislocations results in a sharp break in the dislocation line. These breaks can be of two kinds: a break in dislocation

line moving it out of slip plane is called jog whereas a kink is break in dislocation line that remains in slip plane.

Other hindrances to dislocation motion include interstitial and substitutional atoms, foreign particles, grain boundaries, external grain surface, and change in structure due to phase change. Important practical consequences of hindrance of dislocation motion are that, dislocations are still movable but at higher stresses (or forces), and in most instances leads to generation of more dislocations. Dislocations can spawn from existing dislocations, and from defects, grain boundaries and surface irregularities. Thus, the number of dislocations increases dramatically during plastic deformation. As additional movement of dislocations requires increase of stress, material can be said to be strengthened i.e. materials can be strengthened by controlling the motion of dislocation.

6.1.1 Mechanisms of Plastic Deformation in Metals

Plastic deformation, as explained in earlier section, involves motion of dislocations. There are two prominent mechanisms of plastic deformation, namely ***slip*** and ***twinning***.

Slip is the prominent mechanism of plastic deformation in metals. It involves sliding of blocks of crystal over one another along definite crystallographic planes, called slip planes. In physical words it is analogous to a deck of cards when it is pushed from one end. Slip occurs when shear stress applied exceeds a critical value. During slip each atom usually moves same integral number of atomic distances along the slip plane producing a step, but the orientation of the crystal remains the same. Steps observable under microscope as straight lines are called slip lines.

Slip occurs most readily in specific directions (slip directions) on certain crystallographic planes. This is due to limitations imposed by the fact that single crystal remains homogeneous after deformation. Generally slip plane is the plane of greatest atomic density, and the slip direction is the close packed direction within the slip plane. It turns out that the planes of the highest atomic density are the most widely spaced planes, while the close packed directions have the smallest translation distance. Feasible combination of a slip plane together with a slip direction is considered as a ***slip system***. The possible slip systems for BCC, FCC and HCP crystal structures are listed in Table 6.1. For each of these structures, slip is possible on more than one family of planes (e.g., {110}, {211}, and {321} for BCC). For metals having BCC and HCP crystal structures, some slip systems are often operable only at elevated temperatures.

In a single crystal, plastic deformation is accomplished by the process known as slip, and sometimes by twinning. The extent of slip depends on many factors including external load and the corresponding value of shear stress produced by it, the geometry of crystal structure, and the orientation of active slip planes with the direction of shearing stresses generated. Schmid first recognized that, single crystals at different orientations having the same material require different stresses to produce slip.

Although an applied stress may be pure tensile (or compressive), shear components exist at all but parallel or perpendicular alignments to the stress direction. These are termed *resolved shear stresses*, and their magnitudes depend not only on the applied stress, but also on the orientation of both the slip plane and direction within that plane.

Table 6.1 Slip systems for metals with different crystal structures

Metals	*Slip Plane*	*Slip Direction*	*Number of Slip Systems*
	Face-Centered Cubic		
Cu, Al, Ni, Ag, Au	{111}	$\langle 1\bar{1}0 \rangle$	12
	Body-Centered Cubic		
α-Fe, W, Mo	{110}	$\langle \bar{1}11 \rangle$	12
α-Fe,W	{211}	$\langle \bar{1}11 \rangle$	12
α-Fe, K	{321}	$\langle \bar{1}11 \rangle$	24
	Hexagonal Close-Packed		
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg, Zr	$\{10\bar{1}0\}$	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg	$\{10\bar{1}1\}$	$\langle 11\bar{2}0 \rangle$	6

As indicated in Figure 6.2, F is the external load applied, A is cross-sectional area over which the load applied, ϕ represent the angle between the normal to the slip plane and the applied stress direction, and λ is the angle between the slip direction and stress directions (tensile axis). The resolved shear stress, τ_R can be determined as;

$$\tau_R = \frac{F cos\lambda}{A/cos\phi} = \frac{F cos\lambda cos\phi}{A} = \sigma cos\lambda cos\phi = \sigma m \qquad (6.1)$$

Where $m = cos\lambda cos\phi$ is the Schmid factor and σ *is* is the applied stress.

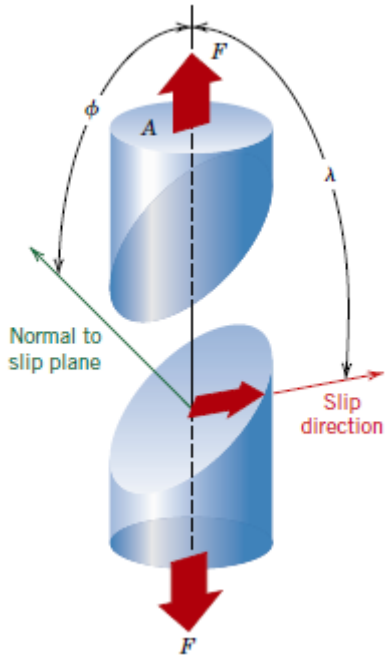


Figure 6.2 Geometrical relationships between the tensile axes, slip plane, and slip direction used in calculating the resolved shear stress for a single crystal

In general, $\phi + \lambda \neq 90°$, because it need not be the case that the tensile axis, the slip plane normal, and the slip direction all lie in the same plane.

Shear stress is maximum for the condition where $\lambda = \phi = 45°$. If either of the angles are equal to 90°, resolved shear stress will be zero, and thus no slip occurs. If the conditions are such that either of the angles is close to 90°, crystal will tend to fracture rather than slip. Single crystal metals and alloys are used mainly for research purpose and only in a few cases of engineering applications.

Almost all engineering alloys are polycrystalline in nature. Gross plastic deformation of a polycrystalline specimen corresponds to the comparable distortion of the individual grains by means of slip. Although some grains may be oriented favorably for slip, yielding cannot occur unless the unfavorably oriented neighboring grains can also slip. Thus in a polycrystalline aggregate, individual grains provide a mutual geometrical constraint on one another, and this precludes plastic deformation at low applied stresses. Hence, to initiate plastic deformation, polycrystalline metals require higher stresses than for equivalent single crystals, where stress depends on orientation of the crystal. Much of this increase is attributed to geometrical reasons.

Slip in polycrystalline material involves generation, movement and (re-)arrangement of dislocations. Because of dislocation motion on different planes in various directions, they may interact as well. This interaction can cause dislocation immobile or mobile at higher stresses. During deformation, mechanical integrity and coherency are maintained along the grain boundaries; that is, the grain boundaries are constrained, to some degree, in the shape it may assume by its neighboring grains. Once yielding has occurred, continued plastic deformation is possible only if enough slip systems are simultaneously operative so as to accommodate grain shape changes while maintaining grain boundary integrity.

According to Von Mises criterion, a minimum of five independent slip systems must be operative for a polycrystalline solid to exhibit ductility and maintain grain boundary integrity. This arises from the fact that an arbitrary deformation is specified by the six components of strain tensor, but because of requirement of constant volume, there are only independent strain components. Crystals which do not possess five independent slip systems are never ductile in polycrystalline form, although small plastic elongation may be noticeable because of twinning or a favorable preferred orientation.

The second important mechanism of plastic deformation is ***twinning***. It results when a portion of crystal takes up an orientation that is related to the orientation of the rest of the untwined lattice in a definite, symmetrical way. The twinned portion of the crystal is a mirror image of the parent crystal. The plane of symmetry is called twinning plane. Each atom in the twinned region moves by a homogeneous shear a distance proportional to its distance from the twin plane. The lattice strains involved in twinning are small, usually in order of fraction of interatomic distance, thus resulting in very small gross plastic deformation. The important role of twinning in plastic deformation is that it causes changes in plane orientation so that further slip can occur. If the surface is polished, the twin would still be visible after etching because it possesses a different orientation from the untwined region. This is in contrast with slip, where slip lines can be removed by polishing the specimen.

Twinning also occurs in a definite direction on a specific plane for each crystal structure. However, it is not known if there exists resolved shear stress for twinning. Twinning generally occurs when slip is restricted, because the stress necessary for twinning is usually higher than that for slip. Thus, some HCP metals with limited number of slip systems may preferably twin. Also, BCC metals twin at low temperatures because slip is difficult. Of course, twinning and

slip may occur sequentially or even concurrently in some cases. Twinning systems for some metals are given in Table 6.2. Figure 6.3 presents schematic movement of atoms during plastic deformation in slip and during twinning. In Table 6.3, the mechanisms of plastic deformations are compared with respect to their characteristics.

Table 6.2 Twin systems for different crystal structures

Crystal structure	Examples	Twin plane	Twin direction
FCC	Ag, Au, Cu	(111)	[112]
BCC	α-Fe, Ta	(112)	[111]
HCP	Zn, Cd, Mg, Ti	$(10\bar{1}2)$	$[\bar{1}011]$

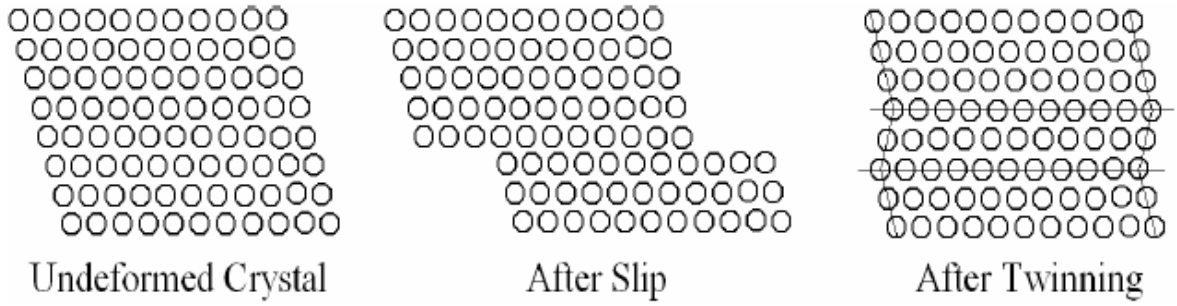


Figure 6.3 Schematic presentation of different plastic deformation mechanism.

Table 6.3 Comparison of mechanism of plastic deformation.

	During/In Slip	During/In Twinning
Crystal orientation	Same above and below the slip plane	Differ across the twin plane
Size (in terms of inter-atomic distance)	Multiples	Fractions
Occurs on	Widely spread planes	Every plane of region involved
Time required	Milli seconds	Micro seconds
Occurrence	On many slip systems simultaneously	On a particular plane for each crystal

6.2 Strengthening Mechanisms in Metals

The ability of a metal to deform plastically depends on ease of dislocation motion under applied external stresses. As mentioned in earlier section, strengthening of a metal consist hindering dislocation motion. The dislocation motion can be hindered in many ways, by using strengthening mechanisms. Strengthening by methods of grain-size reduction, solid-solution alloying and strain hardening are applied in single-phase metals. Precipitation hardening, dispersion hardening, fiber strengthening and Martensitic strengthening are applicable to multi-phase metallic materials.

6.2.1 Strengthening by Grain Size Reduction

This mechanism is based on the fact that crystallographic orientation changes abruptly in passing from one grain to the next across the grain boundary. Thus it is difficult for a dislocation moving on a common slip plane in one crystal to pass over to a similar slip plane in another grain, especially if the orientation is very misaligned. In addition, the crystals are separated by a thin non-crystalline region, which is the characteristic structure of a large angle grain boundary. As shown in Figure 6.4, during plastic deformation, slip or dislocation motion must take place across this common boundary; say, from grain A to grain B. The grain boundary acts as a barrier to dislocation motion for two reasons. First, because the two grains are of different orientations, a dislocation passing into grain B will have to change its direction of motion; this becomes more difficult as the crystallographic misorientation increases and lastly, the atomic disorder within a grain boundary region will result in a discontinuity of slip planes from one grain into the other.

Hence dislocations are stopped by a grain boundary and pile up against it. The smaller the grain size, the more frequent is the pile up of dislocations. A twin boundary can also act as an obstacle to dislocation motion.

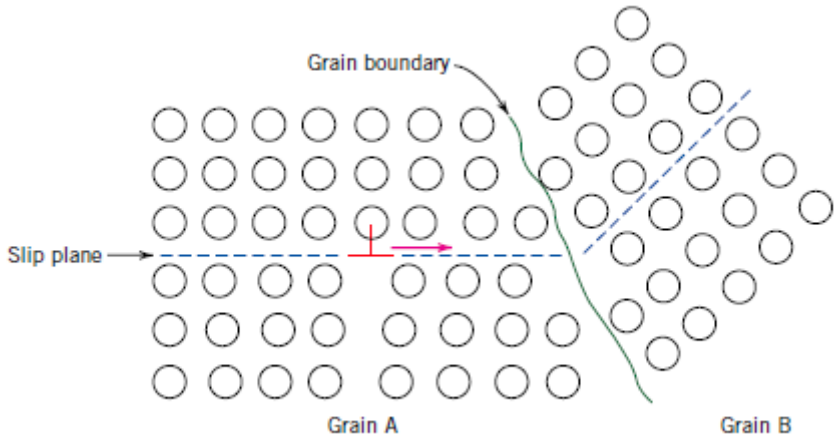


Figure 6.4 Dislocation motion as it encounters a grain boundary

Effectiveness of grain boundary depends on its characteristic misalignment, represented by an angle. The ordinary high-angle grain boundary (misalignment > 5°) represents a region of random misfit between the grains on each side of the boundary. This structure contains grain-boundary dislocations which are immobile. However they group together within the boundary to form a step or grain boundary ledge. These ledges can act as effective sources of dislocations as the stress at end of slip plane may trigger new dislocations in adjacent grains. Small angle grain boundaries (misalignment < 1°) are considered to be composed of a regular array of dislocations, and are not effective in blocking dislocations.

With decrease in grain size, the mean distance a dislocation can travel decreases, and soon starts pile up of dislocations at grain boundaries. This leads to increase in yield strength of the material. Hall and Petch have derived the relation, famously known as *Hall–Petch equation*, between yield strength (σ_y) and grain size (d) as;

$$\sigma_y = \sigma_o + k_y d^{-1/2} \qquad (6.2)$$

where σ_o is the 'friction stress', representing the overall resistance of the crystal lattice to dislocation movement, k_y is the 'locking parameter' that measures the relative hardening contribution of the grain boundaries and d is the average grain diameter. Friction stress is interpreted as the stress needed to move unlocked dislocations along the slip plane. It depends strongly on temperature, strain, alloy and impurity content. Locking parameter is known to be independent of temperature. Therefore friction stress and locking parameters are constants for particular material. It is important to note that the above relation is not valid for both very large grain and extremely fine grain sizes. Figure 6.5 demonstrates the yield strength dependence on grain size for a brass alloy.

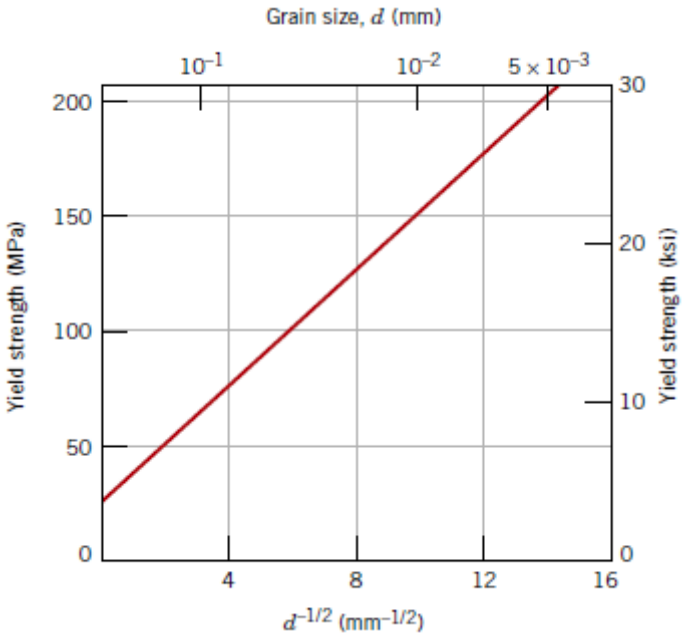


Figure 6.5 The influence of grain size on the yield strength of a brass alloy.

Grain size reduction improves not only strength, but also the toughness of many alloys. Grain size can be controlled by rate of cooling, and also by plastic deformation followed by appropriate heat treatment. Grain size is usually measured using a light microscope to observe a polished specimen by counting the number of grains within a given area, by determining the number of grains that intersect a given length of random line, or by comparing with standard-grain-size charts. If d is average grain diameter, S_v is grain boundary area per unit volume, N_L is mean number of intercepts of grain boundaries per unit length of test line, and N_A is number of grains per unit area on a polished surface; then all these are related as follows:

$$S_v = 2N_L, \quad d = \frac{3}{S_v} = \frac{3}{2N_L} \quad \text{and} \quad d = \sqrt{\frac{6}{\pi N_A}} \tag{6.3}$$

Another common method of measuring the grain size is by comparing the grains at a fixed magnification with standard grain size charts. Charts are coded with ASTM grain size number, G, and is related with n_a – number of grains per mm^2 at 1X magnification as

$$G = -2.9542 + 1.4427 \ln n_a \tag{6.4}$$

The higher the ASTM grain number, smaller is the grain diameter. Grain diameter, *D* (in mm), and ASTM number, *G*, can be related as follows:

$$D = \frac{1}{100}\sqrt{\frac{645}{2^{G-1}}} \qquad (6.5)$$

Where *G* represents number of grains per square inch (645 mm^2) at a magnification of 100X is equal to 2^{G-1}.

6.2.2 Solid Solution Strengthening

Adding atoms of another element that occupy interstitial or substitutional positions in parent lattice increases the strength of parent material. This is because stress fields generated around the solute atoms interact with the stress fields of a moving dislocation, thereby increasing the stress required for plastic deformation i.e. the impurity atoms cause lattice strain which can 'anchor' dislocations. This occurs when the strain caused by the alloying element compensates that of the dislocation, thus achieving a state of low potential energy. Since the addition of solid-solution alloy affect the entire stress-strain curve, it can be said that solute atoms have more influence on the frictional resistance to dislocation motion than on the static locking of dislocations. Pure metals are almost always softer than their alloys. Solute strengthening effectiveness depends on two factors namely; size difference between solute and parent atoms, and concentration of solute atoms.

Solute atoms are two categories with respect to their relative strengthening effect. Those producing non-spherical distortions, such as interstitial atoms, have a relative strengthening effect per unit concentration of about three times their shear modulus whereas those producing spherical distortion, such as substitutional atoms, have a relative strengthening of about G/10. Solute atoms interact with dislocations in many ways, namely: elastic interaction; modulus interaction; stacking-fault interaction; electrical interaction; short-range order interaction; and long-range order interaction. Elastic, modulus, and long-range order interactions are of long-range i.e. they are relatively insensitive to temperature and continue to act about 0.6 T_m where T_m is the melting temperature in absolute Kelvin degrees.

Elastic interaction results from mutual interaction of elastic stress fields, while *modulus interaction* occurs if the presence of a solute atom locally alters the modulus of the crystal. *Stacking-fault interactions* arise because solute atoms may segregate to the stacking-faults, thus lowering stacking-fault energy and widening partial dislocations. *Electrical interaction* arises if solute atoms of dissimilar valence interact with dislocations which have electrical dipoles. *Short-range order interaction* arises from the tendency for solute atoms to arrange themselves so that they have more than the equilibrium number of dissimilar neighbors. The opposite of short-range order is clustering. *Long-range order interaction* arises in alloys which form super-lattices, in which long-range periodic arrangement of dissimilar atoms gets disturbed, to form anti-phase boundaries, because of dislocation motion which leads to dissociation of dislocation into pairs of ordinary dislocations.

Some polycrystalline metals, such as mild steel, display a discrete yield point type of behavior where a higher stress is necessary to initiate plastic flow than to continue it. Thus, there exists a localized, heterogeneous type of transition from elastic to plastic deformation

which produces a yield point in the stress-strain curve i.e. elastic-plastic transition is very well demarked and occurs abruptly in what is called *yield-point phenomenon*. During loading, the load increases steadily with elastic strain, drops suddenly at the upper yield point where plastic deformation gets initiated. Continued deformation fluctuates slightly about some constant stress value (lower yield point). Subsequently stress increase with increasing strain. Elongation that occurs at constant load is called the yield-point elongation. Yield strength for metals with this phenomenon is taken as average of lower yield point, thus it is not necessary to employ strain offset method. Figure 6.6 presents the exaggerated version of yield-point phenomenon.

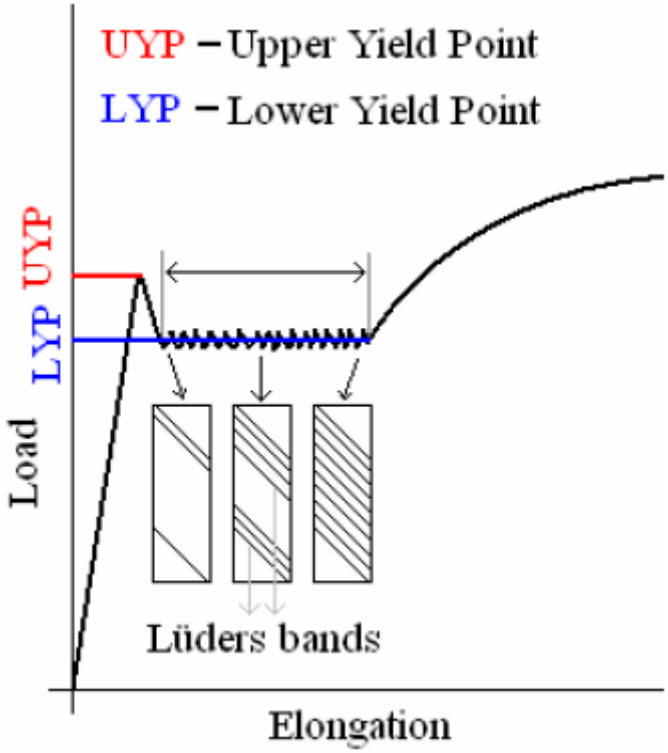


Figure 6.6 Yield point phenomenon.

At the upper yield point, deformed part of metal forms a discrete band usually visible to eye at a stress concentration. Many bands may form at the same time at different stress concentration sites. Each band then propagates along the length of the specimen, causing yield-point elongation. These bands are called Lüders bands / Hartmann lines / stretcher stains, and generally are approximately 45 to the tensile axis. Occurrence of yield point is associated with presence of small amounts of interstitial or substitutional impurities. It's been found that either unlocking of dislocations by a high stress for the case of strong pinning or generation of new dislocations are the reasons for yield-point phenomenon. Thus when the dislocation line is pulled free from the influence of solute atoms, slip can occur at a lower stress. On the other hand, when dislocations are strongly pinned, new dislocations will generate followed by drop in flow stress. This is the origin of upper yield stress. Released dislocations tend to pile-up at grain boundaries, producing stress concentration which in addition to applied stress unlocks sources in next grain. This is the way band propagates. Magnitude of yield-point effect will depend on energy of interaction between solute atoms and dislocations and on the concentration of solute atoms at the dislocations.

6.2.3 Strain Hardening

The two most important industrial processes used to harden metals or alloys are strain hardening and heat treatment. Strain hardening is used for hardening/strengthening materials that are not responsive to heat treatment. The phenomenon whereby ductile metals become stronger and harder when they are deformed plastically is called strain hardening or work hardening.

For some metals and alloys, the region of the true stress–strain curve from the onset of plastic deformation to the point at which necking begins may be approximated by;

$$\sigma_t = K\varepsilon_t^n \qquad (6.6)$$

In the above expression, *K* and *n* are constants; these values vary from alloy to alloy and also depend on the condition of the material (i.e., whether it has been plastically deformed, heat-treated, etc.). The parameter *n* is often termed the *strain hardening exponent* and has a value less than unity. It is a measure of the ability of the metal to strain harden and for a given amount of plastic strain, the higher the value of n, the greater is the strain hardening. Values of *n* and *K* for several alloys are contained in Table 6.4.

Increasing temperature lowers the rate of strain hardening, and thus the treatment is given, usually, at temperatures well below the melting point of the material. Thus the treatment is also known as *cold working*. Most metals strain hardens at room temperature. The consequence of strain hardening a material is improved strength and hardness but material's ductility will be reduced.

It is convenient to express the degree of plastic deformation as percent cold work, defined as:

$$\%CW = \left(\frac{A_o - A_d}{A_o}\right) \times 100 \qquad (6.7)$$

Where A_o is the original cross sectional area that experiences deformation, and A_d is the area after deformation.

Table 6.4 Tabulation of n and K Values for Some Alloys

		K	
Material	***n***	***MPa***	***psi***
Low-carbon steel (annealed)	0.21	600	87,000
4340 steel alloy (tempered @ 315°C)	0.12	2650	385,000
304 stainless steel (annealed)	0.44	1400	205,000
Copper (annealed)	0.44	530	76,500
Naval brass (annealed)	0.21	585	85,000
2024 aluminum alloy (heat-treated—T3)	0.17	780	113,000
AZ-31B magnesium alloy (annealed)	0.16	450	66,000

Figures 6.7*a* and 6.7*b* demonstrate how steel, brass, and copper increase in yield and tensile strength with increasing cold work. The price for this enhancement of hardness and strength is in the ductility of the metal. This is shown in Figure 6.7*c*, in which the ductility, in percent elongation, experiences a reduction with increasing percent cold work for the same three alloys.

The influence of cold work on the stress–strain behavior of a low-carbon steel is shown in Figure 6.7*d*; here stress–strain curves are plotted at 0%CW, 4%CW, and 24%CW.

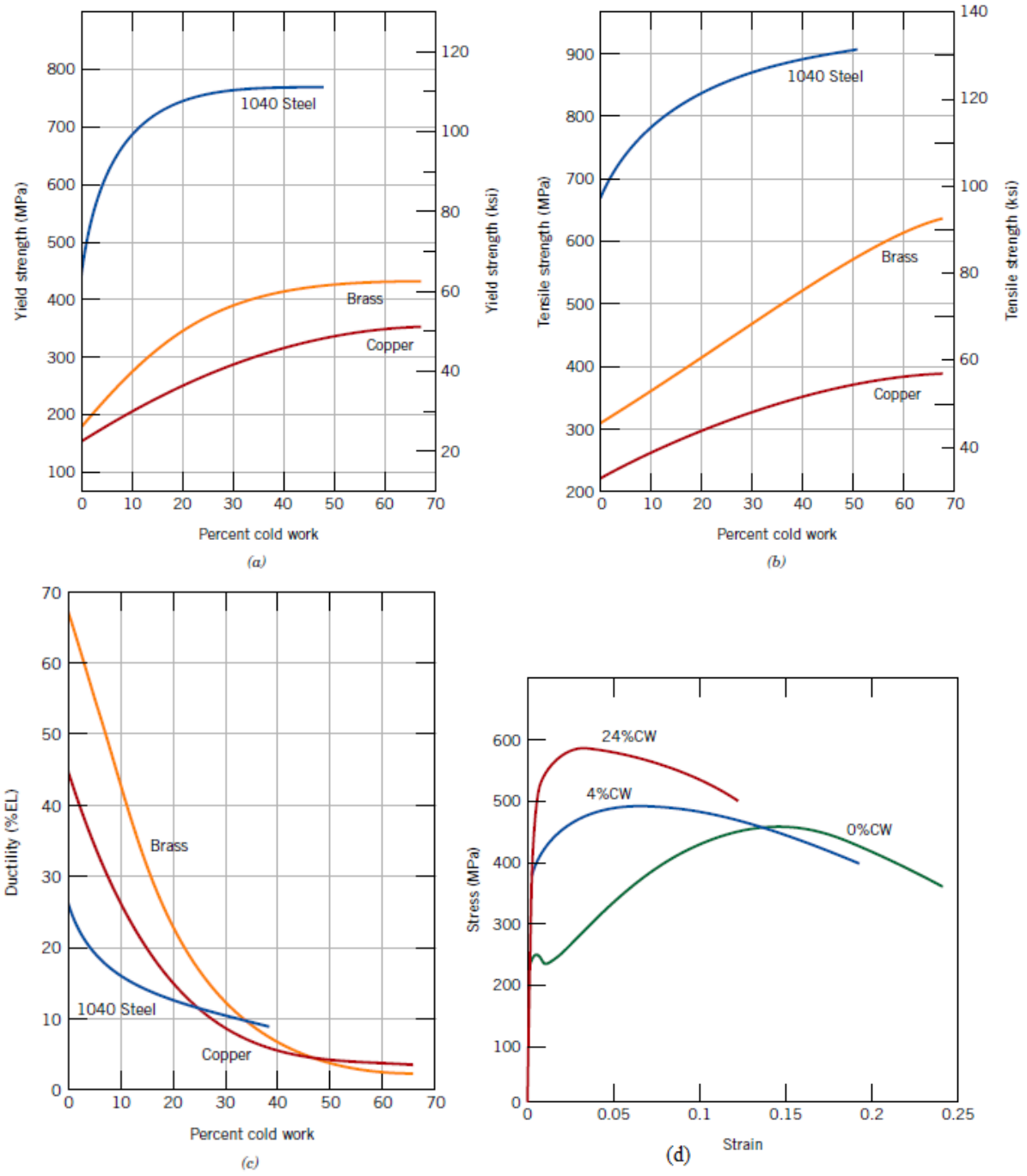


Figure 6.7 The influence of cold work on yield strength, tensile strength, ductility and stress–strain behavior

Strain hardening is used commercially to enhance the mechanical properties of metals during fabrication procedures. In addition to mechanical properties, physical properties of a material also changes during cold working. There is usually a small decrease in density, an appreciable decrease in electrical conductivity, small increase in the thermal coefficient of expansion and increased chemical reactivity (decrease in corrosion resistance). The cold worked state is a condition of higher internal energy than the un-deformed metal. Although

cold worked dislocation cell structure is mechanically stable, it is not thermodynamically stable. With increase in temperature state becomes more unstable, eventually reverts to strain-free condition. This process of heating to attain strain-free condition is called annealing heat treatment where effects of strain hardening may be removed. Annealing process can be divided into three distinct processes: recovery, recrystallization and grain growth. All these steps of the heat treatment process are explained in the later section. It is usual industrial practice to use alternate cycles of strain hardening and annealing to deform most metals to a very great extent.

Worked example 6.1

Compute the tensile strength and ductility (%EL) of a cylindrical copper rod if it is cold worked such that the diameter is reduced from 15.2 mm to 12.2 mm.

Solution

It is first necessary to determine the percent cold work resulting from the deformation. This is possible using equation 6.7.

$$\%CW = \frac{\left(\frac{15.2\ mm}{2}\right)^2 \pi - \left(\frac{12.2\ mm}{2}\right)^2 \pi}{\left(\frac{15.2\ mm}{2}\right)^2 \pi} \times 100 = 35.6\%$$

The tensile strength is read directly from the curve for copper (Figure 6.7*b*) as 340 MPa (50,000 psi). From Figure 6.7*c*, the ductility at 35.6%CW is about 7%EL.

6.2.4 Precipitation Hardening and Dispersion Strengthening

Small second-phase particles distributed in a ductile matrix can hinder the dislocation motion and thus increase the strength of a material. Second-phase particles either can be introduced by mixing and consolidation (dispersion strengthening) or precipitated in solid state (precipitation hardening).

Precipitation hardening or age hardening is produced by solution treating and quenching an alloy. The term ‘age hardening’ is used to describe the process because strength develops with time. Requisite for precipitation hardening to take place is that second phase must be soluble at an elevated temperature but precipitates upon quenching and aging at a lower temperature. This limits the alloy systems which can be strengthened by precipitation hardening. For example: Al-alloys, Cu-Be alloys, Mg-Al alloys, Cu-Sn alloys. If the precipitation occurs at normal ambient temperatures, it is called natural aging. Some alloy systems needed to be aged at higher temperatures and the process is known as artificial aging. Most precipitation hardened alloys are limited in their maximum service temperatures, which may lose their strength at elevated temperatures due to over-aging.

In dispersion strengthening, hard particles are mixed with matrix powder and consolidated and processed by powder metallurgy techniques. Here, second phase shall have very little solubility in the matrix, even at elevated temperatures. Because there is very little solubility, the particles resist growth or over-aging to a much greater extent than the second phase particles in a precipitation hardening system. Theoretically, at least, it is possible to produce infinite number of dispersion-hardened systems by mixing finely divided metallic powders and second phase particles (oxides, carbides, nitrides, borides, etc).

Dislocations moving through the matrix, which is either precipitation hardened or dispersion strengthened, have two alternatives. They can either cut through the precipitate particles *or* bend around and bypass them. The first alternative is possible only when the slip plane is continuous from the matrix through the precipitate particle and when the stress to move a dislocation in precipitate is comparable to that in matrix. Cutting of particles is easier for small particles which can be considered as segregated solute atoms. There are many particle properties that can dictate the ease of shearing, for example: coherency strains, stacking-fault energy, ordered structure, modulus effect, interfacial energy, morphology and lattice friction stress.

The degree of strengthening from second phase particles depends on particle distribution in the matrix. Particle dispersion along with the shape can be expressed by specifying the volume fraction, average particle diameter, and mean interspacing of particles which are interrelated. A simple expression for interspacing of particles is:

$$\lambda = \frac{4(1-f)r}{3f} \qquad (6.8)$$

Where f is the volume fraction of spherical particles of radius r.

The spacing between second phase particles should be typically a few hundred angstroms. Optimum strengthening occurs during aging once the right interspacing of particles is achieved. Smaller the particles, dislocations can cut through them at lower stresses. On the other hand, larger the particles they will be distributed at wider distances. Thus it is necessary to arrest the growth of particles in precipitation hardened alloys at right interspacing. Over-aging allows the fine particles to coalesce into larger and fewer particles. The interspacing is increased thereby and the yield stress decreases.

6.2.5 Fibre Strengthening

Second phase material can also be introduced into matrix in form of fibers to strengthen it. However, the mechanism of strengthening is different from either precipitation hardening or dispersion strengthening where second phase is introduced as fine particles.

Prerequisites are materials to be used as fibers include high strength and/or high strength-to-weight ratio. Fibers usually, thus, have high strength and high modulus while the matrix must be ductile and non-reactive with the fibers. Fibers may be long and continuous or they may be discontinuous. Examples for fiber material: Al_2O_3, boron, graphite, metal, glass, etc. Examples for matrix material: metals, polymers. Fiber reinforced materials are an important group of materials known as composite materials.

In fiber-reinforced materials, high modulus fibers carry essentially the entire load while the matrix serves to transmit the load to the fibers. Matrix also protects fibers from surface damage, serves to blunt cracks which arise from fiber breakage while it also serves to separate the fibers. It is understood that analysis of precipitation/dispersion strengthening is based on dislocation theory. On other hand, the analysis of strength of fiber-reinforced material involves the direct application of continuum principles at the microscopic level as the material behavior is essentially elastic.

To achieve any benefit from presence of fibers, empirically it can be shown that there is a need for critical fiber volume which must be exceeded for fiber strengthening to occur.

$$f_{critical} = \frac{\sigma_{mu} - \sigma_m}{\sigma_{fu} - \sigma_m} \qquad (6.9)$$

Where σ_{mu} is strength of strain hardened matrix, σ_m is the flow stress of matrix at a strain equal to fiber breaking stress, and σ_{fu} is the ultimate tensile strength of the fiber.

However for small values of critical fiber volume, number of fibers present may not be sufficient to effectively restrain the elongation of the matrix. Thus a minimum volume fraction of fiber which must be exceeded to have real reinforcement is defined as follows:

$$f_{min} = \frac{\sigma_{mu} - \sigma_m}{\sigma_{fu} + \sigma_{mu} - \sigma_m} \qquad (6.10)$$

When fiber-reinforced material is loaded, load transmits from matrix to fibers via interface. Thus interface between matrix and fiber play an important role in fiber strengthening.

Usually fibers are arranged in unidirectional manner, thus it is highly anisotropic in nature. One of the consequences of the anisotropy of fiber composites is that they exhibit shear coupling. This means that an axial load produces shear stresses, and shear load produces axial strains. In an isotropic material a uniaxial load applied produces only axial and transverse normal strains. Shear strains in fiber reinforced composites are compensated practically by using a cross-ply laminate.

6.2.6 Martensite Strengthening

This strengthening can be achieved in systems where a diffusion-controlled invariant transformation can be suppressed by rapid cooling. Basically, the martensite strengthening process is a diffusion-less and displacement reaction. The martensitic phase is formed from the retained high temperature phase at temperatures lower than the equilibrium invariant transformation temperature. It occurs by a process of lattices shearing. Martensite under microscope appears as lenticular plates which divide and subdivide the grains of the parent phase, always touching but never crossing one another. The characteristic lenticular shape minimizes the elastic distortion in the matrix. These platelets grow at about one-third the velocity of sound. This high speed means activation energy for growth is very low, and thus activation energy for nucleation determines the amount of martensite that can form under given conditions.

Martensite platelets attain their shape by two successive shear displacements contained in boundaries coherent with the parent phase. The first displacement is a homogeneous shear throughout the plate which occurs parallel to a specific plane in the parent phase known as the habit plane. The second displacement, the lesser of the two, can take place by one of two mechanisms: slip as in Fe-C martensite or twinning as in Fe-Ni martensite. Martensite formation occurs in many systems like Fe-C, Fe-Ni, Fe-Ni-C, Cu-Zn, Au-Cd, and even in pure metals like Li, Zr and Co. However, only the alloys based on Fe and C show a pronounced strengthening effect.

High strength of martensite is attributed to its typical crystal structure i.e. effective barriers to slip are provided by the fine twin structure or the high dislocation density. In Fe-C system, carbon atoms are also involved in strengthening. Super saturated carbon atoms strain the ferrite lattice, and this strain can be relieved by redistribution of carbon atoms by diffusion at room temperature. One result is that a strong binding is set up between dislocations and the carbon atoms. This hinders the motion of dislocations, thus increasing the strength.

This strengthening mechanism is one of the most common processes used in engineering materials. The name basically arises from the phase martensite that forms in steels when they are quenched from the solutionizing temperature.

6.3 Recovery, Recrystallization and Grain Growth

As mentioned in earlier sections, annealing is an important industrial process to relieve the stresses from cold working. During cold working, grain shape changes while material strain hardens because of increase in dislocation density. Between 1-10% of the energy of plastic deformation is stored in material in the form of strain energy associated with point defects and dislocations. On annealing i.e. on heating the deformed material to higher temperatures and holding, material tends to lose the extra strain energy and revert to the original condition before deformation by the processes of recovery and recrystallization. Grain growth may follow these in some instances.

6.3.1 Recovery

This is the first stage of restoration after cold working where physical properties of the cold-worked material are restored without any observable change in microstructure. The properties that are mostly affected by recovery ate those sensitive to point defects, for example – thermal and electrical conductivities. During recovery, which takes place at low temperatures of annealing, some of the stored internal energy is relieved by virtue of dislocation motion as a result of enhanced atomic diffusion. There is some reduction, though not substantial, in dislocation density as well apart from formation of dislocation configurations with low strain energies. Excess point defects that are created during deformation are annihilated either by absorption at grain boundaries or dislocation climbing process. Stored energy of cold work is the driving force for recovery.

6.3.2 Recrystallization

This stage of annealing follows after recovery stage. Here also driving force is stored energy of cold work. Even after complete recovery, the grains are still in relatively high strain energy state. This stage, thus, involves replacement of cold-worked structure by a new set of strain-free, approximately equi-axed grains i.e. it is the process of nucleation and growth of new, strain-free crystals to replace all the deformed crystals. It starts on heating to temperatures in the range of 0.3-0.5 T_m, which is above the recovery stage. There is no crystal structure change during recrystallization. This process is characterized by recrystallization temperature which is defined as the temperature at which 50% of material recrystallizes in one hour time. The recrystallization temperature is strongly dependent on the purity of a material. Pure materials may recrystallizes around 0.3 T_m, while impure materials may recrystallizes around 0.5-0.7 T_m.

There are many variables that influence recrystallization behavior, namely amount of prior deformation, temperature, time, initial grain size, composition and amount of recovery prior to the start of the recrystallization. This dependence leads to following empirical laws:

- A minimum amount of deformation is needed to cause recrystallization.
- Smaller the degree of deformation, higher will be the recrystallization temperature.
- The finer is the initial grain size; lower will be the recrystallization temperature.
- The larger the initial grain size, the greater degree of deformation is required to produce an equivalent recrystallization temperature.
- Greater the degree of deformation and lower the annealing temperature, the smaller will be the recrystallized grain size.
- The higher is the temperature of cold working, the less is the strain energy stored and thus recrystallization temperature is correspondingly higher.
- The recrystallization rate increases exponentially with temperature.

During recrystallization, the mechanical properties that were changed during deformation are restored to their pre-cold-work values. Thus material becomes softer, weaker and ductile. During this stage of annealing, impurity atoms tend to segregate at grain boundaries, and retard their motion and obstruct the processes of nucleation and growth. This *solute drag effect* can be used to retain cold worked strength at higher service temperatures. Presence of second phase particles causes slowing down of recrystallization – *pinning action of the particles*.

6.3.3 Grain growth

This stage follows complete crystallization if the material is left at elevated temperatures. However, grain growth does not need to be preceded by recovery and recrystallization; it may occur in all polycrystalline materials. During this stage newly formed strain-free grains tend to grow in size. This grain growth occurs by the migration of grain boundaries. Driving force for this process is reduction in grain boundary energy i.e. decreasing in free energy of the material. As the grains grow larger, the curvature of the boundaries becomes less. This results in a tendency for larger grains to grow at the expense of smaller grains. In practical applications, grain growth is not desirable. Incorporation of impurity atoms and insoluble second phase particles are effective in retarding grain growth.

Because the driving force for grain growth is lower than that for recrystallization, grain growth occurs slowly at a temperature where recrystallization occurs at substantially high speeds. However, grain growth is strongly temperature dependent.

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CHAPTER 7. PHASE DIAGRAMS AND TRANSFORMATIONS

The understanding of phase diagrams is important to the engineer because it relates to the design and control of heat-treating procedures; some properties of materials are functions of their microstructures, and, consequently, of their thermal histories. Even though most phase diagrams represent stable (or equilibrium) states and microstructures, they are nevertheless useful in understanding the development and preservation of non-equilibrium structures and their attendant properties; it is often the case that these properties are more desirable than those associated with the equilibrium state.

Concepts discussed in this chapter provide a foundation that is necessary to understand phase transformations that occur in steel alloys.

Learning objectives

After studying this chapter, you should be able to do the following:

- Explain the basic concepts and definitions in relation to alloys and phase equilibrium
- Describe unary and binary phase diagrams on the basis of components in a system
- State and apply the Gibbs phase rule in binary phase diagrams.
- Explain invariant and eutectic (eutectoid) reactions in binary systems and describe the formation process and distinctive feature of each microstructure formed by the reactions.
- Describe and illustrate the phase transformations that occur in an iron-carbon alloy using transformation diagrams.

7.0 Introduction

Many engineering materials possess mixtures of phases, e.g. steel, paints, and composites. The mixture of two or more phases may permit interaction between the phases, and results in properties usually different from the properties of individual phases. Different components can be combined into a single material by means of solutions or mixtures. *A solution (liquid or solid) is phase with more than one component; a mixture is a material with more than one phase*. Solute does not change the structural pattern of the solvent, and the composition of any solution can be varied. In mixtures, there are different phases, each with its own atomic arrangement. It is possible to have a mixture of two different solutions.

A pure substance, under equilibrium conditions, may exist as either a vapor, liquid or solid phase, depending upon the conditions of temperature and pressure. A *phase* can be defined as a homogeneous portion of a system that has uniform physical and chemical characteristics. Thus, it is physically distinct from other phases, chemically homogeneous and mechanically separable portion of a system. In other words, a phase is a structurally homogeneous portion of matter. When two phases are present in a system, it is not necessary that there be a difference in both physical and chemical properties; a disparity in one or the other set of properties is sufficient.

There is only one vapor phase no matter how many constituents make it up. For pure substance there is only one liquid phase, however there may be more than one solid phase because of differences in crystal structure. A liquid solution is also a single phase, even as a liquid mixture (e.g. oil and water) forms two phases as there is no mixing at the molecular level. In the solid state, different chemical compositions and/or crystal structures are possible

so a solid may consist of several phases. For the same composition, different crystal structures represent different phases. A solid solution has atoms mixed at atomic level thus it represents a single phase. A single-phase system is termed as homogeneous, and systems composed of two or more phases are termed as mixtures or heterogeneous. Most of the alloy systems and composites are heterogeneous.

It is important to understand the existence of phases under various practical conditions which may dictate the microstructure of an alloy, thus the mechanical properties and usefulness of it. Phase diagrams provide a convenient way of representing which state of aggregation (phase or phases) is stable for a particular set of conditions. In addition, phase diagrams provide valuable information about melting, casting, crystallization, and other phenomena.

7.1 Definitions and Concepts

It is necessary to establish a foundation of definitions and basic concepts relating to alloys, phases, and equilibrium before delving into the interpretation and utilization of phase diagrams.

Component is either pure metal and/or compounds of which an alloy is composed. The components of a system may be elements, ions or compounds. They refer to the independent chemical species that comprise the system. ***System*** refers to a specific body of material under consideration or it may relate to the series of possible alloys consisting of the same components but without regard to alloy composition. ***Solid solution*** consists of atoms having at least two different types where solute atoms occupy either substitutional or interstitial positions in the solvent lattice and the crystal structure of the solvent is maintained.

For almost all alloy systems and at a specific temperature, there is a maximum concentration of solute atoms that may dissolve in the solvent phase to form a solid solution. This limit is known as ***solubility limit***. Generally, solubility limit changes with temperature. The addition of solute in excess of this solubility limit results in the formation of different phase, either a solid solution or compound. ***Phase equilibrium*** refers to the set of conditions where more than one phase may exist. It can be reflected by constancy with time in the phase characteristics of a system. In most metallurgical and materials systems, phase equilibrium involves just solid phases. However, the state of equilibrium is never completely achieved because of very slow rate of approach of equilibrium in solid systems. This leads to non-equilibrium or meta-stable state, which may persist indefinitely and has more practical significance than equilibrium phases. An equilibrium state of solid system can be reflected in terms of characteristics of the microstructure, phases present and their compositions, relative phase amounts and their spatial arrangement or distribution.

Variables of a system include number of external variables (N) such as temperature and pressure with internal variable such as composition, *C* and number of phases, *P*. The number of independent variables among these gives the degrees of freedom, *F* or variance. According to *Gibbs Phase rule*, all these are related for a chosen system as;

$$P + F = C + N \qquad (7.1)$$

The degrees of freedom cannot be less than zero so that an upper limit to the number of phases can exist in equilibrium for a given system. For practical purpose, in metallurgical and

materials field, pressure can be considered as a constant and temperature becomes the only external variable, and thus the *condensed phase rule* is given as follows:

$$P + F = C + 1 \qquad (7.2)$$

7.2 Equilibrium Phase Diagrams

A diagram that depicts existence of different phases of a system under equilibrium is termed as ***phase diagram***. It is also known as *equilibrium* or *constitutional diagram*. There are three externally controllable parameters that will affect phase structure; temperature, pressure, and composition. Equilibrium phase diagram represent relationships between these parameters and the quantities of phases at equilibrium. In general practice it is sufficient to consider only solid and liquid phases, thus pressure is assumed to be constant (1 *atm*.) in most applications. These diagrams do not indicate the dynamics when one phase transforms into another. However, it depicts information related to microstructure and phase structure of a particular system in a convenient and concise manner.

Important information, useful for scientists and engineers who are involved with materials development, selection, and application in product design, obtainable from a phase diagram can be summarized as follows:

- To show phases are present at different compositions and temperatures under slow cooling (equilibrium) conditions.
- To indicate equilibrium solid solubility of one element/compound in another.
- To indicate temperature at which an alloy starts to solidify and the range of solidification.
- To indicate the temperature at which different phases start to melt.
- Amount of each phase in a two-phase mixture can be obtained.

A phase diagram is a collection of solubility limit curves. The phase fields in equilibrium diagrams depend on the particular systems being depicted. The set of solubility curves that represents locus of temperatures above which all compositions are liquid are called *liquidus*, while *solidus* represents set of solubility curves that denotes the locus of temperatures below which all compositions are solid. Every phase diagram for two or more components must show a liquidus, solidus, and an intervening freezing range, except for pure system, as melting of a phase occurs over a range of temperature. Whether the components are metals or nonmetals, there are certain locations on the phase diagram where the liquidus and solidus meet. For a pure component, a contact point lies at the edge of the diagram. The liquidus and solidus also meet at the other invariant positions on the diagram. Each invariant point represents an invariant reaction that can occur only under a particular set of conditions between particular phases, so is the name for it.

Phase diagrams are classified based on the number of components in the system. Single component systems have unary diagrams, two-component systems have binary diagrams, three-component systems are represented by ternary diagrams, and so on. When more than two components are present, phase diagrams become extremely complicated and difficult to represent. This chapter deals mostly with binary phase diagrams.

7.2.1 Unary Phase Diagrams

In these systems there is no composition change ($C = 1$), the only variables are temperature and pressure. Thus, in region of single phase, two variables (temperature and pressure) can be varied independently. If two phases coexist then, according to Phase rule, either temperature or pressure can be varied independently, but not both. At triple points, three phases can coexist at a particular set of temperature and pressure. At these points, neither temperature nor the pressure can be changed without disrupting the equilibrium; one of the phases may disappear.

Figure 7.1 depicts phase diagram for water. It may be noted that regions for three different phases; solid, liquid, and vapor are delineated on the plot. Each phase exist under equilibrium conditions over the temperature–pressure ranges of its corresponding area. Furthermore, the three curves shown on the plot (labeled *aO*, *bO*, and *cO*) are phase boundaries; at any point on one of these curves, the two phases on either side of the curve are in equilibrium (or coexist) with one another. That is, equilibrium between solid and vapor phases is along curve *aO;* for the solid–liquid, curve *bO*, and the liquid–vapor, curve *cO*.

Also, one phase transforms to another upon crossing a boundary as the temperature and/or pressure is altered. For example, at 1 atm pressure, during heating, the solid phase transforms to the liquid phase (i.e., melting occurs) at point 2. This point corresponds to a temperature of 0 °C. The reverse transformation (liquid-to-solid, or solidification) takes place at the same point upon cooling. Similarly, at the intersection of the dashed line with the liquid–vapor phase boundary (point 3 at 100 °C) liquid transforms to the vapor phase upon heating; condensation occurs for cooling. And, finally, solid ice sublimes or vaporizes upon crossing the curve labeled *aO*.

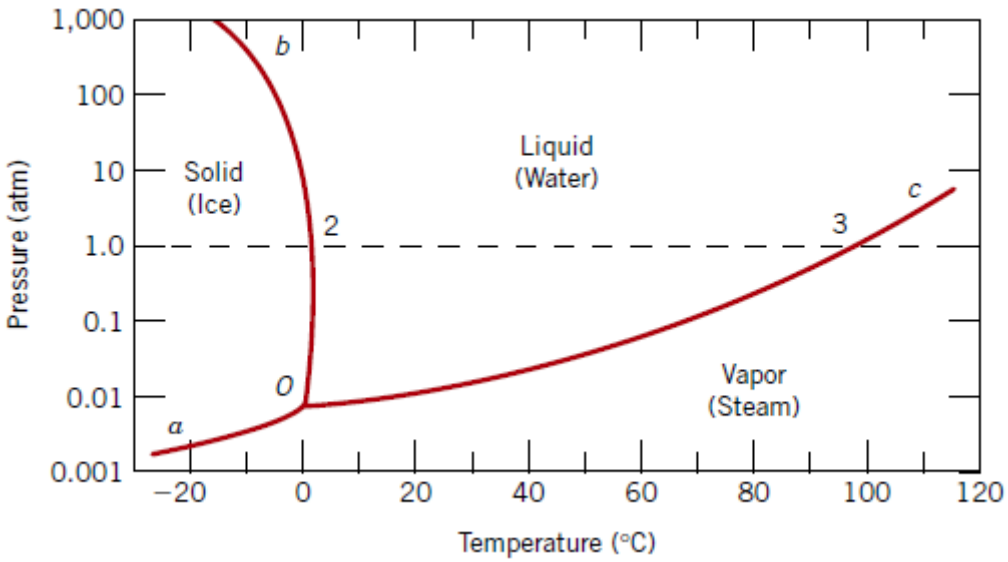


Figure 7.1 Pressure–temperature unary phase diagram for water.

7.2.2 Binary Phase Diagrams

These diagrams constitutes two components, e.g.: two metals (Cu and Ni), or a metal and a compound (Fe and Fe_3C), or two compounds (Al_2O_3 and Si_2O_3), etc. In most engineering applications, condensed phase rule is applicable. It is assumed that the same is applicable for all binary diagrams, thus the presentation of binary diagrams becomes less complicated. Hence,

binary diagrams are usually drawn showing variations in temperature and composition only. It is also to be noted that all binary systems consist only one liquid phase i.e. a component is completely soluble in the other component when both are in liquid state. Therefore, binary systems are classified according to their solid solubility. If both the components are completely soluble in each other, the system is called ***isomorphous system***. E.g.: Cu-Ni, Ag-Au, Ge-Si, Al_2O_3-Cr_2O_3. The extent of solid solubility for a system of two metallic components can be predicted based on Hume-Rothery conditions, summarized in the following:

- Crystal structure of each element of solid solution must be the same.
- Size of atoms of each two elements must not differ by more than 15%.
- Elements should not form compounds with each other i.e. there should be no appreciable difference in the electro-negativities of the two elements.
- Elements should have the same valence.

All the Hume-Rothery rules are not always applicable for all pairs of elements which show complete solid solubility.

In systems other than isomorphous systems i.e. in case of limited solid solubility, there exist solid state miscibility gaps; number of invariant reactions can take place; intermediate phases may exist over a range of composition (*intermediate solid solutions*) or only at relatively fixed composition (*compound*). The intermediate phases may undergo polymorphic transformations, and some may melt at a fixed temperature (*congruent transformations*, in which one phase changes to another of the same composition at definite temperature). A solid solution based on a pure component and extending to certain finite compositions into a binary phase diagram is called a *terminal solid solution*, and the line representing the solubility limit of a terminal solid solution with respect to a two-phase solid region is called a *solvus* line (Figure 7.4).

7.2.2.1 Isomorphous system

Figure 7.2 illustrates a typical phase diagram for an isomorphous system made of two metallic elements ***A*** and ***B***. As pointed out earlier, any phase diagram can be considered as a map. A set of coordinates; a temperature and a composition is associated with each point in the diagram. If the alloy composition and temperature are specified, then phase diagram allows determination of the phase or phases that will be present under equilibrium conditions. There are only two phases in the phase diagram, the liquid and the solid phases. These single-phases regions are separated by a two-phase region where both liquid and solid co-exist. The area in Figure 7.2 above the line marked liquidus (***A'bB'***) corresponds to the region of stability of the liquid phase, and the area below the solidus line (***A'dB'***) represents the stable region for the solid phase.

For interpretation of the phase diagram, consider the vertical line ***ae*** drawn corresponding to composition of 50%***A*** +50%***B*** and assume the system is undergoing equilibrium cooling. The point ***a*** on the line ***ae*** signifies that for that particular temperature and composition, only liquid phase is stable. This is true up to the point ***b*** which lies on the liquidus line, representing the start of solidification. Completion of solidification of the alloy is represented by the point, ***d***. Point ***e*** corresponds to single-phase solid region up to the room temperature. Point ***c*** lies in the two-phase region made of both liquid and solid phases. Corresponding micro-structural changes are also shown. As shown in Figure 7.2, above liquidus only a liquid phase exists, and

below the solidus single solid phase exists as completely solidified grains. Between these two lines, system consist both solid crystals spread in liquid phase. It is customary to use ***L*** to represent liquid phase(s) and Greek alphabets (***α***, ***β***, ***γ***) for representing solid phases.

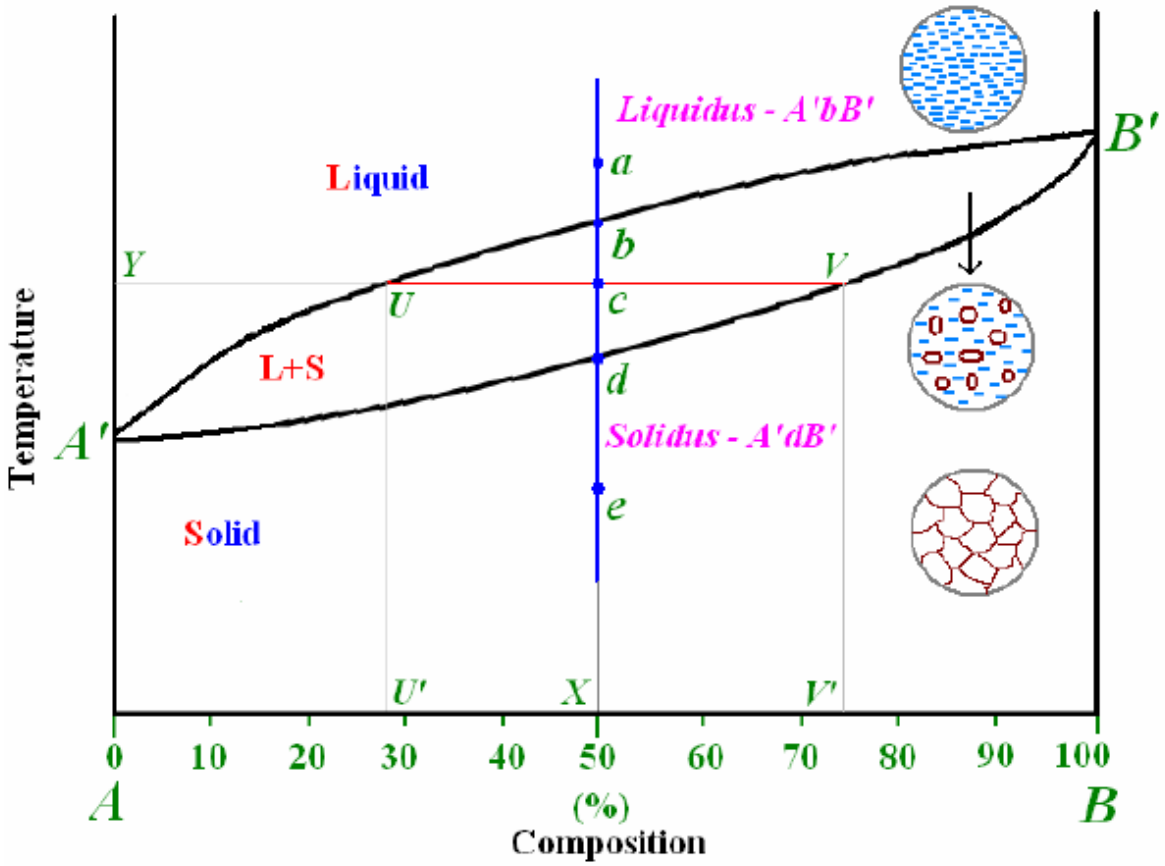


Figure 7.2 Phase diagram for typical isomorphous binary system.

Between two extremes of the horizontal axis of the diagram, cooling curves for different alloys are shown in Figure 7.3 as a function of time and temperature. Cooling curves shown in Figure 7.3 represent ***A***, ***U'***, ***X***, ***V'*** and ***B*** correspondingly in Figure 7.2. Change in slope of the cooling curve is caused by heat of fusion. In fact these changes in slope are nothing but points on either solidus or liquidus of a phase diagram. An experimental procedure where repeated cooling/heating of an alloy at different compositions, and corresponding changes in slope of cooling curves will be used to construct the phase diagram.

Another important aspect of interpreting phase diagrams along with phases present is finding the relative amount of phases present and their individual composition. The procedure to find equilibrium concentrations of phases:

- A *tie-line* or *isotherm* (***UV***) is drawn across two-phase region to intersect the boundaries of the region.
- Perpendicular lines are dropped from these intersections to the composition axis, represented by ***U'*** and ***V'*** in Figure 7.2, from which each phase is read. ***U'*** represents composition of the liquid phase and ***V'*** represents composition of the solid phase as intersection ***U*** meets liquidus line and ***V*** meets solidus line.

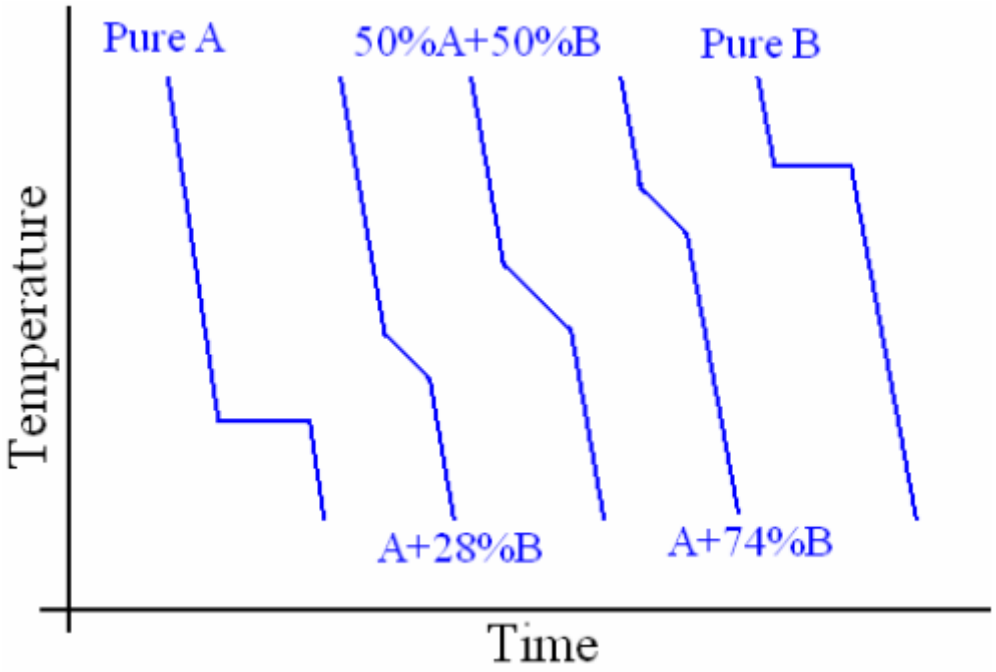


Figure 7. 3 Cooling curves for isomorphous binary system.

Procedure to find equilibrium relative amounts of phases (*lever rule*):

- A tie-line is constructed across the two phase region at the temperature of the alloy to intersect the region boundaries.
- The relative amount of a phase is computed by taking the length of tie-line from overall composition to the phase boundary for the other phase, and dividing by the total tie-line length. In Figure 7.2, relative amounts of liquid and solid phases is given respectively by

$$C_L = \frac{cV}{UV}, \quad C_S = \frac{Uc}{UV}, \quad C_L + C_s = 1 \qquad (7.3)$$

7.2.2.2 Eutectic system

Many binary systems have components which have limited solid solubility, e.g.: Cu-Ag, Pb-Sn. The regions of limited solid solubility at each end of a phase diagram are called *terminal solid solutions* as they appear at ends of the diagram.

Many of the binary systems with limited solubility are of eutectic type, which consists of specific alloy composition known as *eutectic composition* that solidifies at a lower temperature than all other compositions. This low temperature which corresponds to the lowest temperature at which the liquid can exist when cooled under equilibrium conditions is known as *eutectic temperature*. The corresponding point on the phase diagram is called *eutectic point*. When the liquid of eutectic composition is cooled, at or below eutectic temperature this liquid transforms simultaneously into two solid phases (two terminal solid solutions, represented by $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$). This transformation is known as *eutectic reaction* and is written symbolically as:

Liquid ($\boldsymbol{L}$) ↔ solid solution-1 ($\boldsymbol{\alpha}$) + solid solution-2 ($\boldsymbol{\beta}$)

This eutectic reaction is called invariant reaction as it occurs under equilibrium conditions at a specific temperature and specific composition which cannot be varied. Thus, this reaction

is represented by a thermal horizontal arrest in the cooling curve of an alloy of eutectic composition. A typical eutectic type phase diagram is shown in Figure 7.4 along with a cooling curve.

As shown in Figure 7.4, there exist three single phase regions, namely liquid (***L***), ***α*** and ***β*** phases. There also exist three two-phase regions: ***L*+*α***, ***L*+*β*** and ***α*+*β***. These three two-phase regions are separated by horizontal line corresponding to the eutectic temperature. Below the eutectic temperature, the material is fully solid for all compositions. Compositions and relative amount of the phases can be determined using tie-lines and lever rule. Compositions that are on left-hand-side of the eutectic composition are known as *hypo-eutectic compositions* while compositions on the right-hand side of the eutectic composition are called *hyper-eutectic compositions*. Development of micro-structure and respective cooling curves for eutectic alloys are shown in Figures 7.5, 7.6, 7.7 and 7.8 for different compositions. The phase that forms during cooling but before reaching eutectic temperature is called *pro-eutectic phase*.

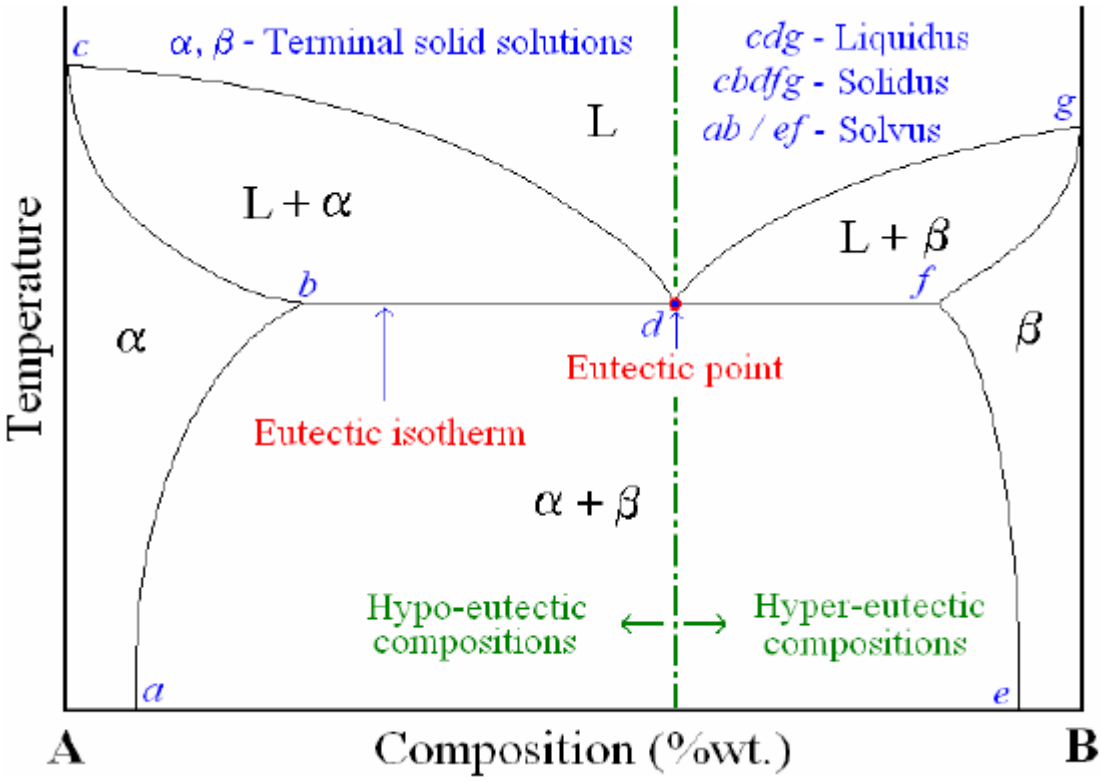


Figure 7.4 Typical phase diagram for a binary eutectic system.

In many systems, solidification in the solid + liquid region may lead to formation of layered (cored) grains, even at very slow cooling rates. This is as a result of very slow or no-diffusion in solid state compared with very high diffusion rates in liquids. The composition of the liquid phase evolves by diffusion, following the equilibrium values that can be derived from the tie-line method. However, new layers that solidify on top of the grains have the equilibrium composition at that temperature but once they are solid their composition does not change.

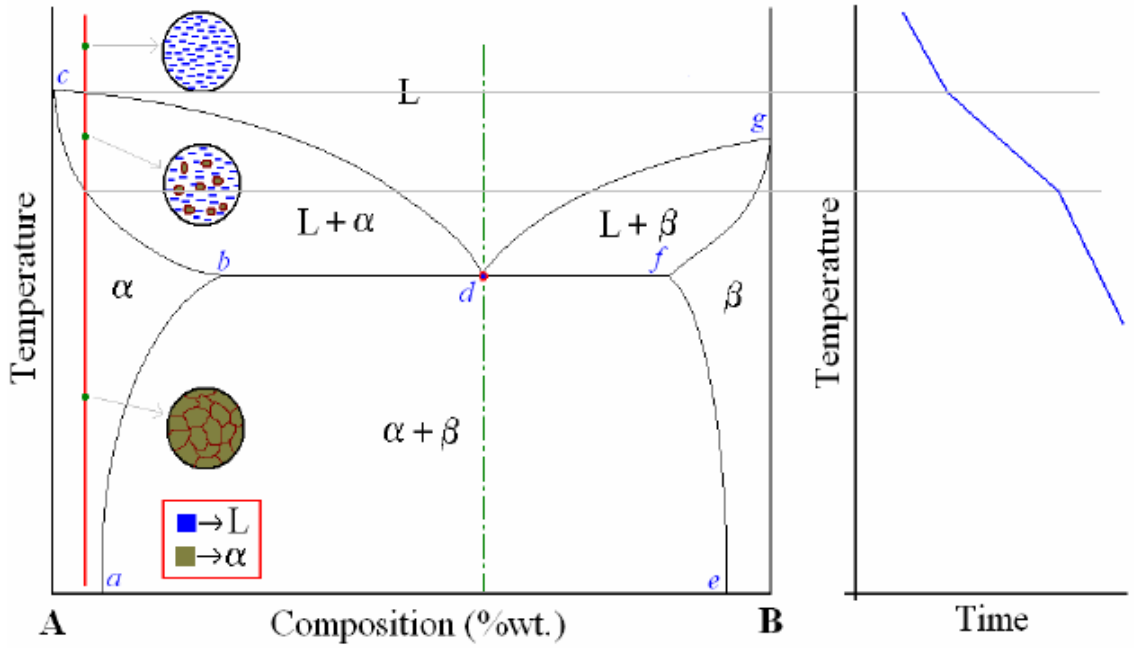


Figure 7.5 Cooling curve and micro-structure development for eutectic alloy that passes mainly through terminal solid solution.

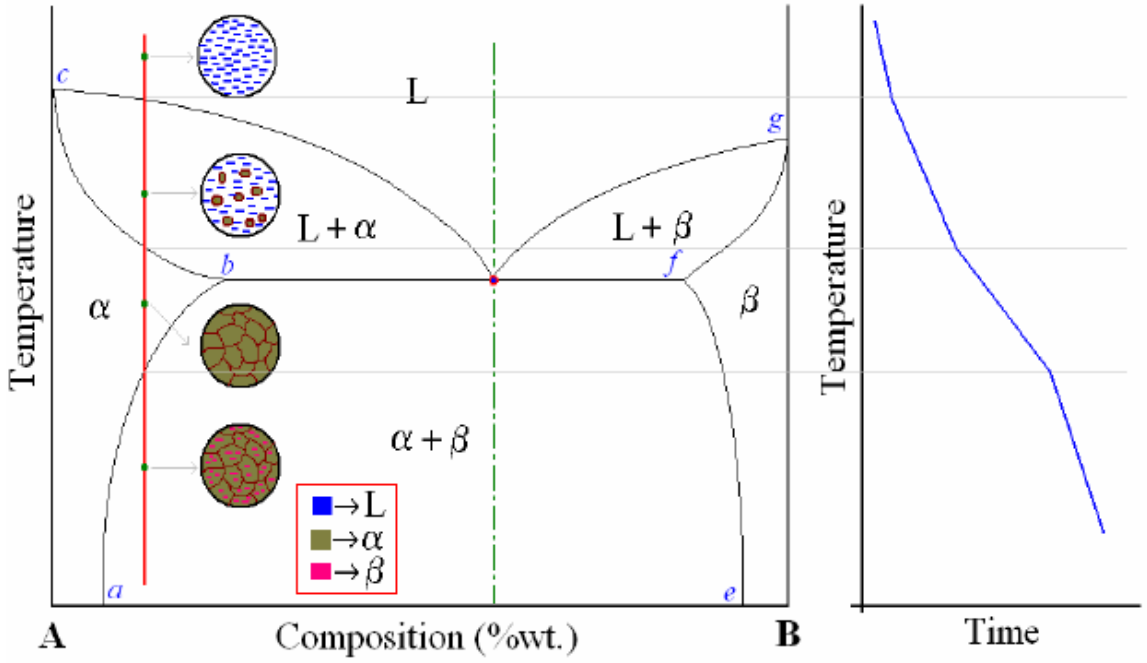


Figure 7.6 Cooling curve and micro-structure development for eutectic alloy that passes through terminal solid solution without formation of eutectic solid.

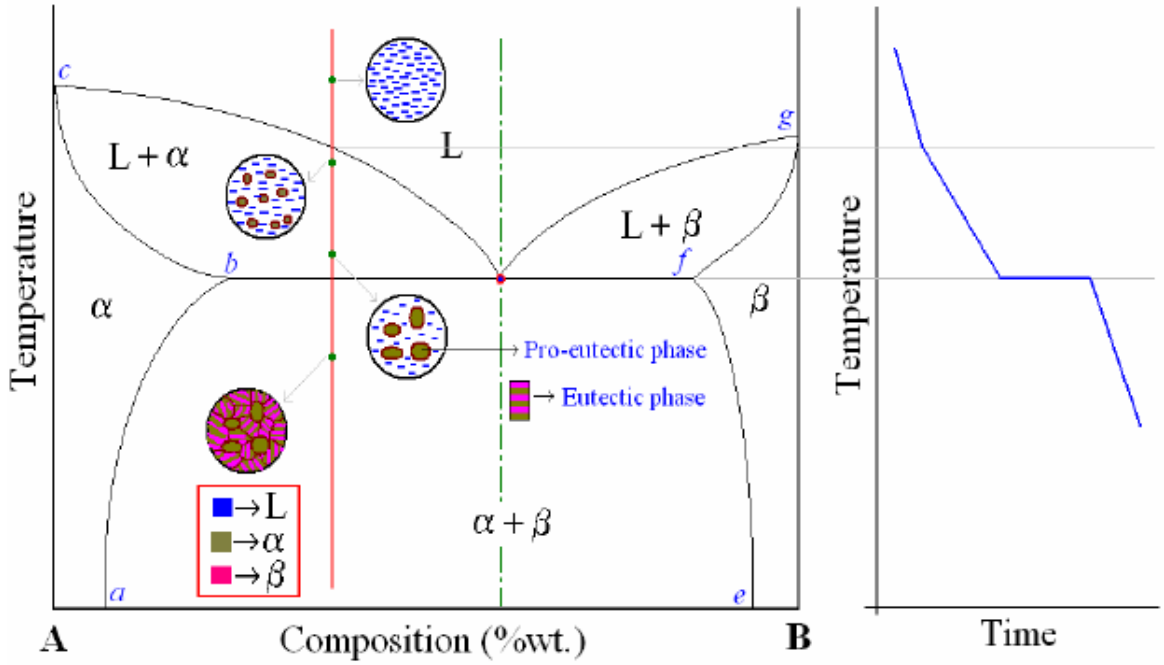


Figure 7.7 Cooling curve and micro-structure development for eutectic alloy that passes through hypo-eutectic region

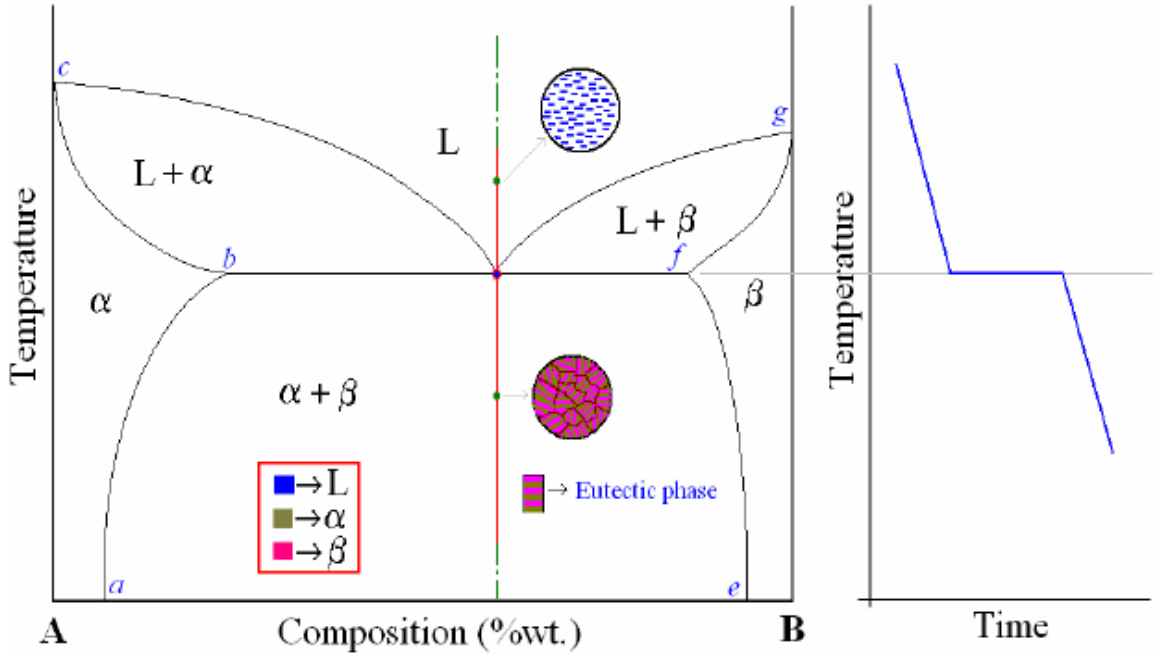


Figure 7.8 Cooling curve and micro-structure development for eutectic alloy that passes through eutectic-point.

7.2.2.3 Invariant Reactions

The *eutectic reaction*, in which a liquid transforms into two solid phases, is just one of the possible three-phase invariant reactions that can occur in non-isomorphous binary systems.

Schematically it can be shown as in Figure 7.9. It represents that a liquid phase, ***L***, transforms into two different solids phases (***α*** and ***β***) upon cooling during eutectic reaction.

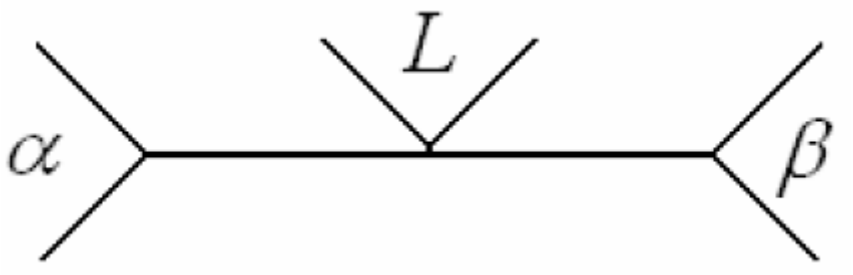


Figure 7.9 Schematic of eutectic invariant reaction.

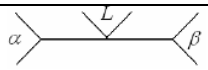
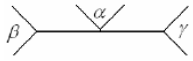
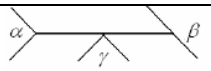
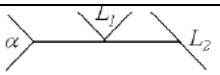
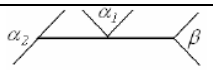
In solid state analogy of a eutectic reaction, called a ***eutectoid reaction***, one solid phase having eutectoid composition transforms into two different solid phases. Another set of invariant reactions that occur often in binary systems are ***peritectic reaction*** where a solid phase reacts with a liquid phase to produce a new solid phase, and in ***peritectoid reaction***, two solid phases react to form a new solid phase. Peritectic reaction is commonly present as part of more-complicated binary diagrams, particularly if the melting points of the two components are quite different. Peritectic and peritectoid reactions do not give rise to micro-constituents as the eutectic and eutectoid reactions do. Another invariant reaction that involves liquid phase is ***monotectic reaction*** in which a liquid phase transforms into a solid phase and a liquid phase of different composition. Over a certain range of compositions the two liquids are immiscible like oil and water and so constitute individual phases, thus monotectic reaction can said to be associated with miscibility gaps in the liquid state. Example system for monotectic reaction: Cu-Pb at 954 °C and 36% Pb. Analogous to monotectic reaction in solid state is ***monotectoid reaction*** in which a solid phase transforms to two solid phases of different compositions. Another notable invariant reaction that is associated with liquid immiscibility is ***syntectic reaction*** in which two liquid phases react to form a solid phase. All the invariant reactions are summarized in Table 7.1 showing both symbolic reaction and schematic part of phase diagram.

7.2.2.4 Intermediate Phases and Congruent Phase Transformations

An intermediate phase may occur over a composition range (intermediate solid solution) or at a relatively fixed composition (compound) inside the phase diagram and are separated from other two phases in a binary diagram by two phase regions. Many phase diagrams contain intermediate phases whose occurrence cannot be readily predicted from the nature of the pure components. Intermediate solid solution often have higher electrical resistivity and hardness than either of the two components. The intermediate compounds form at a relatively fixed composition when there exists a stoichiometric relationship between the components, for example: Mg_2Ni and $MgNi_2$ in Mg-Ni system. These are called *inter-metallic compounds,* and differ from other chemical compounds in that the bonding is primarily metallic rather than ionic or covalent, as would be found with compounds in certain metal-nonmetal or ceramic systems. Some metal-nonmetal compounds, Fe_3C, are metallic in nature, whereas in others, MgO and

Mg_2Si, bonding is mainly covalent. When using the lever rules, inter-metallic compounds are treated like any other phase, except they appear not as a wide region but as a vertical line.

Table 7.1 Summary of invariant reactions in binary systems

Reaction	Symbolic equation	Schematic presentation	Example
Eutectic	$\boldsymbol{L \leftrightarrow \alpha + \beta}$		Fe-C, 4.27% C, 1147 °C
Eutectoid	$\boldsymbol{\alpha \leftrightarrow \beta + \gamma}$		Fe-C, 0.80% C, 723 °C
Peritectic	$\boldsymbol{L + \alpha \leftrightarrow \beta}$		Fe-C, 0.16% C, 1495 °C
Peritectoid	$\boldsymbol{\alpha + \beta \leftrightarrow \gamma}$		
Monotectic	$\boldsymbol{L_1 \leftrightarrow L_2 + \alpha}$		Fe-C, 0.51% C, 1495 °C
Monotectoid	$\boldsymbol{\alpha_1 \leftrightarrow \alpha_2 + \beta}$		
Synthetic	$\boldsymbol{L_1 + L_2 \leftrightarrow \alpha}$		

Number of phase transformations may takes place for each system. Phase transformations in which there are no compositional alternations are said to be *congruent transformations*, and during *incongruent transformations* at least one of the phases will experience a change in composition. Examples of congruent transformations include allotropic transformations, and melting of pure materials. All invariant (eutectic and eutectoid) reactions, and also melting of alloy that belongs to an isomorphous system represent incongruent transformations.

Intermediate phases are sometimes classified on the basis of whether they melt congruently or incongruently. $MgNi_2$, for example, melts congruently whereas Mg_2Ni melts incongruently since it undergoes peritectic decomposition.

7.3 The Iron – Carbon System, Phase Transformations

A study of iron-carbon system is useful and important in many respects. This is because steels constitute the greatest amount of metallic materials being used and also, solid state transformations that occur in steels are varied and interesting. These are similar to those that occur in many other systems and helps explain the properties.

Iron-carbon phase diagram in Figure 7.10 is not a complete diagram. Part of the diagram after 6.67wt% C is ignored as it has little commercial significance. The 6.67% C represents the composition where an inter-metallic compound, cementite (Fe_3C), with solubility limits forms. In addition, the phase diagram is not true equilibrium diagram because cementite is not an equilibrium phase. However, in ordinary steels, decomposition of cementite into graphite is never observed because nucleation of cementite is much easier than that of graphite. Thus cementite can be treated as an equilibrium phase for practical purposes.

The Fe-Fe_3C is characterized by five individual phases and four invariant reactions. Five phases that exist in the diagram are: ***α*–ferrite** (BCC) Fe-C solid solution, ***γ*-austenite** (FCC) Fe-C solid solution, ***δ*-ferrite** (BCC) Fe-C solid solution, Fe_3C (iron carbide) or **cementite**; an

inter-metallic compound and **liquid** Fe-C solution. The four invariant reactions that cause transformations in the system are namely eutectoid, eutectic, monotectic and peritectic.

As depicted by left axes, pure iron upon heating exhibits two allotropic changes in the crystal structure before it melts. One involves ***α***–ferrite of BCC crystal structure transforming to FCC austenite, ***γ***-iron, at 910 °C. At 1400 °C, austenite changes to BCC phase known as ***δ***-ferrite, which finally melts at 1536 °C.

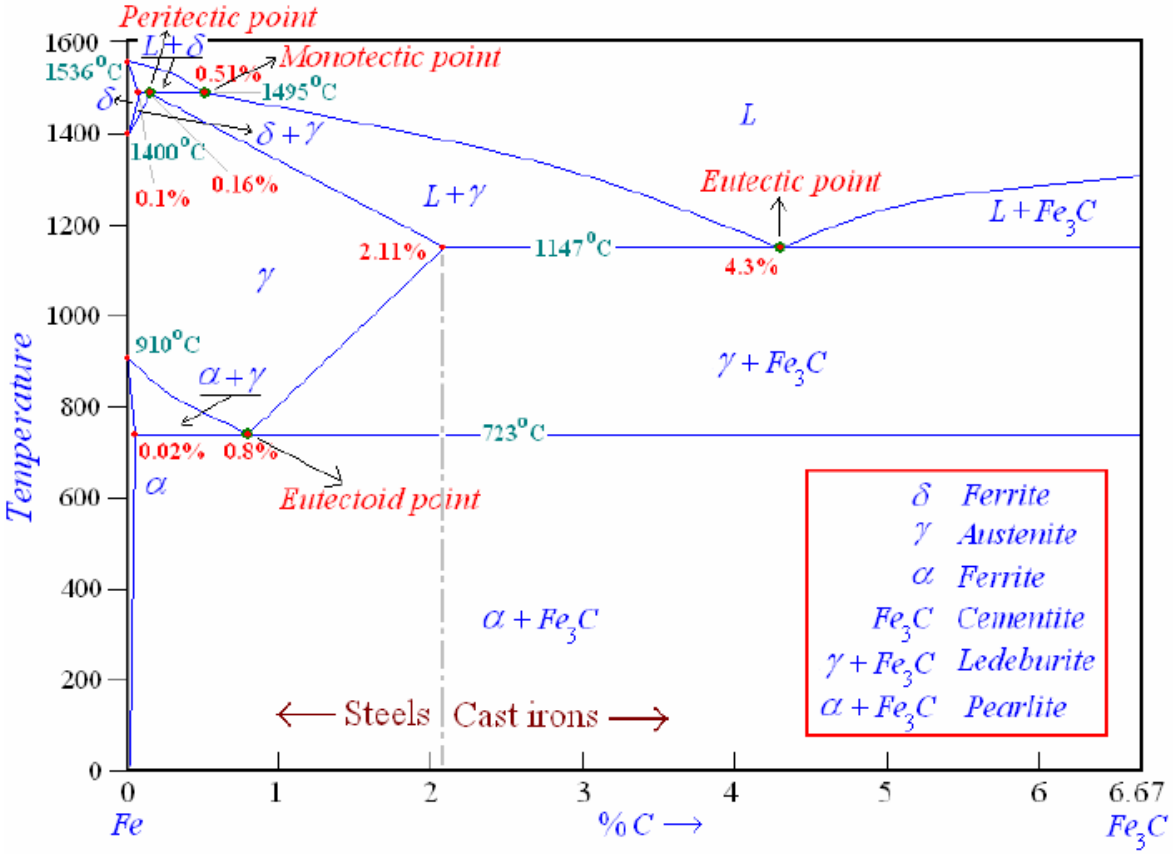


Figure 7. 10 Iron – Iron carbide phase diagram.

Carbon exist in solid iron as interstitial impurity, and forms solid solution with ferrites or austenite as depicted by three single fields represented by ***α***, ***γ*** and ***δ***. Carbon dissolves least in ***α***–ferrite in which maximum amount of carbon soluble is 0.02% at 723 °C. This limited solubility is attributed to shape and size of interstitial position in BCC ***α***–ferrite. However, carbon present greatly influences the mechanical properties of ***α***–ferrite. Below 768 °C, ***α***–ferrite can be used as magnetic material. Solubility of carbon in ***γ***-iron reaches its maximum, 2.11%, at a temperature of 1147 °C. Higher solubility of carbon in austenite is attributed to FCC structure and corresponding interstitial sites. Phase transformations involving austenite play very significant role in heat treatment of different steels. Austenite itself is non-magnetic. Carbon solubility in ***δ***-ferrite is 0.1% maximum at 1495 °C. As this ferrite exists only at elevated temperatures, it is of no commercial importance. Cementite, Fe_3C an inter-metallic compound forms when the amount of carbon present exceeds its solubility limit at respective temperatures. Out of these four solid phases, cementite is hardest and brittle and it is used in different forms to increase the strength of steels. On the other hand, ***α***–ferrite is softest and act

as matrix of a composite material. By combining these two phases in a solution, a material's properties can be varied over a large range.

For technological convenience, based on % C dissolved in it, a Fe-C solution is classified as: *commercial pure irons* with less than 0.008% C; *steels* having % C between 0.008-2.11; while *cast irons* have carbon in the range of 2.11%-6.67%. Thus commercial pure iron is composed of exclusively ***α***–ferrite at room temperature. Most of the steels and cast irons contain both ***α***–ferrite and cementite. However, commercial cast irons are not simple alloys of iron and carbon as they contain large quantities of other elements such as silicon, thus better consider them as ternary alloys. The presence of Si promotes the formation of graphite instead of cementite. Thus cast irons may contain carbon in form of both graphite and cementite, while steels will have carbon only in combined from as cementite.

As shown in Figure 7.10, and mentioned earlier, Fe-C system constitutes four invariant reactions:

- peritectic reaction at 1495 °C and 0.16% C, ***δ***-ferrite + ***L*** ↔ ***γ***-iron (austenite)
- monotectic reaction 1495 °C and 0.51% C, ***L*** ↔ ***L*** + ***γ***-iron (austenite)
- eutectic reaction at 1147 °C and 4.3 % C, ***L*** ↔ ***γ***-iron + Fe3C (cementite) [ledeburite]
- eutectoid reaction at 723 °C and 0.8% C, ***γ***-iron ↔ ***α***–ferrite + Fe3C (cementite) [pearlite]

Product phase of eutectic reaction is called ledeburite, while product from eutectoid reaction is called pearlite. During cooling to room temperature, ledeburite transforms into pearlite and cementite. At room temperature, thus after equilibrium cooling, Fe-C diagram consists of either ***α***–ferrite, pearlite and/or cementite. Pearlite is not a single phase, but a micro-constituent having alternate thin layers of ***α***–ferrite (~88%) and Fe_3C, cementite (~12%). Steels with less than 0.8% C (mild steels up to 0.3% C, medium carbon steels with C between 0.3%-0.8% i.e. hypo-eutectoid Fe-C alloys) consists of pro-eutectoid ***α***–ferrite in addition to pearlite, while steels with carbon higher than 0.8% (high-carbon steels i.e. hyper-eutectoid Fe-C alloys) consists of pearlite and pro-eutectoid cementite. Phase transformations involving austenite i.e. processes those involve eutectoid reaction are of great importance in heat treatment of steels.

In practice, steels are almost always cooled from the austenitic region to room temperature. During the cooling upon crossing the boundary of the single phase ***γ***-iron, first pro-eutectoid phase (either ***α***–ferrite or cementite) forms up to eutectoid temperature. With further cooling below the eutectoid temperature, remaining austenite decomposes to eutectoid product called pearlite, mixture of thin layers of ***α***–ferrite and cementite. Though pearlite is not a phase, nevertheless, a constituent because it has a definite appearance under the microscope and can be clearly identified in a structure composed of several constituents. The decomposition of austenite to form pearlite occurs by nucleation and growth. Nucleation, usually, occurs heterogeneously and rarely homogeneously at grain boundaries. When it is not homogeneous, nucleation of pearlite occurs both at grain boundaries and in the grains of austenite. When austenite forms pearlite at a constant temperature, the spacing between adjacent lamellae of cementite is very nearly constant. For a given colony of pearlite, all cementite plates have a common orientation in space, and it is also true for the ferrite plates. Growth of pearlite colonies occurs not only by the nucleation of additional lamellae but also through an advance at the ends of the lamellae. Pearlite growth also involves the nucleation of new colonies at the interfaces between established colonies and the parent austenite. The thickness ratio of the ferrite and

cementite layers in pearlite is approximately 8 to 1. However, the absolute layer thickness depends on the temperature at which the isothermal transformation is allowed to occur.

The temperature at which austenite is transformed has a strong effect on the inter-lamellar spacing of pearlite. The lower the reaction temperature, the smaller will be inter-lamellar spacing. For example, pearlite spacing is in order of 10-3 mm when it formed at 700 °C, while spacing is in order of 10-4 mm when formed at 600 °C. The spacing of the pearlite lamellae has a practical significance because the hardness of the resulting structure depends upon it; the smaller the spacing, the harder the metal. The growth rate of pearlite is also a strong function of temperature. At temperatures just below the eutectoid, the growth rate increases rapidly with decreasing temperature, reaching a maximum at 600 °C, and then decreases again at lower temperatures.

Additions of alloying elements to Fe-C system bring changes (alternations to positions of phase boundaries and shapes of fields) depends on that particular element and its concentration. Almost all alloying elements causes the eutectoid concentration to decrease, and most of the alloying elements (e.g.: Ti, Mo, Si, W, Cr) causes the eutectoid temperature to increase while some other (e.g.: Ni, Mn) reduces the eutectoid temperature. Thus alloying additions alters the relative amount of pearlite and pro-eutectoid phase that form.

Fe-C alloys with more than 2.11% C are called cast irons. Phase transformations in cast irons involve formation of pro-eutectic phase on crossing the liquidus. Upon further cooling, liquid of eutectic composition decomposes into mixture of austenite and cementite, known as ledeburite. On further cooling through eutectoid temperature, austenite decomposes to pearlite. The room temperature microstructure of cast irons thus consists of pearlite and cementite. Because of presence of cementite, which is hard, brittle and white in color, product is called *white cast iron*. However, depending on cooling rate and other alloying elements, carbon in cast iron may be present as graphite or cementite. *Gray cast iron* contains graphite in form of flakes. These flakes are sharp and act as stress risers. Brittleness arising because of flake shape can be avoided by producing graphite in spherical nodules, as in *malleable cast iron* and *SG (spheroidal graphite) cast iron*. Malleable cast iron is produced by heat treating white cast iron (Si < 1%) for prolonged periods at about 900 °C and then cooling it very slowly. The cementite decomposes and temper carbon appears approximately as spherical particles. SG iron is produced by adding inoculants to molten iron. In these Si content must be about 2.5%, and no subsequent heat treatment is required.

To determine the relative amounts of ferrite, cementite, and pearlite micro constituents in alloys, the lever rule in conjunction with a tie-line that extends from the $\alpha - (\alpha + Fe_3C)$ phase boundary (0.022 wt% C) to the eutectoid composition (0.76 wt% C), inasmuch as pearlite is the transformation product of austenite having this composition. For example, let us consider an alloy of composition C_o' in Figure 7.11. Thus, the fractions of pearlite, C_p and proeutectoid ferrite, $C_{\alpha'}$ may be determined from equations (7.4) and (7.5) respectively.

$$C_p = \frac{T}{T+U} = \frac{C_o' - 0.022}{0.76 - 0.022} = \frac{C_o' - 0.022}{0.74} \qquad (7.4)$$

$$C_{\alpha'} = \frac{U}{T+U} = \frac{0.76 - C_o'}{0.76 - 0.022} = \frac{0.76 - C_o'}{0.74} \qquad (7.5)$$

Fractions of both total α (eutectoid and proeutectoid) and cementite are determined using the lever rule and a tie line that extends across the entirety of the $\alpha + Fe_3C$ phase region, from 0.022 to 6.70 wt% C.

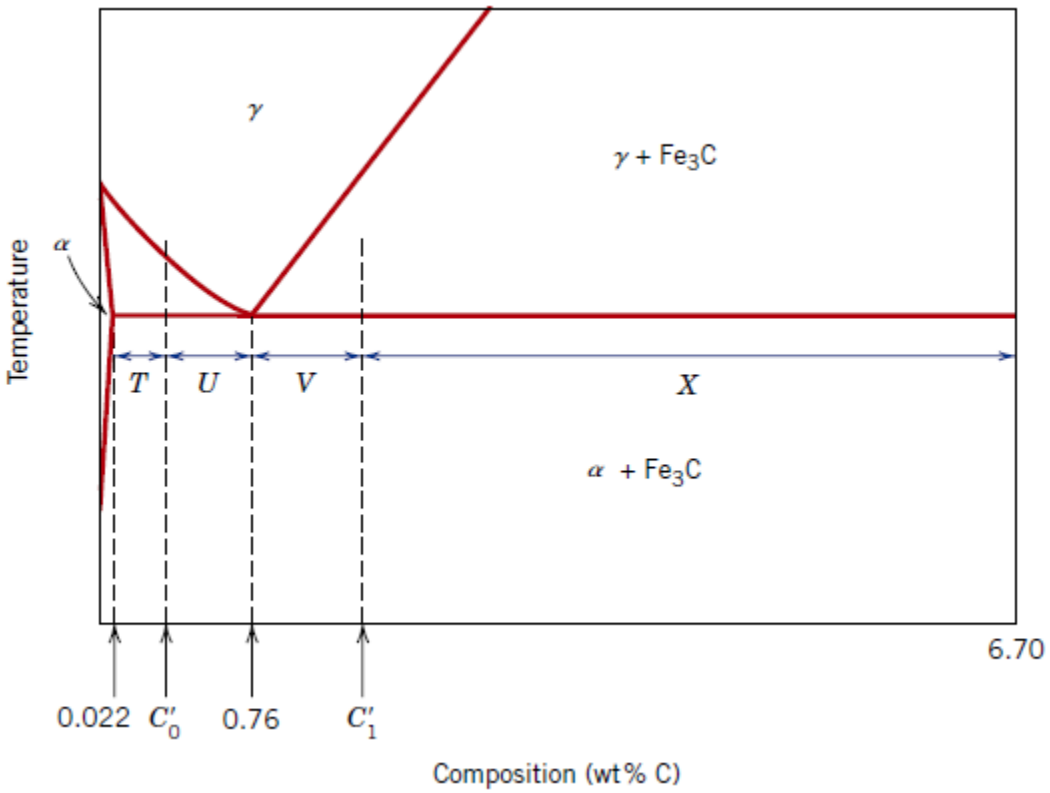


Figure 7.11 Fe–Fe3C phase diagram used in determining phase amounts

For an alloy having composition, C_1', fractions of pearlite, C_p and proeutectoid cementite, $C_{Fe_3C'}$ are determined from the following lever rule expressions:

$$C_p = \frac{X}{V+X} = \frac{6.70 - C_1'}{6.70 - 0.76} = \frac{6.70 - C_1'}{5.94} \qquad (7.6)$$

$$C_{Fe_3C'} = \frac{V}{V+X} = \frac{C_1' - 0.76}{6.70 - 0.76} = \frac{C_1' - 0.76}{5.94} \qquad (7.7)$$

Worked Example 7.1

For a 99.65 wt% Fe–0.35 wt% C alloy at a temperature just below the eutectoid, determine the following:

(a) The fractions of total ferrite and cementite phases

(b) The fractions of the proeutectoid ferrite and pearlite

(c) The fraction of eutectoid ferrite

Solution

(a) This part of the problem is solved by application of the lever rule expressions employing a tie line that extends all the way across the $\alpha + Fe_3C$ phase field. Thus, C_o' is 0.35 wt% C, and the fractions of total ferrite and cementite phases are;

$$C_\alpha = \frac{U+V+X}{T+U+V+X} = \frac{6.70 - C_o'}{6.70 - 0.022} = \frac{6.70 - 0.35}{5.94} = 0.95$$

$$C_{Fe_3C} = \frac{T}{T+U+V+X} = \frac{C_o' - 0.022}{6.70 - 0.022} = \frac{0.35 - 0.022}{5.94} = 0.05$$

(b) The fractions of proeutectoid ferrite and pearlite are determined by using the lever rule and a tie line that extends only to the eutectoid composition (i.e., equations 7.4 and 7.5)

$$C_{\alpha'} = \frac{U}{T+U} = \frac{0.76 - C_o'}{0.76 - 0.022} = \frac{0.76 - 0.35}{0.74} = 0.56$$

$$C_p = \frac{T}{T+U} = \frac{C_o' - 0.022}{0.76 - 0.022} = \frac{0.35 - 0.022}{0.74} = 0.44$$

(c) All ferrite is either as proeutectoid or eutectoid (in the pearlite). Therefore, the sum of these two ferrite fractions will equal the fraction of total ferrite; that is,

$$C_{\alpha'} + C_{\alpha e} = C_\alpha$$

where $C_{\alpha e}$ denotes the fraction of the total alloy that is eutectoid ferrite. Values for C_α and $C_{\alpha'}$ were determined in parts (a) and (b) as 0.95 and 0.56, respectively. Therefore,

$$C_{\alpha e} = C_\alpha - C_{\alpha'} = 0.95 - 0.56 = 0.39$$

7.4 Isothermal Transformation Diagrams

The eutectoid reaction for the Fe-Fe_3C system shown in equation 7.8 is fundamental to the development of microstructure in steel alloys. Upon cooling, austenite, having an intermediate carbon concentration, transforms to a ferrite phase, having a much lower carbon content, and also cementite, with much higher carbon concentration. Pearlite is one microstructural product of this transformation.

$$\gamma\ (0.76\ wt\%\ C) \rightleftharpoons \alpha\ (0.022\ wt\%\ C) + Fe_3C\ (6.70\ wt\%\ C) \qquad (7.8)$$

Solid state transformations, which are very important in steels, are known to be dependent on time at a particular temperature. Isothermal transformation diagram, also known as TTT diagram, measures the rate of transformation at a constant temperature. Thus, it shows time relationships for the phases during isothermal transformation. Information regarding the time to start the transformation and the time required to complete the transformation can be obtained from set of TTT diagrams. One such set of diagram for reaction of austenite to pearlite in steel is shown in Figure 7.12. The diagram is not complete in the sense that the transformations of austenite that occur at temperatures below about 550 °C are not shown.

As mentioned in previous section, thickness of layers in pearlite depends on the temperature at which the transformation occurred. If the transformation took place at a temperature that is just below the eutectoid temperature, relatively thick layers of ***α***–ferrite and cementite are produced in what is called *coarse pearlite*. This is because of high diffusion rates of carbon atoms. Thus with decreasing transformation temperature, sluggish movement of carbon results in thinner layers of ***α***–ferrite and cementite i.e. *fine pearlite* is produced.

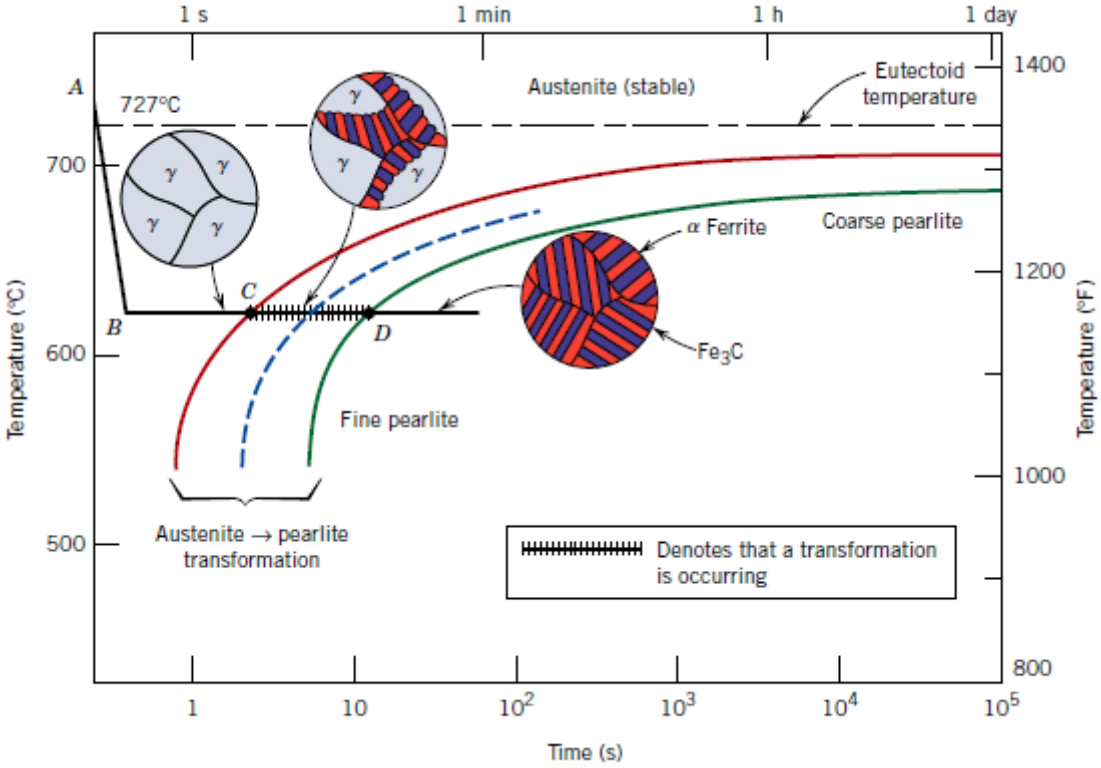


Figure 7.12 Isothermal transformation diagram for a eutectoid iron–carbon alloy

At transformation temperatures below 550 °C, austenite results in different product known as ***bainite***. Bainite also consists of ***α***–ferrite and cementite phases i.e. transformation is again diffusion controlled but morphologically it consists of very small particles of cementite within or between fine ferrite plates. Bainite forms as needles or plates, depending on the temperature of the transformation; the microstructural details of bainite are so fine that their resolution is only possible using electron microscope. It differs from pearlite in the sense that different mechanism is involved in formation of bainite which does not have alternating layers of ***α***–ferrite and cementite. In addition, because of equal growth rates in all directions, pearlite tends to form spherical colonies, whereas bainite grows as plates and has a characteristic acicular (needlelike) appearance. *Upper bainite*, formed at the upper end of the temperature range (550 °C-350 °C), is characterized by relatively coarse, irregular shaped cementite particles in ***α***–ferrite plates. If the transformation is taking place at lower temperatures (350 °C-250 °C), the ***α***–ferrite plates assume a more regular needlelike shape, and the transformation product is called *lower bainite*. At the same time carbide particles become smaller in size and appear as cross-striations making an angle of about 55 °C to the axis of the ***α***–ferrite plate. Upper bainite has large rod-like cementite regions, whereas lower bainite has much finer cementite particles

as a result of sluggish diffusion of carbon atoms at lower temperatures. Lower bainite is considerably harder than upper bainite. Another characteristic of bainite is that as it has crystallographic orientation that is similar to that found in simple ferrite nucleating from austenite, it is believed that bainite is nucleated by the formation of ferrite. This is in contrast to pearlite which is believed to be nucleated by formation of cementite.

Basically, bainite is a transformation product that is not as close to equilibrium as pearlite. The most puzzling feature of the bainite reaction is its dual nature. In a number of respects, it reveals properties that are typical of a nucleation and growth type of transformation such as occurs in the formation of pearlite and also a mixture of $\boldsymbol{\alpha}$–ferrite and cementite though of quite different morphology (no alternate layers), but at the same time, it differs from martensite as bainite formation is athermal and diffusion controlled though its microstructure is characterized by acicular (needlelike) appearance.

The micro constituent or phase called ***martensite*** is formed when austenitized iron–carbon alloys are rapidly cooled (or quenched) to a relatively low temperature. Martensite is a non-equilibrium single-phase structure resulting from a diffusion-less transformation of austenite. It may be thought of as a transformation product that is competitive with pearlite and bainite. The martensitic transformation occurs when the quenching rate is rapid enough to prevent carbon diffusion. Any diffusion whatsoever will result in the formation of ferrite and cementite phases.

The time-temperature dependence of the bainite transformation can also be presented using TTT diagram as shown in Figure 7.13. It occurs at temperatures below those at which pearlite forms. Thus, it does not form until the transformation temperature falls below a definite temperature, designated as B_s. Above this temperature austenite does not form bainite except under external stresses. Below B_s, austenite does not transform completely to bainite. The amount of bainite formed increases as the isothermal reaction temperature is lowered. By reaching a lower limiting temperature, B_f, it is possible to transform austenite completely to bainite. The B_s and B_f temperatures are equivalent to the $M_{(start)}$ and $M_{(90\%)}$ temperatures for martensite.

In simple eutectoid steels, pearlite and bainite transformations overlap, thus transition from the pearlite to bainite is smooth and continuous i.e. knees of individual pearlite and bainite curves are merged together. However, each of the transformations has a characteristic C-curve, which can be distinguishable in presence of alloying elements. As shown in the complete TTT diagram for eutectoid steel in Figure 7.13, above approximately 550 °C-600 °C, austenite transforms completely to pearlite. Below this range up to 450 °C, both pearlite and bainite are formed. Finally, between 450 °C and 210 °C the reaction product is bainite only. Thus bainite transformation is favored at a high degree of super cooling, and the pearlite transformation at a low degree of super cooling. In the middle region, pearlite and bainite transformations are competitive with each other.

As explained in earlier section, martensitic transformation can dominate the proceedings if steel is cooled rapid enough so that diffusion of carbon can be arrested. Transformation of austenite to martensite is diffusion-less, time independent and the extent of transformation depends on the transformation temperature. Martensite is a *meta-stable* phase and decomposes into ferrite and pearlite but this is extremely slow (and not noticeable) at room temperature.

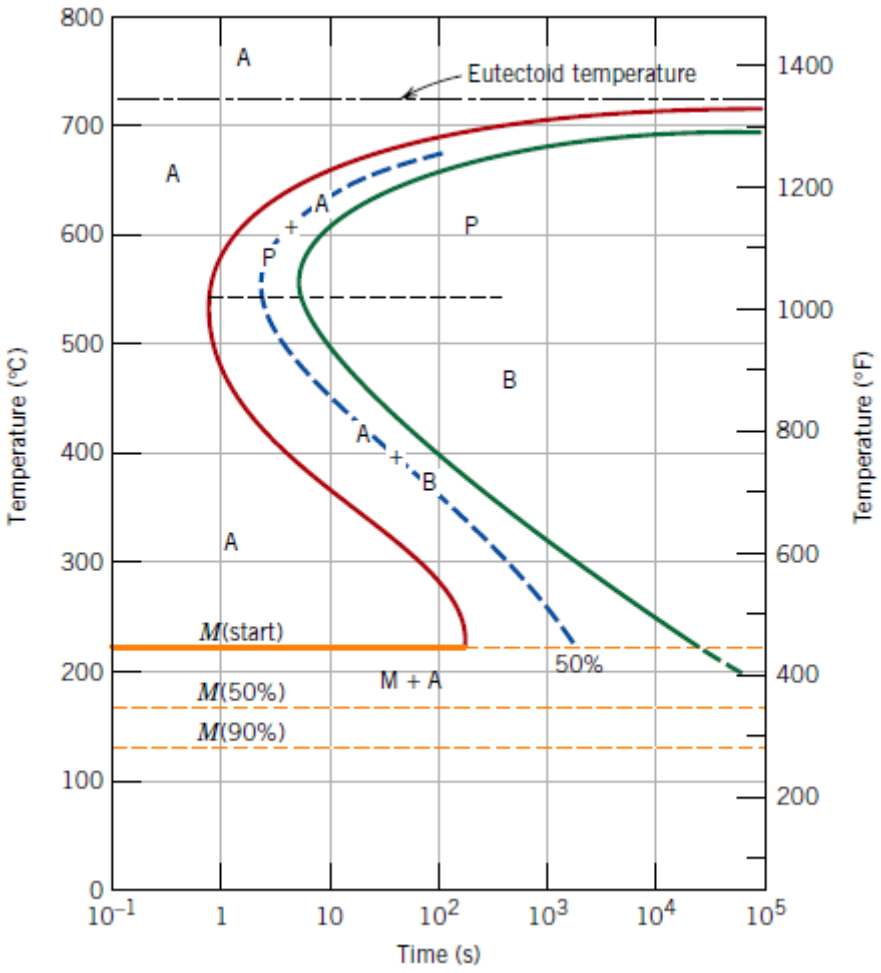


Figure 7.13 Complete TTT (isothermal transformation) diagram for eutectoid steel.

Alloying additions retard the formation rate of pearlite and bainite, thus rendering the martensitic transformation more competitive. Start of the transformation is designated by $M_{(start)}$, while the completion is designated by $M_{(90\%)}$ in a transformation diagram. Martensite formed in steels possesses a body centered tetragonal crystal structure with carbon atoms occupying one of the three interstitial sites available. This is the reason for characteristic structure of steel martensite instead of general BCC. Tetragonal distortion caused by carbon atoms increases with increasing carbon content and so is the hardness of martensite.

Austenite is slightly denser than martensite, and therefore, during the phase transformation upon quenching, there is a net volume increase. If relatively large pieces are rapidly quenched, they may crack as a result of internal stresses, especially when carbon content is more than about 0.5%. Mechanically, martensite is extremely hard, thus its applicability is limited by brittleness associated with it. Characteristics of steel martensite render it unusable for structural applications in the as-quenched form. However, structure and thus the properties can be altered by *tempering*, heat treatment observed below eutectoid temperature to permit diffusion of carbon atoms for a reasonable period of time. During tempering, carbide particles attain spherical shape and are distributed in ferrite phase structure called ***spheroidite***. Spheroidite is the softest yet toughest structure that steel may have. At lower tempering temperature, a structure called ***tempered martensite*** forms with similar microstructure as that of spheroidite

except that cementite particles are much, much smaller. The tempering heat treatment is also applicable to pearlite and bainite structures. This mainly results in improved machinability. The mechanism of tempering appears to be first the precipitation of fine particles of hexagonal ε-carbide of composition about $Fe_{2.4}C$ from martensite, decreasing its tetragonality. At higher temperatures or with increasing tempering times, precipitation of cementite begins and is accompanied by dissolution of the unstable ε-carbide. Eventually the martensite loses its tetragonality and becomes BCC ferrite, the cementite coalesces into spheres. A schematic of possible transformations involving austenite decomposition are shown in Figure 7.14.

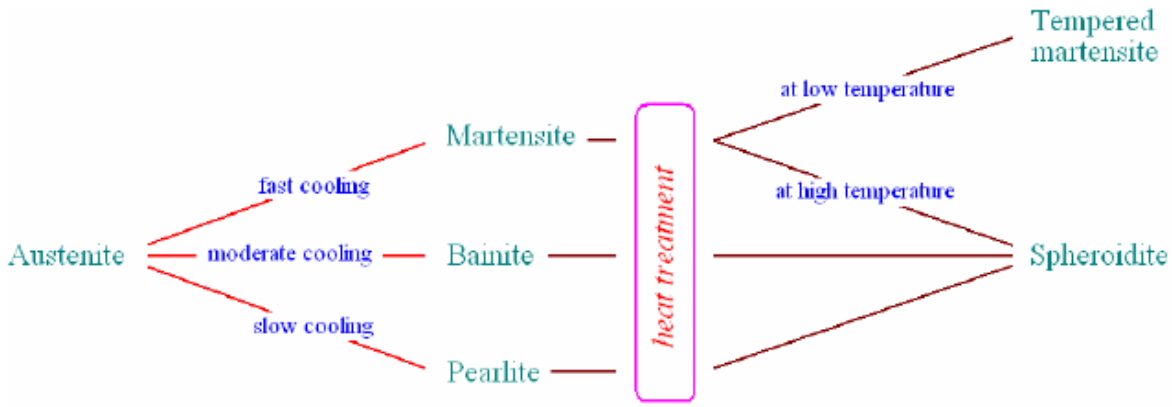


Figure 7. 14 Possible transformation involving austenite decomposition.

Tempering of some steels may result in a reduction of toughness what is known as *temper embrittlement*. This may be avoided by compositional control, and/or tempering above 575 °C or below 375 °C, followed by quenching to room temperature. The effect is greatest in martensite structures, less severe in bainite structures and least severe in pearlite structures. It appears to be associated with the segregation of solute atoms to the grain boundaries lowering the boundary strength. Impurities responsible for temper brittleness are: P, Sn, Sb and As. Si reduces the risk of embrittlement by carbide formation. Mo has a stabilizing effect on carbides and is also used to minimize the risk of temper brittleness in low alloy steels.

TTT diagrams are less of practical importance since an alloy has to be cooled rapidly and then kept at a temperature to allow for respective transformation to take place. However, most industrial heat treatments involve continuous cooling of a specimen to room temperature. Hence, continuous cooling transformation (CCT) diagrams are generally more appropriate for engineering applications as components are cooled (air cooled, furnace cooled, quenched etc.) from a processing temperature as this is more economic than transferring to a separate furnace for an isothermal treatment. CCT diagrams measure the extent of transformation as a function of time for a continuously decreasing temperature. For continuous cooling, the time required for a reaction to begin and end is delayed, thus the isothermal curves are shifted to longer times and lower temperatures.

Both TTT and CCT diagrams are, in a sense, phase diagrams with added parameter in form of time. Each is experimentally determined for alloy of specified composition. These diagrams allow prediction of the microstructure after some time period for constant temperature and continuous cooling heat treatments, respectively.

Normally, bainite will not form during continuous cooling because all the austenite will have transformed to pearlite by the time the bainite transformation has become possible. Thus, as shown in Figure 7.15, region representing austenite-pearlite transformation terminates just below the nose.

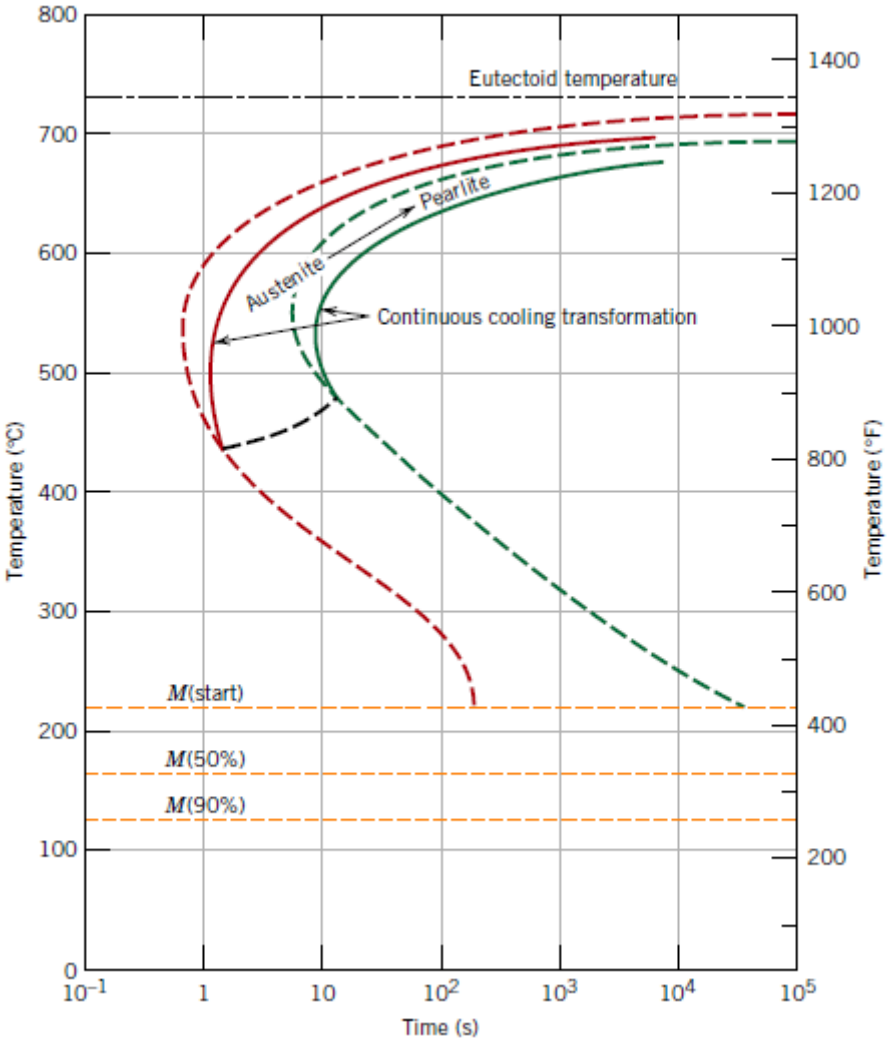


Figure 7.15 Superimposition of TTT and CCT diagrams for a eutectoid steel.

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CHAPTER 8. THERMAL PROPERTIES

Engineering materials are important in everyday life because of their versatile structural properties. Other than these properties, they do play an important role because of their physical properties. Prime physical properties of materials include electrical, thermal magnetic and optical properties. Selection of materials for use at elevated temperatures and/or temperature changes require an engineer to know and understand their thermal properties. This chapter deals with the thermal properties of materials. Physical property of a solid body related to application of heat energy is defined as a thermal property.

This chapter describe thermal properties like heat capacity, thermal expansion, thermal conductivity, and thermal stresses. It is important to know and understand the concept of thermal expansion which is the root cause for thermal stresses. Thermal stresses are stresses leading to failure of engineering structures at elevated temperatures.

Learning objectives

After studying this chapter, you should be able to do the following:

- Define heat capacity and specific heat
- Determine the linear coefficient of thermal expansion in relation to thermal stress
- Explain the phenomenon of thermal expansion from an atomic perspective.
- Define thermal conductivity and heat flux

8.1 Heat Capacity

Many engineering solids, when heated, experiences an increase in temperature indicating that some energy has been absorbed. The property of a material that indicates the material's ability to absorb heat energy from the external surroundings is called ***heat capacity***, C. It is defined as the energy required to change the temperature of the material by one degree. Mathematically, it is expressed as:

$$C = \frac{dQ}{dT} \qquad (8.1)$$

Where dQ is the energy required to produce a temperature change equal to dT.

Ordinarily, heat capacity is specified per mole of material (e.g., J/mol K, or cal/mol K). Heat capacity is not an intrinsic property i.e. the total heat a material can absorb depends on its volume or mass. Hence another parameter called *specific heat*, *c*, it defined as heat capacity per unit mass (J/kg-K, Cal/kg-K).

As internal energy increase, geometrical changes may occur. Accordingly, heat capacity is measured either at constant volume, C_v, or at constant external pressure, C_p. The magnitude of C_p is always greater than C_v but only marginally.

Heat energy absorption of a (solid, liquid or gaseous) material exists in mode of thermal energy vibrations of constituent atoms or molecules, apart from other mechanisms of heat absorption such as electronic contribution. With increase of energy, atoms vibrate at higher frequencies. However, the vibrations of adjacent atoms are coupled through atomic bonding, which may lead to movement of lattices. This may be represented as elastic waves (phonon) or sound waves.

The variation with temperature of the vibrational contribution to the heat capacity at constant volume for many relatively simple crystalline solids is shown in Figure 8.1. The C_v is zero at

0 K, but it rises rapidly with temperature; this corresponds to an increased ability of atomic vibrations or enhanced energy of lattice waves with ascending temperature.
At low temperatures, the relationship between C_v and the absolute temperature T is

$$C_v = AT^3 \qquad (8.2)$$

Where A is a constant which is independent of temperature. Above a temperature called the *Debye temperature*, θ_D, dependence of volumetric heat capacity value reaches saturation and becomes essentially independent of temperature at a value of approximately 3R (≈ 6 cal/mol-K), R being the gas constant. Thus, even though the total energy of the material is increasing with temperature, the quantity of energy required to produce a one-degree temperature change is constant. The value of θ_D is below room temperature for many solid materials, and 25 J/mol. K is a reasonable room-temperature approximation for C_v. Table 8.1 presents experimental specific heats for a number of materials.

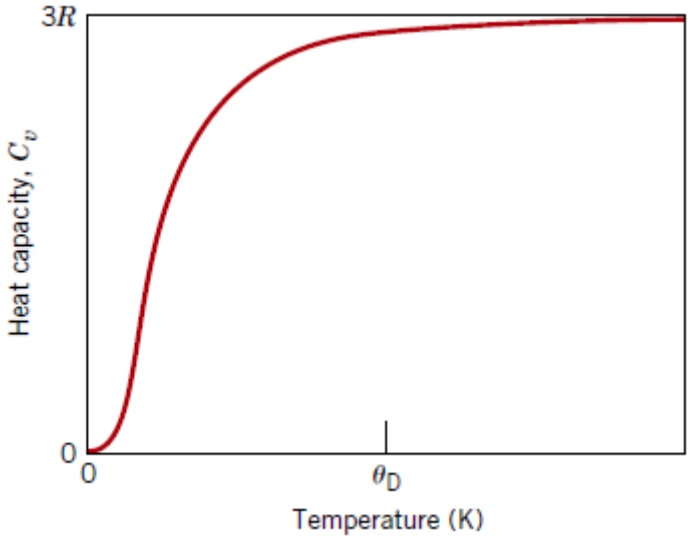


Figure 8.1 Heat capacity as a function of temperature.

8.2 Thermal Expansion

After heat absorption, vibrating atoms behave as though they have larger atomic radius, which leads to increase in material dimensions. The phenomenon is called thermal expansion. It is quantified in terms of thermal expansion coefficient. The linear coefficient of thermal expansion (α_l) defined as the change in the dimensions of the material per unit length, and is expressed as:

$$\alpha_l = \frac{l_f - l_o}{l_o(T_f - T_o)} = \frac{\Delta l}{l_o \Delta T} = \frac{\varepsilon}{\Delta T} \qquad (8.3)$$

Where l_o and l_f represent, respectively, initial and final lengths with the temperature change from T_o to T_f and ε is the strain. The parameter α_l is called the ***linear coefficient of thermal expansion*;** it is a material property that is indicative of the extent to which a material expands

upon heating and has units of reciprocal temperature [$(°C)^{-1}$ or $(°F)^{-1}$]. Of course, heating or cooling affects all the dimensions of a body, with a resultant change in volume.

Table 8.1 Thermal properties for some metals

Material	c_p $(J/kg \cdot K)^a$	α_l $[(°C)^{-1} \times 10^{-6}]^b$	k $(W/m \cdot K)^c$	L $[\Omega \cdot W/(K)^2 \times 10^{-8}]$
		Metals		
Aluminum	900	23.6	247	2.20
Copper	386	17.0	398	2.25
Gold	128	14.2	315	2.50
Iron	448	11.8	80	2.71
Nickel	443	13.3	90	2.08
Silver	235	19.7	428	2.13
Tungsten	138	4.5	178	3.20
1025 Steel	486	12.0	51.9	—
316 Stainless steel	502	16.0	15.9	—
Brass (70Cu–30Zn)	375	20.0	120	—
Kovar (54Fe–29Ni–17Co)	460	5.1	17	2.80
Invar (64Fe–36Ni)	500	1.6	10	2.75
Super Invar (63Fe–32Ni–5Co)	500	0.72	10	2.68

A volume coefficient of thermal expansion, α_v is used to describe the volume change with temperature.

$$\alpha_v = \frac{\Delta V}{V_o \Delta T} \qquad (8.4)$$

Where ΔV and V_o are the volume change and original volume, respectively, and symbolizes the volume coefficient of thermal expansion. In many materials, the value of α_v is anisotropic; that is, it depends on the crystallographic direction along which it is measured. For materials in which the thermal expansion is isotropic, α_v is approximately $3\alpha_l$.

An instrument known as dilatometer is used to measure the thermal expansion coefficient. It is also possible to trace thermal expansion using XRD.

At the microscopic level, thermal expansion can be attributed to the increase in the average distance between the atoms. Thus, the coefficient of thermal expansion of a material is related to the strength of the atomic bonds. The relation between inter-atomic distance and potential energy is shown in the Figure 8.2a. The curve is in the form of a potential energy trough, and the equilibrium interatomic spacing at 0 K, r_o, corresponds to the trough minimum. Heating to successively higher temperatures (T_1, T_2, T_3, etc.) raises the vibrational energy from E_1 to E_2 to E_3, and so on. The average vibrational amplitude of an atom corresponds to the trough width at each temperature, and the average interatomic distance is represented by the mean position, which increases with temperature from r_0 to r_1 to r_2, and so on.

Thermal expansion is really due to the asymmetric curvature of this potential energy trough, rather than the increased atomic vibrational amplitudes with rising temperature. If the potential energy curve were symmetric (Figure 8.2*b*), there would be no net change in interatomic separation and, consequently, no thermal expansion.

For every material, the greater the atomic bonding energy, the deeper and more narrow this potential energy trough. As a result, the increase in interatomic separation with a given rise in temperature will be lower, yielding a smaller value of α_l. Table 8.1 lists the linear coefficients of thermal expansion for some metals. With regards to temperature dependence, the magnitude

of the coefficient of expansion increases with rising temperature. The values in are taken at room temperature unless indicated otherwise.

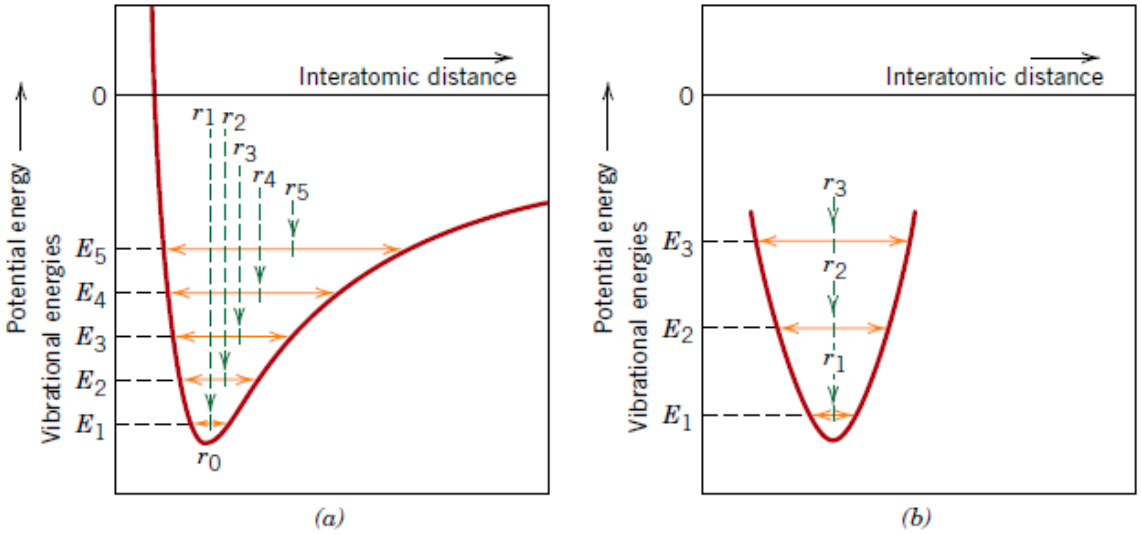


Figure 8.2 Change of inter-atomic distance with temperature.

8.3 Thermal conductivity

Thermal conduction is the phenomenon by which heat is transported from high to low temperature regions of a substance. The property that characterizes the ability of a material to transfer heat energy is called ***thermal conductivity***. Similar to diffusion coefficient, thermal conductivity is a microstructure sensitive property. It is expressed as;

$$q = -k\frac{dT}{dx} \qquad (8.5)$$

Where q denotes the *heat flux*, or heat flow, per unit time per unit area (area taken as that perpendicular to the flow direction), k is thermal conductivity, and dT/dx is the *temperature gradient* through the conducting medium. The units of q and k are respectively, W/m^2 and W/m. K. Equation 8.5 is valid only for steady-state heat flow; that is, for situations in which the heat flux does not change with time. Also, the minus sign in the expression indicates that the direction of heat flow is from hot to cold, or down the temperature gradient. Metals have k values in the range 20-400 W/m. K.

Heat energy in solids is transported by two mechanisms: lattice vibrations (phonons) and free electrons. However, usually only one or other predominates the proceedings. Valence electrons gain energy, move toward the colder areas of the material, and transfer their energy to other atoms. The amount of energy transported depends on number of excited electrons, their mobility i.e. type of material, lattice imperfections, and temperature. The thermal energy associated with phonons is transported in the direction of their motion.

In metals, since the valence band is not completely filled, with little thermal excitation number of electron move and contribute to the transfer of heat energy. Thus thermal conduction in metals is primarily due to movement of electrons. It is the same for electrical conduction. Both conductivities are related through the following relation:

$$\frac{k}{\sigma T} = L \qquad (8.6)$$

Where L is the Lorentz constant, 5.5×10^{-9} cal.ohm/sec.K^2 or 2.44×10^{-8} W.ohm/ K^2. The relation is termed *Wiedemann-Franz law*. Lorentz constant is supposed to be independent of temperature and the same for all metals if the heat energy is transported entirely by free electrons. Thus, the relationship is followed to a limited extension in many metals.

With increase in temperature, both number of carrier electrons and contribution of lattice vibrations increase. Thus thermal conductivity of a metal is expected to increase. However, because of greater lattice vibrations, electron mobility decreases. The combined effect of these factors leads to very different behavior for different metals. For example: thermal conductivity of iron initially decreases then increases slightly; thermal conductivity decreases with increase in temperature for aluminium; while it increases for platinum.

8.4 Thermal stresses

Thermal stresses are stresses induced in a body as a result of changes in temperature. An understanding of the origins and nature of thermal stresses is important because these stresses can lead to fracture or undesirable plastic deformation.

Apart from thermal shock, another instance of problem exists with thermal expansion of a material where there is no scope of dimensional changes. Thus due to temperature changes, material may experience thermal stresses ($\sigma_{thermal}$). The magnitude of the stress resulting from a temperature change from is given by

$$\sigma_{thermal} = E\alpha_l \Delta T \qquad (8.7)$$

Where E is the modulus of elasticity and α_l is the linear coefficient of thermal expansion.

Thermal stresses in a constrained body will be of compressive nature if it is heated, and vice versa. Another source for thermal stresses is thermal gradient within the body when a solid body is heated or cooled. It is because temperature distribution will depend on its size and shape. These thermal stresses may be established as a result of temperature gradients across a body, which are frequently caused by rapid heating or cooling.

Worked example 8.1

A brass rod is to be used in an application requiring its ends to be held rigid. If the rod is stress free at room temperature (20 °C), what is the maximum temperature to which the rod may be heated without exceeding a compressive stress of 172 MPa? Assume a modulus of elasticity of 100 GPa for brass.

Solution

Equation 8.7 is used to solve this problem, where the stress of 172 MPa is taken to be negative. Also, the initial temperature T_0 is 20 °C, and the magnitude of the linear coefficient of thermal expansion from Table 8.1 is 20.0×10^{-6} °C^{-1}. Thus, solving for the final temperature T_f yields

$$\Delta T = T_o - T_f = \frac{\sigma_{thermal}}{E\alpha_l} \Rightarrow T_f = T_o - \frac{\sigma_{thermal}}{E\alpha_l}$$

$$T_f = 20\ °C - \frac{-172\ MPa}{(100 \times 10^3)\ MPa \times\ 20.0 \text{ x } 10^{6}\ °\text{C}^{-1}} = 106\ °C$$

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CHAPTER 9. FERROUS AND NON-FERROUS MATERIALS

Selection of material for a specific purpose depends on many factors. Some of the important ones are strength, resistance to environmental degradation, etc. Another dimension an engineer should be aware of it is how to tailor the required properties of materials. As introduced in one of the earlier chapters, materials can be broadly classified as metals, ceramics and plastics. This chapter introduces different classes of metallic materials

Learning objectives

After studying this chapter, you should be able to do the following:

- Distinguish between ferrous and non-ferrous materials
- Explain various types of steel based on their compositional differences and distinctive properties.
- Describe the cast iron types and, for each, note its general mechanical characteristics.
- Explain the various alloys that constitute non-ferrous materials

9.0 Introduction

Metallic materials are broadly classified into ***ferrous*** and ***non-ferrous*** materials. This classification is primarily based on tonnage of materials used all around the world. Ferrous materials are those in which iron (***Fe***) is the principle constituent. All other materials are categorized as non-ferrous materials. Another classification is made based on their formability. If materials are hard to form, components with these materials are fabricated by casting, thus they are called cast alloys. If material can be deformed, they are known as wrought alloys.

Materials are usually strengthened by cold work and heat treatment. Strengthening by heat treatment involves either precipitation hardening or martensitic transformation, both of which constitute specific heat treating procedure. When a material cannot be strengthened by heat treatment, it is referred as non-heat-treatable alloy.

9.1 Ferrous Materials

Ferrous materials are produced in larger quantities than any other metallic material. Three factors; availability of abundant raw materials combined with economical extraction, ease of forming and their versatile mechanical and physical properties, accounts for it. One main drawback of ferrous alloys is their environmental degradation i.e. poor corrosion resistance. Other disadvantages include: relatively high density and comparatively low electrical and thermal conductivities.

In ferrous materials, the main alloying element is carbon and depending on the amount of carbon present, these alloys will have different properties, especially when the carbon content is either less/higher than 2.14%. Below this amount of carbon, material undergoes eutectoid transformation, while above that limit ferrous materials undergo eutectic transformation. Thus, ferrous alloys with less than 2.14% carbon are termed as ***steels***, and those with higher than 2.14% carbon are termed as ***cast irons***.

9.1.1 Steels

Steels are alloys of iron and carbon plus other alloying elements. In steels, carbon is present in atomic form, and occupies interstitial sites of iron microstructure. Alloying additions are necessary for many reasons including: improving properties, improving corrosion resistance, etc. Arguably steels are well known and most used materials than any other materials.

Mechanical properties of steels are very sensitive to carbon content. Hence, it is practical to classify steels based on their carbon content. Thus steels are basically three kinds: low-carbon steels (%wt of C < 0.3), medium carbon steels (0.3 < %wt of C < 0.6) and high-carbon steels (%wt of C > 0.6). The other parameter available for classification of steels is the amount of alloying additions, and based on this steels are two kinds: (plain) carbon steels and alloy-steels.

9.1.1.1 Low carbon steels

These are arguably produced in the greatest quantities than other alloys. Carbon present in these alloys is limited, and is not enough to strengthen these materials by heat treatment; hence these alloys are strengthened by cold work. Their microstructure consists of ferrite and pearlite, and these alloys are thus relatively soft, ductile combined with high toughness. Therefore these materials are easily machined and welded. Typical applications include: structural shapes, tin cans, automobile body components, buildings, etc.

A special group of ferrous alloys with noticeable amount of alloying additions are known as HSLA (high-strength low-alloy) steels. Common alloying elements are: copper (Cu), vanadium (V), Nickel (Ni), tungsten (W), Chromium (Cr), Molybdenum (Mo), etc. These alloys can be strengthened by heat treatment, and yet the same time they are ductile, formable. Typical applications of HSLA steels include: support columns, bridges, pressure vessels.

9.1.1.2 Medium carbon steels

These are stronger than low carbon steels. However, these are of less ductile than low carbon steels. These alloys can be heat treated to improve their strength. Usual heat treatment cycle consists of austenitizing, quenching, and tempering at suitable conditions to acquire required hardness. They are often used in tempered condition. As hardenability of these alloys is low, only thin sections can be heat treated using very high quench rates. Ni, Cr and Mo alloying additions improve their hardenability. Typical applications include: railway tracks and wheels, gears, other machine parts which may require good combination of strength and toughness.

9.1.1.3 Stainless steel

The name comes from their high resistance to corrosion i.e. they are rust-less (stain-less). Steels are made highly corrosion resistant by addition of special alloying elements, especially a minimum of 12% Cr along with Ni and Mo. Stainless steels are mainly three kinds: ferritic and hardenable Cr steels, austenitic and precipitation hardenable (martensitic, semi-austenitic) steels. This classification is based on the prominent constituent of the microstructure. Typical applications include cutlery, razor blades, surgical knives, etc.

Ferritic stainless steels are principally Fe-Cr-C alloys with 12-14% Cr. They also contain small additions of Niobium (Nb), Mo, V, and Ni. Austenitic stainless steels usually contain 18% Cr and 8% Ni in addition to other minor alloying elements. Ni stabilizes the austenitic

phase assisted by C and nitrogen (N). Other alloying additions include titanium (Ti), Nb, Mo (prevent weld decay), manganese (Mn) and Cu (helps in stabilizing austenite).

By alloying additions, martensitic steel is made to be above the room temperature. These alloys are heat treatable. Major alloying elements are: Cr, Mn and Mo.

Ferritic and austenitic steels are hardened and strengthened by cold work because they are not heat treatable. On the other hand martensitic steels are heat treatable. Austenitic steels are most corrosion resistant, produced in large quantities and are non-magnetic as against ferritic and martensitic steels, which are magnetic.

9.1.2 Cast Irons

Though ferrous alloys with more than 2.14 wt% C are designated as cast irons, commercially cast irons contain about 3.0-4.5% C along with some alloying additions. Alloys with this carbon content melt at lower temperatures than steels i.e. they are responsive to casting. Hence casting is the most used fabrication technique for these alloys.

Hard and brittle constituent presented in these alloys, cementite is a meta-stable phase, and can readily decompose to form α-ferrite and graphite. In this way disadvantages of brittle phase can easily be overcome. Tendency of cast irons to form graphite is usually controlled by their composition and cooling rate. Based on the form of carbon present, cast irons are categorized as gray, white, nodular and malleable cast irons.

9.1.2.1 Gray cast iron

These alloys consists carbon in form graphite flakes, which are surrounded by either ferrite or pearlite. Because of presence of graphite, fractured surface of these alloys look grayish, and so is the name for them. Alloying addition of Si (1-3 wt%) is responsible for decomposition of cementite, and also high fluidity. Thus castings of intricate shapes can be easily made. Due to graphite flakes, gray cast irons are weak and brittle. However they possess good damping properties, and thus typical applications include: base structures, bed for heavy machines, etc. they also show high resistance to wear.

9.1.2.2 White cast iron

When Si content is low (< 1%) in combination with faster cooling rates, there is no time left for cementite to get decomposed, thus most of the brittle cementite retains. Because of presence of cementite, fractured surface appear white, hence the name. They are very brittle and extremely difficult to machine. Hence their use is limited to wear resistant applications such as rollers in rolling mills. Usually white cast iron is heat treated to produce malleable iron.

9.1.2.3 Nodular (or ductile) cast iron

Alloying additions are of prime importance in producing these materials. Small additions of magnesium (Mg) / cerium (Ce) to the gray cast iron melt before casting can result in graphite to form nodules or sphere-like particles. Matrix surrounding these particles can be either ferrite or pearlite depending on the heat treatment. These are stronger and ductile than gray cast irons. Typical applications include: pump bodies, crank shafts, automotive components, etc.

9.1.2.4 Malleable cast iron

These formed after heat treating white cast iron. Heat treatments involve heating the material up to 800-900 °C, keep it for long hours, before cooling to room temperature. High temperature

incubation causes cementite to decompose and form ferrite and graphite. Thus these materials are stronger with appreciable amount of ductility. Typical applications include: railroad, connecting rods, marine and other heavy-duty services.

9.2 Non Ferrous Materials

Non-ferrous materials have specific advantages over ferrous materials in that, they can be fabricated with ease, high relatively low density, and high electrical and thermal conductivities. However different materials have distinct characteristics, and are used for specific purposes.

This section introduces some typical non-ferrous metals and their alloys of commercial importance.

9.2.1 Aluminium alloys

These are characterized by low density, high thermal and electrical conductivities, and good corrosion resistant characteristics. As aluminium (Al) has FCC crystal structure, these alloys are ductile even at low temperatures and can be formed easily. However, the great limitation of these alloys is their low melting point (660 °C), which restricts their use at elevated temperatures. Strength of these alloys can be increased by both cold and heat treatment. Based on these, alloys are designated in to two groups, cast and wrought. Chief alloying elements include: Cu, Si, Mn, Mg, Zn. Recently, alloys of Al and other low-density metals like Li, Mg, Ti gained much attention as there is much concern about vehicle weight reduction. Al-Li alloys enjoy much more attention especially as they are very useful in aircraft and aerospace industries. Common applications of Al alloys include: beverage cans, automotive parts, bus bodies, aircraft structures, etc. Some of the alloys are capable of strengthening by precipitation, while others have to be strengthened by cold work or solid solution methods.

9.2.2 Copper alloys

As history goes by, bronze has been used for thousands of years. It is actually an alloy of Cu and tin (Sn). Unalloyed Cu is soft, ductile thus hard to machine, and has virtually unlimited capacity for cold work. One special feature of most of these alloys is their corrosion resistant in diverse atmospheres. Most of these alloys are strengthened by either cold work or solid solution method. The most common Cu alloys are Brass, alloys of Cu and Zn where Zn is substitutional addition (e.g.: yellow brass, catridge brass, muntz metal, gilding metal); Bronze, alloys of Cu and other alloying additions like Sn, Al, Si and Ni. Bronzes are stronger and more corrosion resistant than brasses. Beryllium coppers who possess combination of relatively high strength, excellent electrical and corrosion properties, wear resistance, can be cast, hot worked and cold worked. Applications of copper alloys include costume jewelry, coins, musical instruments, electronics, springs, bushes, surgical and dental instruments, radiators, etc.

9.2.3 Magnesium alloys

The most sticking property of Mg is its low density among all structural metals. Mg has HCP structure, thus Mg alloys are difficult to form at room temperatures. Hence Mg alloys are usually fabricated by casting or hot working. As in case of Al, alloys are cast or wrought type, and some of them are heat treatable. Major alloying additions are: Al, Zn, Mn and rare earths.

Common applications of Mg alloys include: hand-held devices like saws, tools, automotive parts like steering wheels, seat frames, electronics like casing for laptops, camcorders, cell phones etc.

9.2.4 Titanium alloys

Titanium (Ti) and its alloys are of relatively low density, high strength and have very high melting point. At the same time they are easy to machine and forge. However, the major limitation is its chemical reactivity at high temperatures, which necessitated special techniques to extract. Thus these alloys are expensive. They also possess excellent corrosion resistance in diverse atmospheres, and wear properties. Most common applications include: space vehicles, airplane structures, surgical implants, and petroleum and chemical industries.

9.2.5 Refractory metals

These are metals of very high melting points. For example: Nb, Mo, W and Ta. They also possess high strength and high elastic modulus. Common applications include: space vehicles, x-ray tubes, welding electrodes, and where there is a need for corrosion resistance.

9.2.6 Noble metals

These are eight all together namely argon (Ag), gold (Au), platinum (Pt), protactinium (Pa), rhodium (Rh), ruthenium (Ru), iridium (Ir) and osmium (Os). All these possess some common properties such as: expensive, soft and ductile, oxidation resistant.

Ag, Au and Pt are used extensively in jewelry, alloys are Ag and Au are employed as dental restoration materials; Pt is used in chemical reactions as a catalyst and in thermocouples.

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CHAPTER 10. CERAMICS AND POLYMER MATERIALS

Selection of material for a specific purpose depends on many factors. Some of the important ones are strength, resistance to environmental degradation, etc. Another dimension an engineer should be aware of it is how to tailor the required properties of materials. As introduced in one of the earlier chapters, materials can be broadly classified as metals, ceramics and polymers. This chapter introduces ceramic and polymer materials.

Learning objectives

After studying this chapter, you should be able to do the following:

- Distinguish between ceramics and polymer materials
- Explain various types and application of ceramic materials
- Explain the various types and application of polymer materials

10.0 Introduction

Ceramics and polymers form an important part of materials group. Ceramics are compounds between metallic and nonmetallic elements for which the inter-atomic bonds are either ionic or predominantly ionic. The term ceramics comes from the Greek word *keramikos* which means 'burnt stuff'. The characteristic properties of ceramics are, in fact, optimized through thermal treatments. Ceramic materials exhibit physical properties different from metallic materials. Thus, metallic materials, ceramics, and polymers tend to complement each other in service.

Polymers play a very important role in human life. In fact, our body possesses a lot of polymers, e.g. Proteins, enzymes, etc. Other naturally occurring polymers like wood, rubber, leather and silk are serving the humankind for many centuries now. Modern scientific tools revolutionized the processing of polymers thus available synthetic polymers like useful plastics, rubbers and fiber materials. As with other engineering materials (metals and ceramics), the properties of polymers are related their constituent structural elements and their arrangement. The suffix in polymer 'mer' is originated from Greek word *meros* – which means part. The word polymer is thus coined to mean material consisting of many parts/mers. Most of the polymers are basically organic compounds, however they can be inorganic (e.g. silicones based on Si-O network).

10.1 Types and Application of Ceramic Materials

Ceramics greatly differ in their basic composition. The properties of ceramic materials also vary greatly due to differences in bonding, and thus found a wide range of engineering applications. The two important classification of ceramic materials is based on their specific applications and composition. Based on their composition, ceramics are classified as: Oxides, carbides, nitrides, sulfides, fluorides, etc. The classification based on their application, such as: glasses, clay products, refractories, abrasives, cements, advanced ceramics.

In general, ceramic materials used for engineering applications can be divided into two groups: traditional ceramics, and the engineering ceramics. Typically, traditional ceramics are made from three basic components: clay, silica (flint) and feldspar. For example bricks, tiles and porcelain articles. However, engineering ceramics consist of highly pure compounds of aluminium oxide (Al_2O_3), silicon carbide (SiC) and silicon nitride (Si_3N_4).

10.1.1 Types of Ceramics

10.1.1.1 Glasses

Glasses are a familiar group of ceramics; containers, windows, mirrors, lenses, etc. They are non-crystalline silicates containing other oxides, usually CaO, Na_2O, K_2O and Al_2O_3 which influence the glass properties and its color. Typical property of glasses that is important in engineering applications is its response to heating. There is no definite temperature at which the liquid transforms to a solid as with crystalline materials. A specific temperature, known as glass transition temperature or fictive temperature is defined based on viscosity above which material is named as super cooled liquid or liquid, and below it is termed as glass.

10.1.1.2 Clay products

Clay is the one of most widely used ceramic raw material. It is found in great abundance and popular because of ease with which products are made. Clay products are mainly two kinds – structural products (bricks, tiles, sewer pipes) and white-wares (porcelain, chinaware, pottery, etc.).

10.1.1.3 Refractories

These are described by their capacity to withstand high temperatures without melting or decomposing; and their inertness in severe environments. Thermal insulation is also an important functionality of refractories.

10.1.1.4 Abrasive ceramics

These are used to grind, wear, or cut away other material. Thus the prime requisite for this group of materials is hardness or wear resistance in addition to high toughness. As they may also exposed to high temperatures, they need to exhibit some refractoriness. Diamond, silicon carbide, tungsten carbide, silica sand, aluminium oxide / corundum are some typical examples of abrasive ceramic materials.

10.1.1.5 Cements

Cement, plaster of paris and lime come under this group of ceramics. The characteristic property of these materials is that when they are mixed with water, they form slurry which sets subsequently and hardens finally. Thus it is possible to form virtually any shape. They are also used as bonding phase, for example between construction bricks.

10.1.1.6 Advanced ceramics

These are newly developed and manufactured in limited range for specific applications. Usually their electrical, magnetic and optical properties and combination of properties are exploited. Typical applications: heat engines, ceramic armors, electronic packaging, etc.

10.1.2 Application of ceramics

Some typical ceramics and respective applications are as follows:

Aluminium oxide / Alumina (Al_2O_3): It is one of most commonly used ceramic material. It is used in many applications such as to contain molten metal, where material is operated at very high temperatures under heavy loads, as insulators in spark plugs, and in some unique

applications such as dental and medical use. Chromium doped alumina is used for making lasers.

***Aluminium nitride* (AlN):** Due to its typical properties such as good electrical insulation but high thermal conductivity, it is used in many electronic applications such as in electrical circuits operating at a high frequency. It is also suitable for integrated circuits. Other electronic ceramics include – barium titanate ($BaTiO_3$) and Cordierite ($2MgO\text{-}2Al_2O_3\text{-}5SiO_2$).

***Diamond* (C):** The hardest material known to available in nature. It has many applications such as industrial abrasives, cutting tools, abrasion resistant coatings, etc. it is, of course, also used in jewelry.

***Lead zirconium titanate* (PZT):** It is the most widely used piezoelectric material, and is used as gas igniters, ultrasound imaging, in underwater detectors.

***Silica* (SiO_2):** Is an essential ingredient in many engineering ceramics, the most widely used ceramic material. Silica-based materials are used in thermal insulation, abrasives, laboratory glassware, etc. It also found application in communications media as integral part of optical fibers. Fine particles of silica are used in tires, paints, etc.

***Silicon carbide* (SiC):** It is known as one of best ceramic material for very high temperature applications. It is used as coatings on other material for protection from extreme temperatures. It is also used as abrasive material. It is used as reinforcement in many metallic and ceramic based composites. It is a semiconductor and often used in high temperature electronics. Silicon nitride (Si_3N_4) has properties similar to those of SiC but is somewhat lower, and found applications in such as automotive and gas turbine engines.

***Titanium oxide* (TiO_2)**: It is mostly found as pigment in paints. It also forms part of certain glass ceramics. It is used to making other ceramics like $BaTiO_3$.

***Titanium boride* (TiB_2):** It exhibits great toughness properties and hence found applications in armor production. It is also a good conductor of both electricity and heat.

***Uranium oxide* (UO_2):** It is mainly used as fuel in nuclear reactor. It has exceptional dimensional stability because its crystal structure can accommodate the products of fission process.

***Yttrium aluminium garnet* (YAG, $Y_3Al_5O_{12}$):** it has main application in lasers (Nd-YAG lasers).

***Zirconia* (ZrO_2):** it is also used in producing many other ceramic materials. It is also used in making oxygen gas sensors, as additive in many electronic ceramics. Its single crystals are part of jewelry.

10.2 Types of Polymer Materials

Polymers are classified in several ways; by how the molecules are synthesized, by their molecular structure, or by their chemical family. For example, linear polymers consist of long molecular chains, while the branched polymers consist of primary long chains and secondary chains that stem from these main chains. However, linear does not mean straight lines. The better way to classify polymers is according to their mechanical and thermal behavior. Industrially, polymers are classified into two main classes – ***plastics*** and ***elastomers***.

Plastics are moldable organic resins. These are either natural or synthetic, and are processed by forming or molding into shapes. Plastics are important materials for many reasons. They

have a wide range of properties, some of which are unattainable from any other materials, and in most cases they are relatively low in cost. The properties of plastics are: light weight, wide range of colors, low thermal and electrical conductivity, less brittle, good toughness, good resistance to acids, bases and moisture, high dielectric strength (use in electrical insulation), etc.

Plastics are again classified in two groups depending on their mechanical and thermal behavior as *thermoplasts* (thermoplastic polymers) and *thermosets* (thermosetting polymers).

Thermoplasts: These plastics soften when heated and harden when cooled; processes that are totally reversible and may be repeated. These materials are normally fabricated by the simultaneous application of heat and pressure. They are linear polymers without any cross-linking in structure where long molecular chains are bonded to each other by secondary bonds and/or inter-wined. They have the property of increasing plasticity with increasing temperature which breaks the secondary bonds between individual chains. Common thermoplasts are: acrylics, PVC, nylons, polypropylene, polystyrene, polymethyl methacrylate (plastic lenses or perspex), etc.

Thermosets: These plastics require heat and pressure to mold them into shape. They are formed into permanent shape and cured or 'set' by chemical reactions such as extensive cross-linking. They cannot be re-melted or reformed into another shape but decompose upon being heated to too high a temperature. Thus thermosets cannot be recycled, whereas thermoplasts can be recycled. The term thermoset implies that heat is required to permanently set the plastic. Most thermosets composed of long chains that are strongly cross-linked (and/or covalently bonded) to one another to form 3-D network structures to form a rigid solid. Thermosets are generally stronger, but more brittle than thermoplasts. The advantages of thermosets for engineering design applications include one or more of the following: high thermal stability, high dimensional stability, high rigidity, light weight, high electrical and thermal insulating properties and resistance to creep and deformation under load. There are two methods whereby cross-linking reaction can be initiated; cross-linking can be accomplished by heating the resin in suitable a mold (e.g. bakelite), or resins such as epoxies (e.g. araldite) are cured at low temperature by the addition of a suitable cross-linking agent, an amine. Epoxies, vulcanized rubbers, phenolics, unsaturated polyester resins, and amino resins (ureas and melamines) are examples of thermosets.

Elastomers also known as rubbers, these polymers can undergo large elongations under load, at room temperature, and return to their original shape when the load is released. There are number of man-made elastomers in addition to natural rubber. These consist of coil-like polymer chains those can reversibly stretch by applying a force.

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